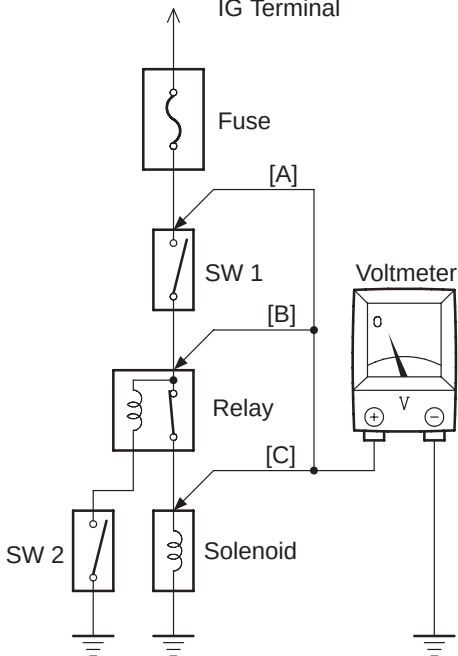


BACK

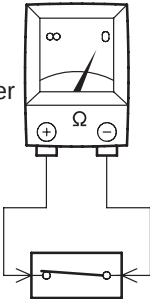
To Ignition SW
IG Terminal



BACK

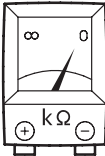
Ohmmeter

SW

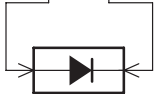
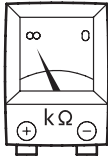
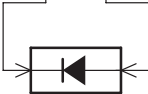


BACK

Ohmmeter



Diode

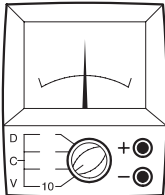


[BACK](#)

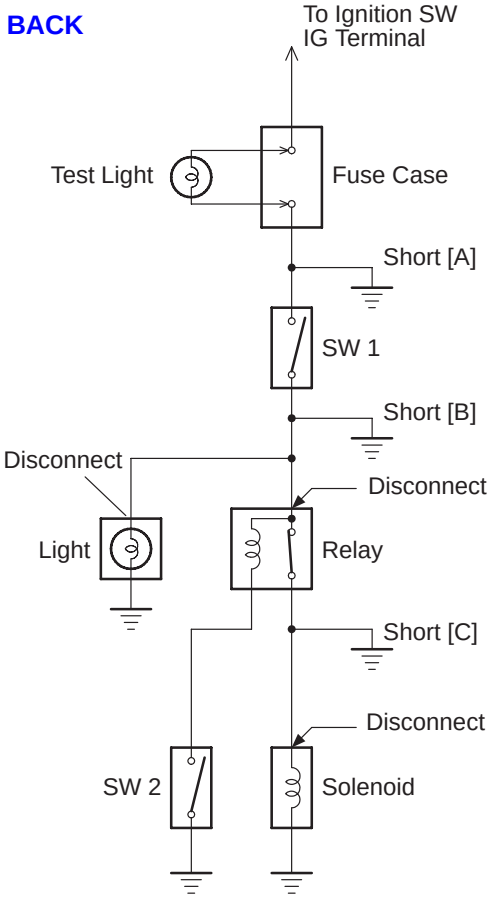
Digital Type



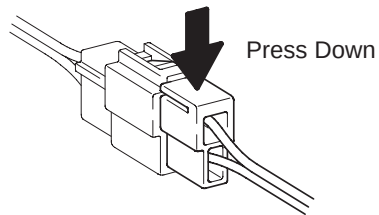
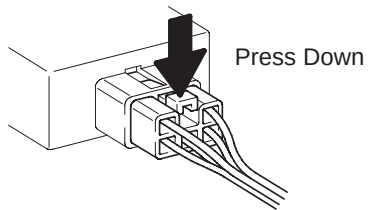
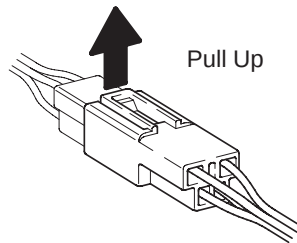
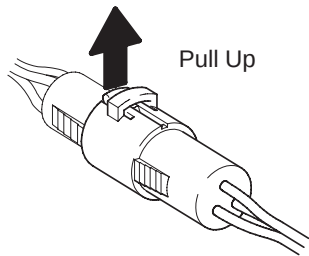
Analog Type



BACK

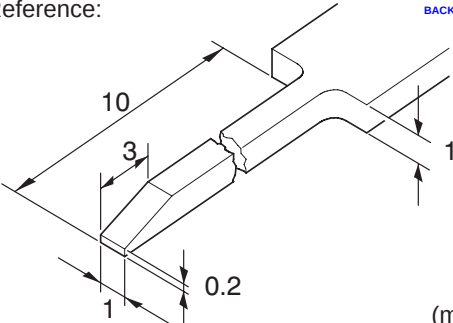


BACK



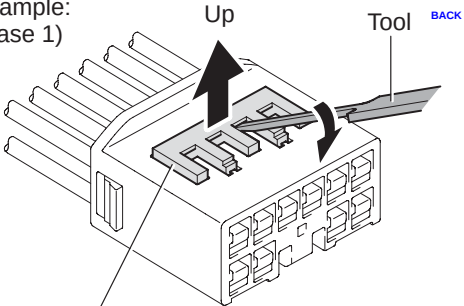
Reference:

BACK



(mm)

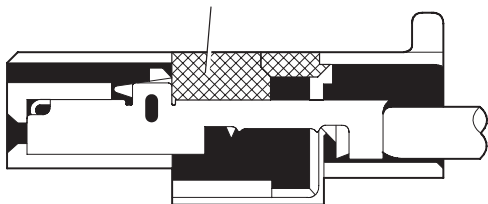
Example:
(Case 1)



Terminal Retainer

BACK

Terminal Retainer

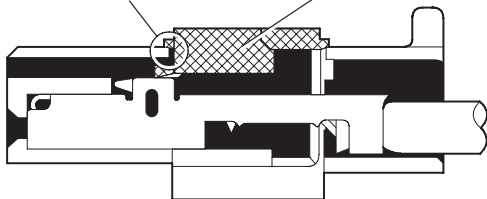


[Retainer at Full Lock Position]



Stopper

Terminal Retainer

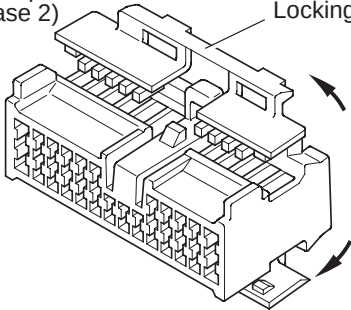


[Retainer at Temporary Lock Position]

Example:
(Case 2)

Secondary
Locking Device

BACK

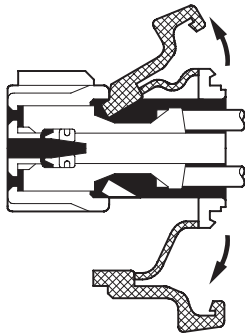


BACK

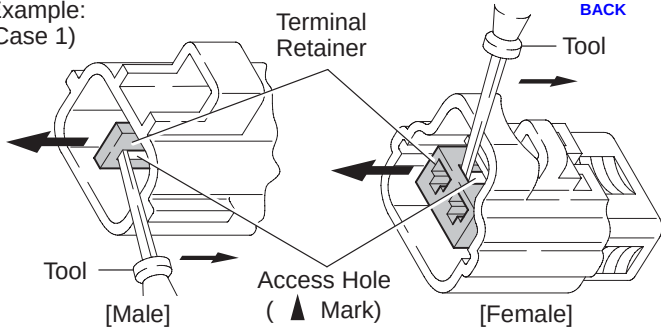
Tool

Tab

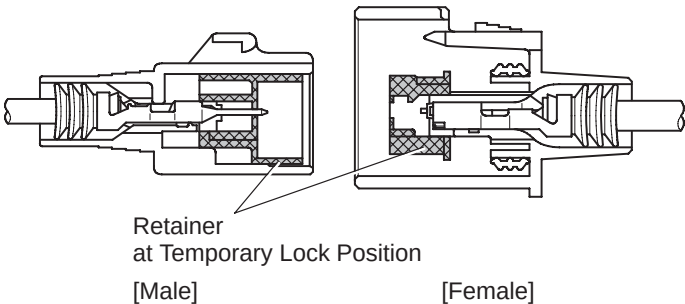
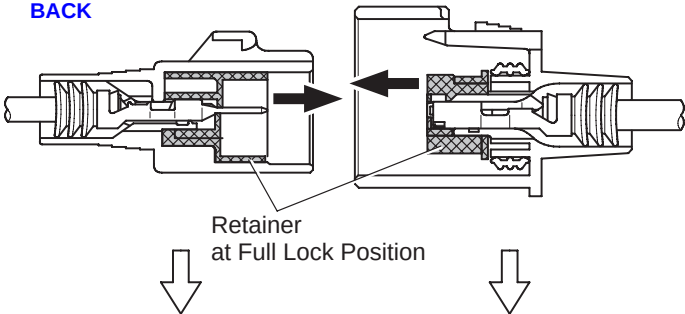
Tab



Example:
(Case 1)



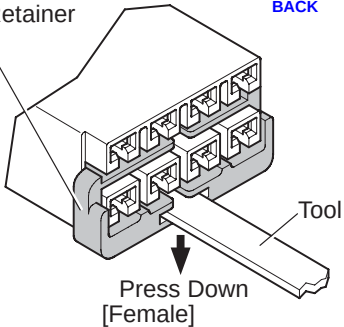
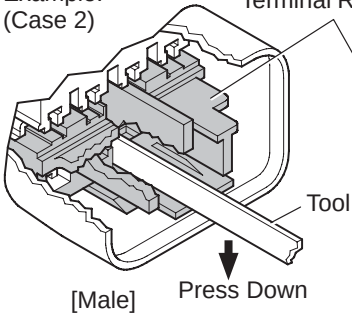
BACK



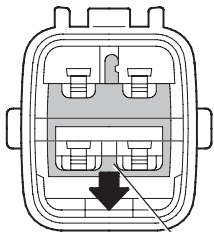
Example:
(Case 2)

Terminal Retainer

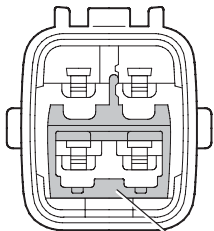
BACK



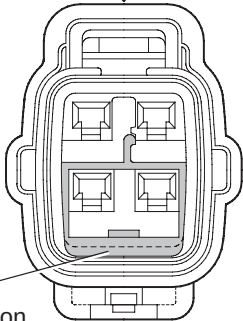
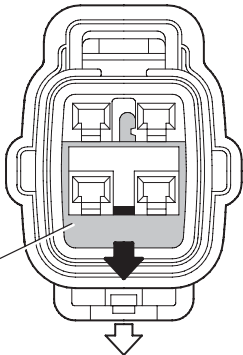
BACK



Retainer at
Full Lock Position



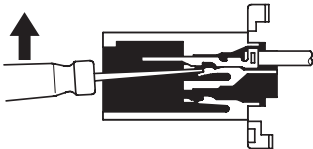
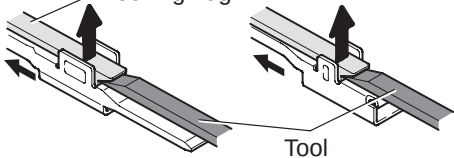
Retainer at
Temporary Lock Position
[Male]



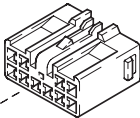
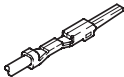
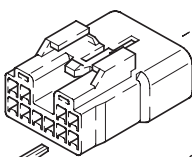
[Female]

BACK

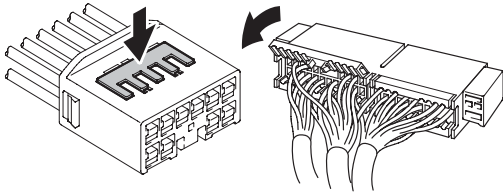
Locking Lug



BACK



BACK



ABBREVIATIONS

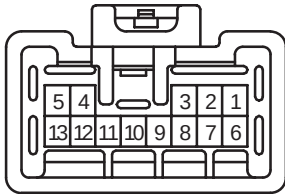
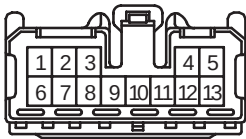
The following abbreviations are used in this manual.

2WD	=	Two Wheel Drive Vehicles
4WD	=	Four Wheel Drive Vehicles
A/C	=	Air Conditioner
A/T	=	Automatic Transmission
ABS	=	Anti-Lock Brake System
ADD	=	Automatic Disconnecting Differential
CAN	=	Controller Area Network
CD	=	Compact Disc
DLC3	=	Data Link Connector 3
ECT	=	Electronic Control Transmission
ECU	=	Electronic Control Unit
EDU	=	Electronic Driving Unit
EGR	=	Exhaust Gas Recirculation
IC	=	Integrated Circuit
INT	=	Intermittent
J/B	=	Junction Block
LCD	=	Liquid Crystal Display
LH	=	Left-Hand
M/T	=	Manual Transmission
R/B	=	Relay Block
RH	=	Right-Hand
SRS	=	Supplemental Restraint System
SW	=	Switch
TEMP.	=	Temperature
VRV	=	Vacuum Regulating Valve
VSV	=	Vacuum Switching Valve
w/	=	With
w/o	=	Without

* The titles given inside the components are the names of the terminals (terminal codes) and are not treated as being abbreviations.

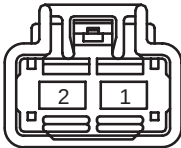
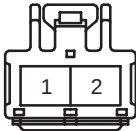
BACK

EA1



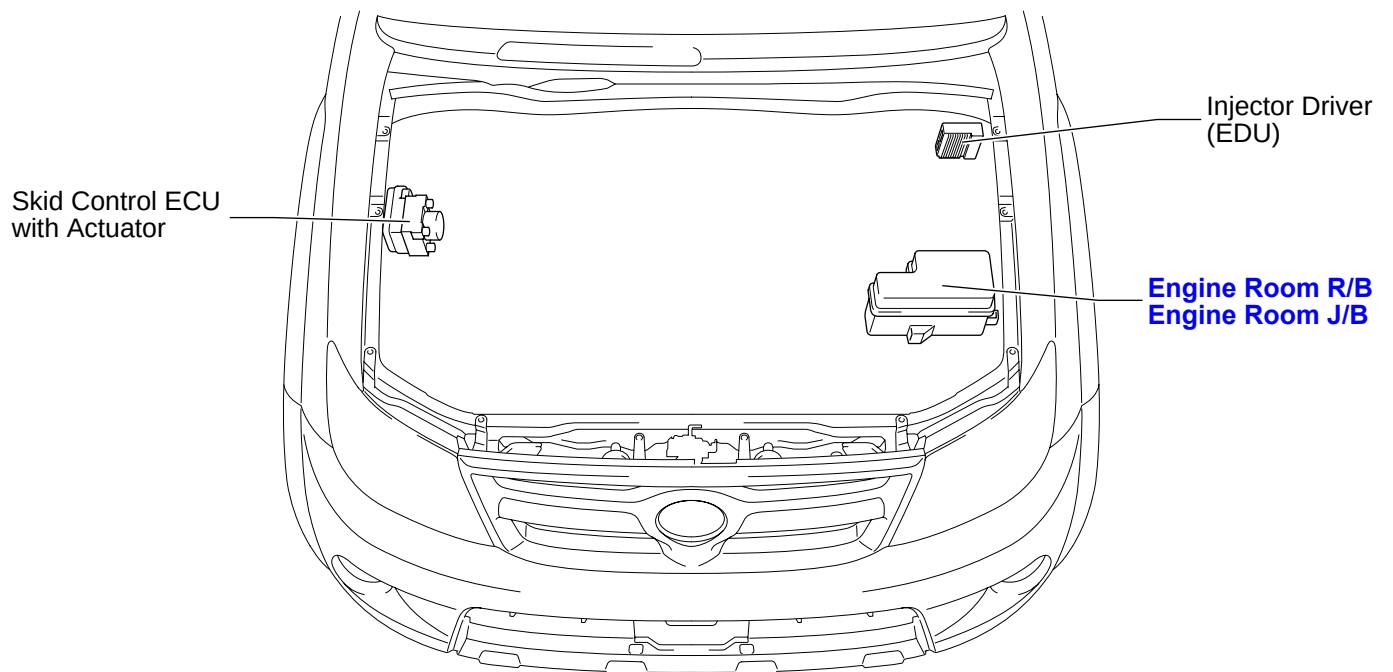
BACK

EA2

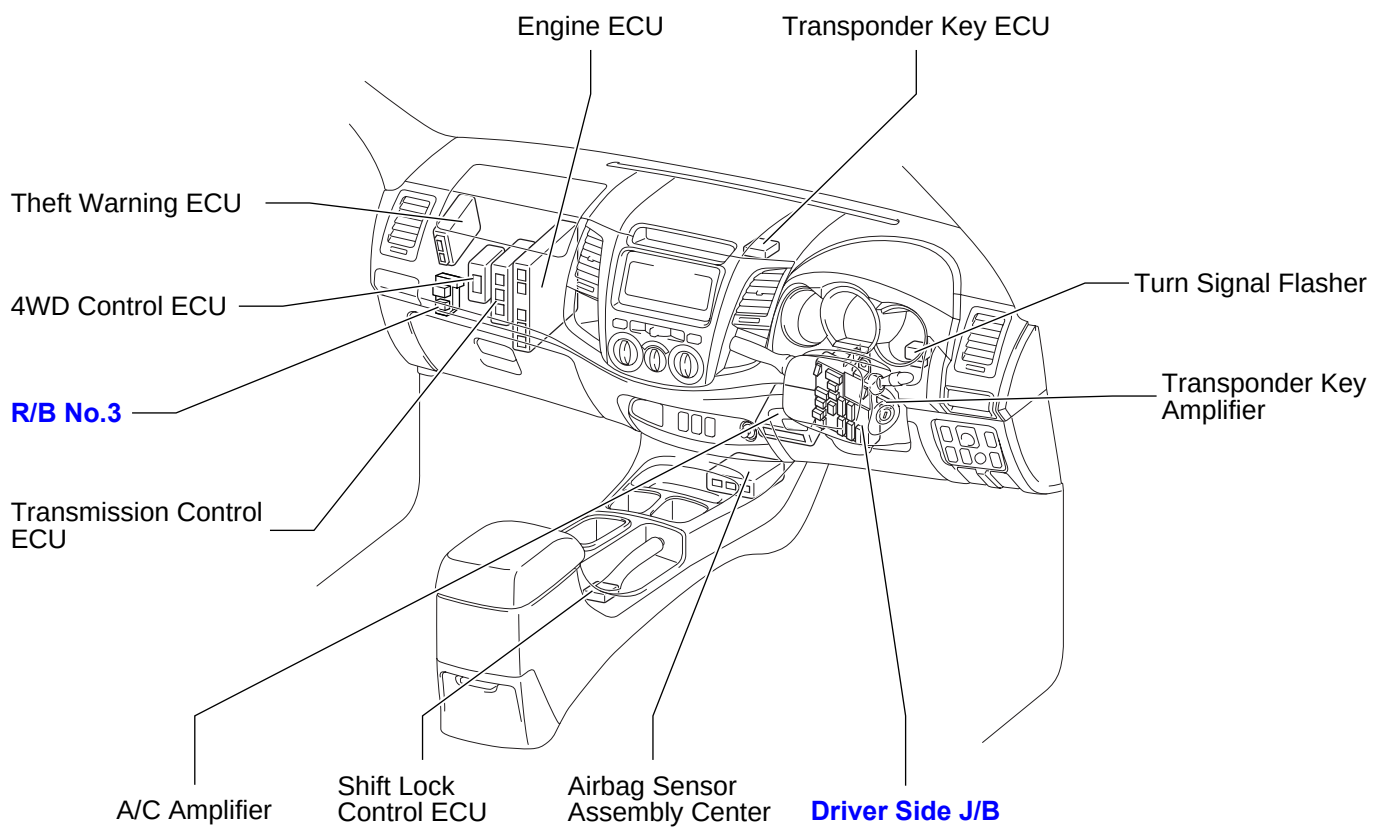


F RELAY LOCATIONS

[Engine Compartment]



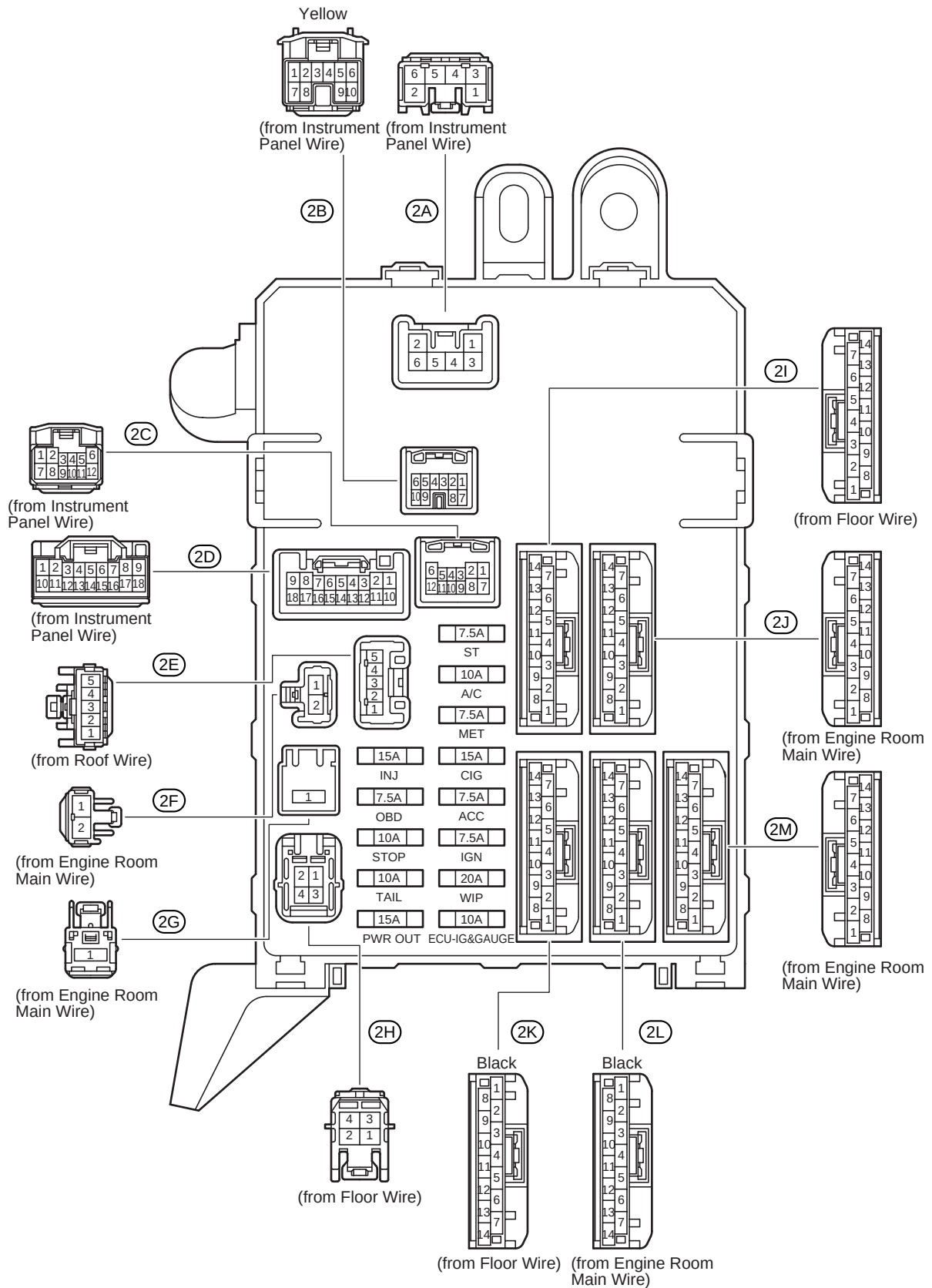
[Instrument Panel]

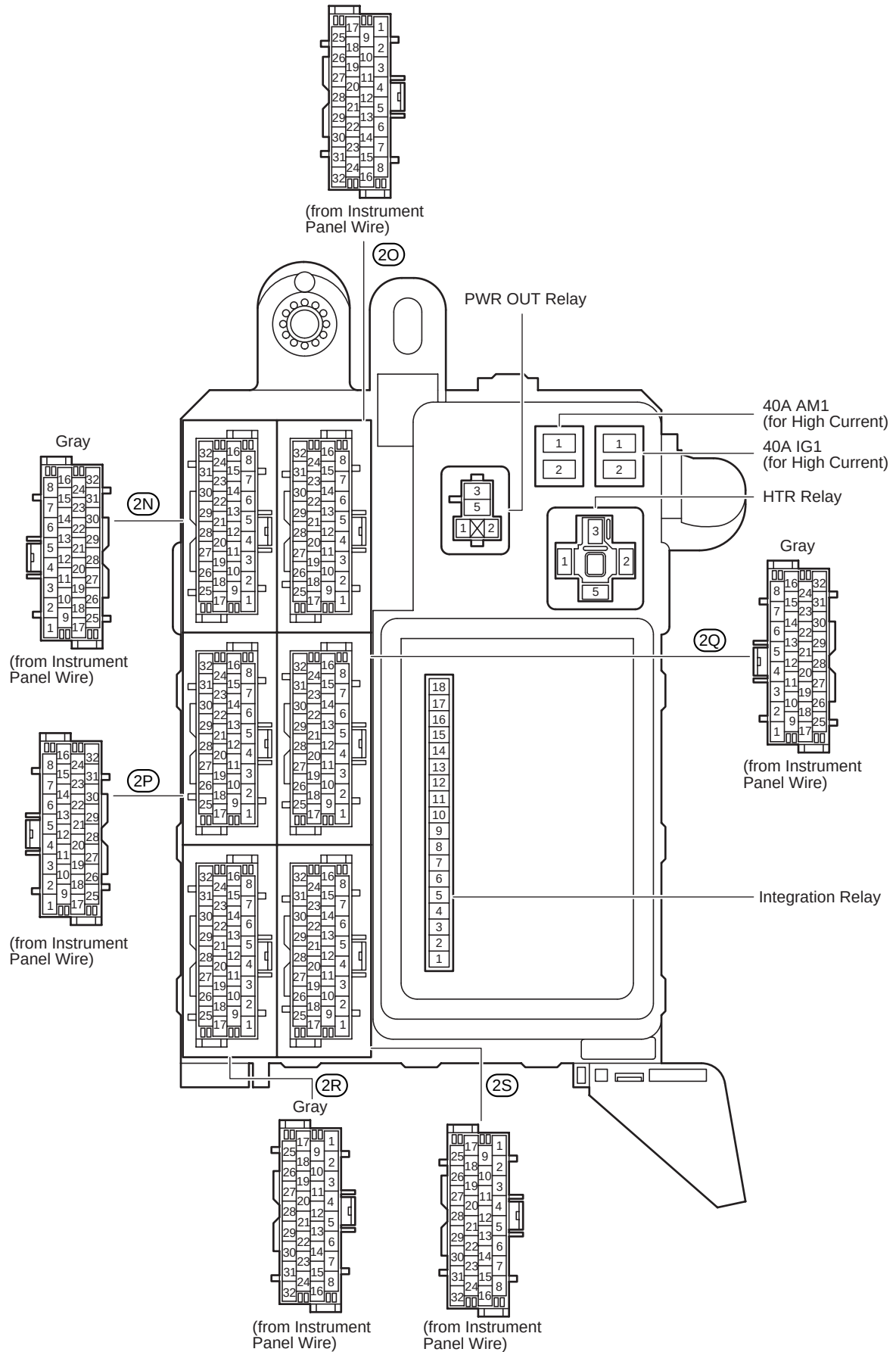


F RELAY LOCATIONS

○ : Driver Side J/B

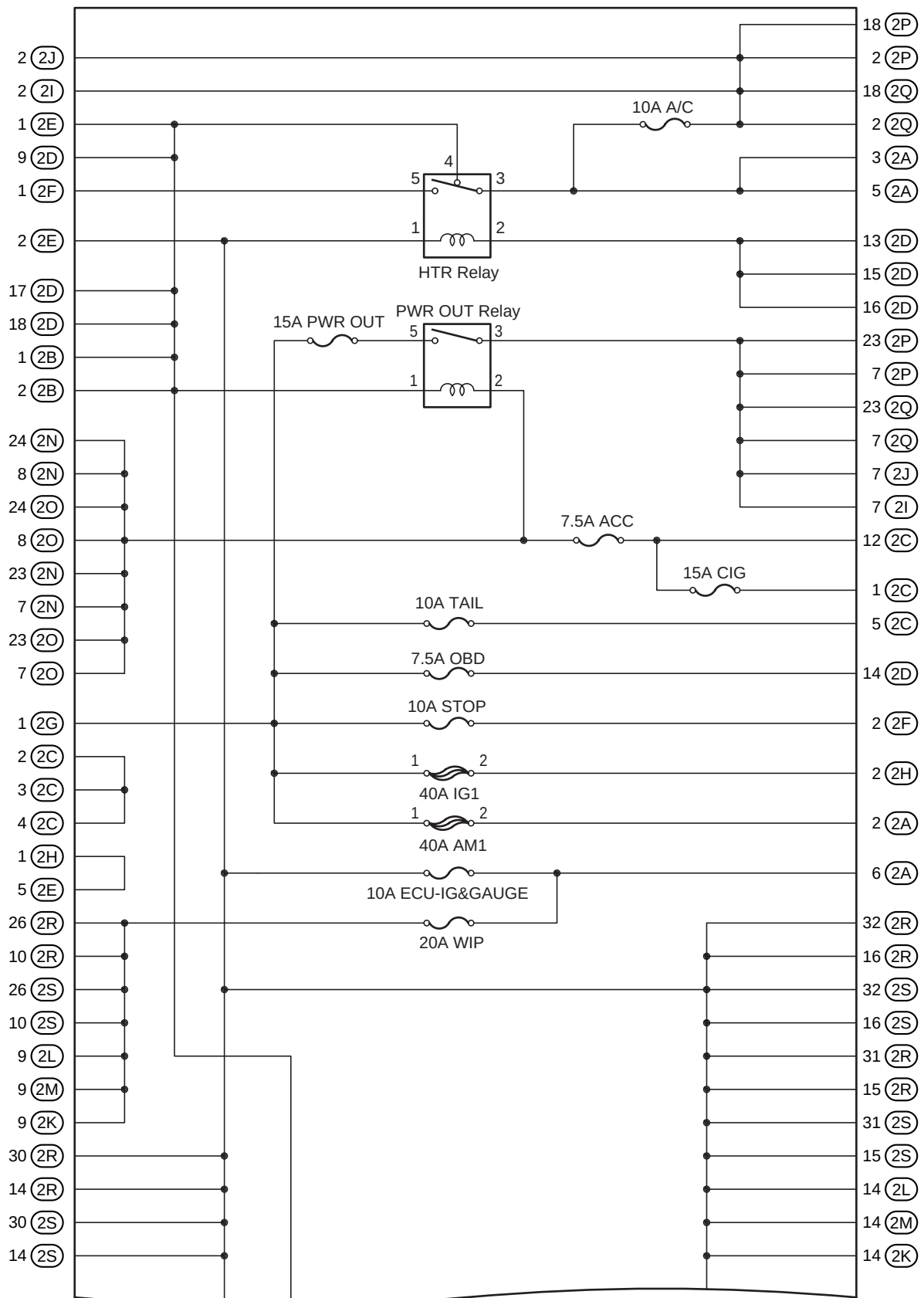
Instrument Panel Brace RH (See Page 21)





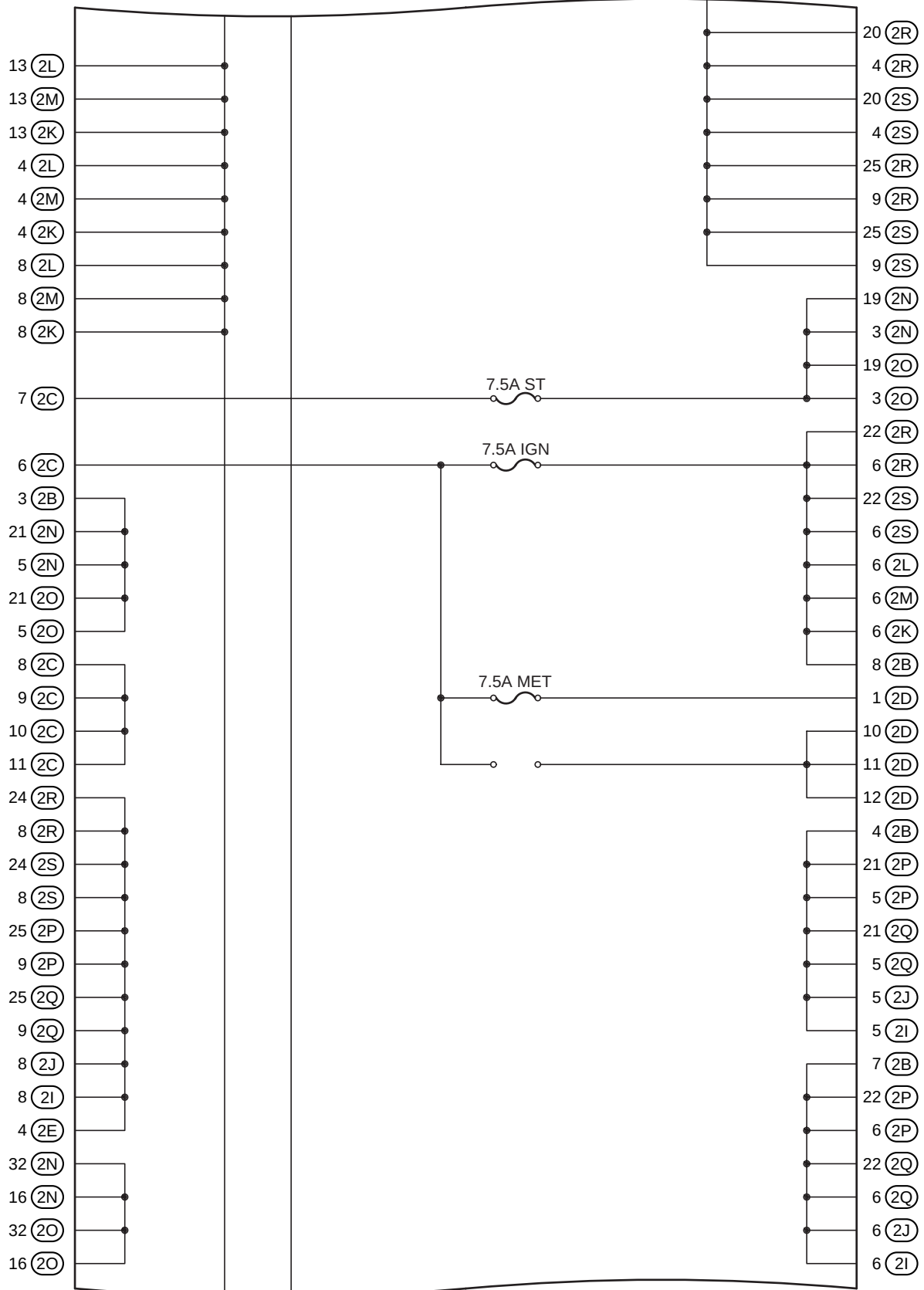
F RELAY LOCATIONS

[Driver Side J/B Inner Circuit]



(Cont. Next Page)

(Cont'd)

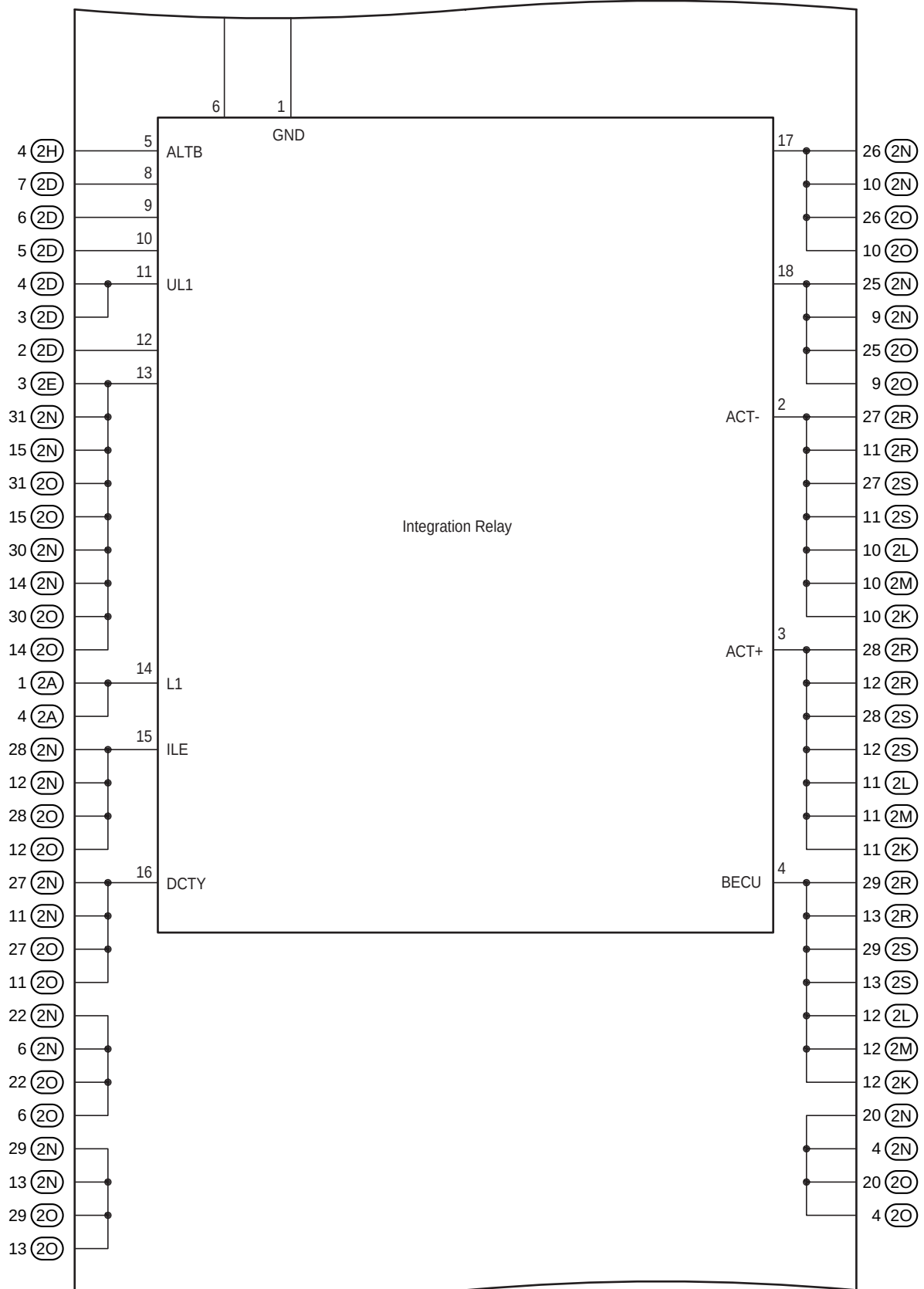


(Cont. Next Page)

F RELAY LOCATIONS

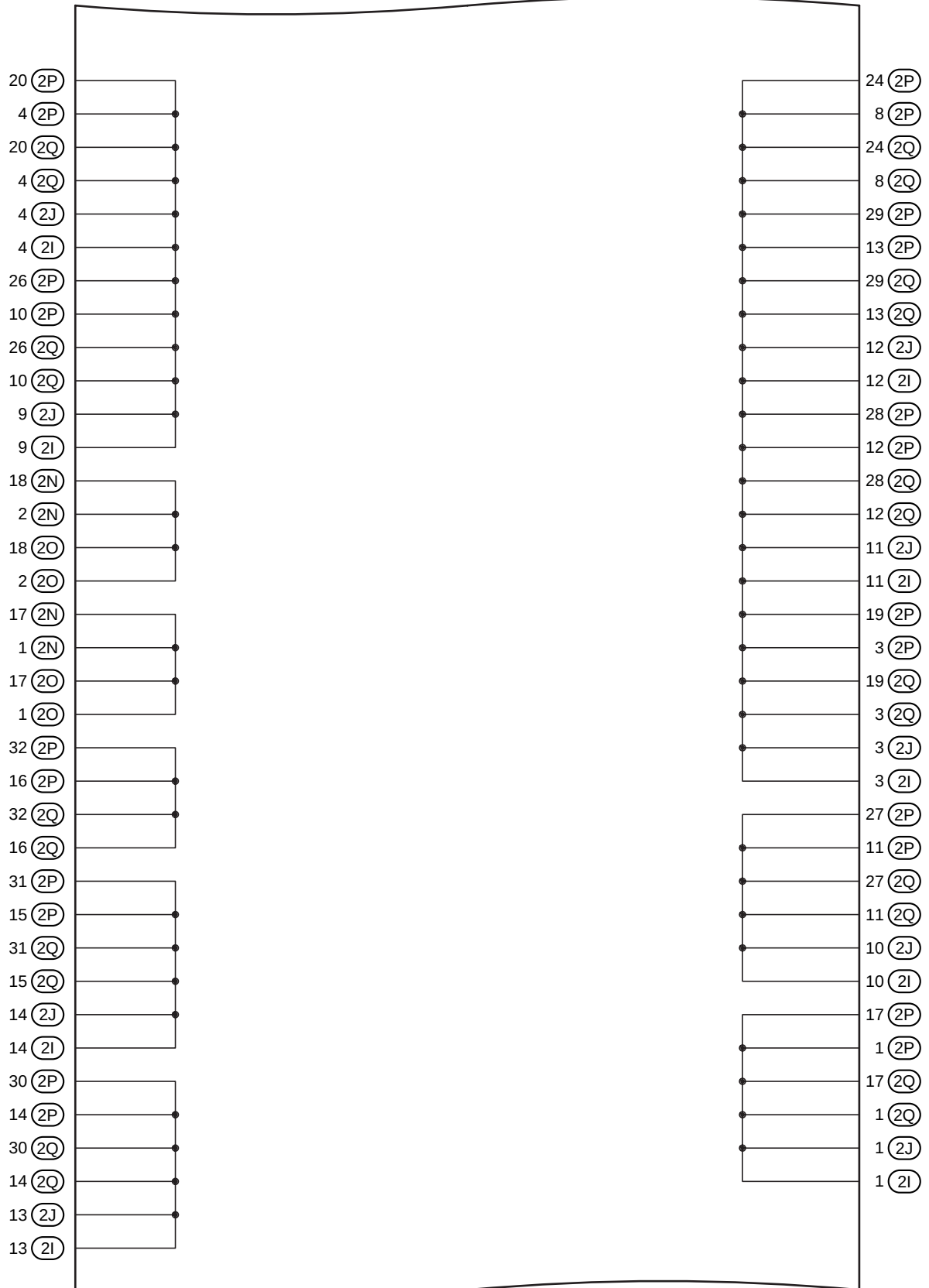
[Driver Side J/B Inner Circuit]

(Cont'd)



(Cont. Next Page)

(Cont'd)

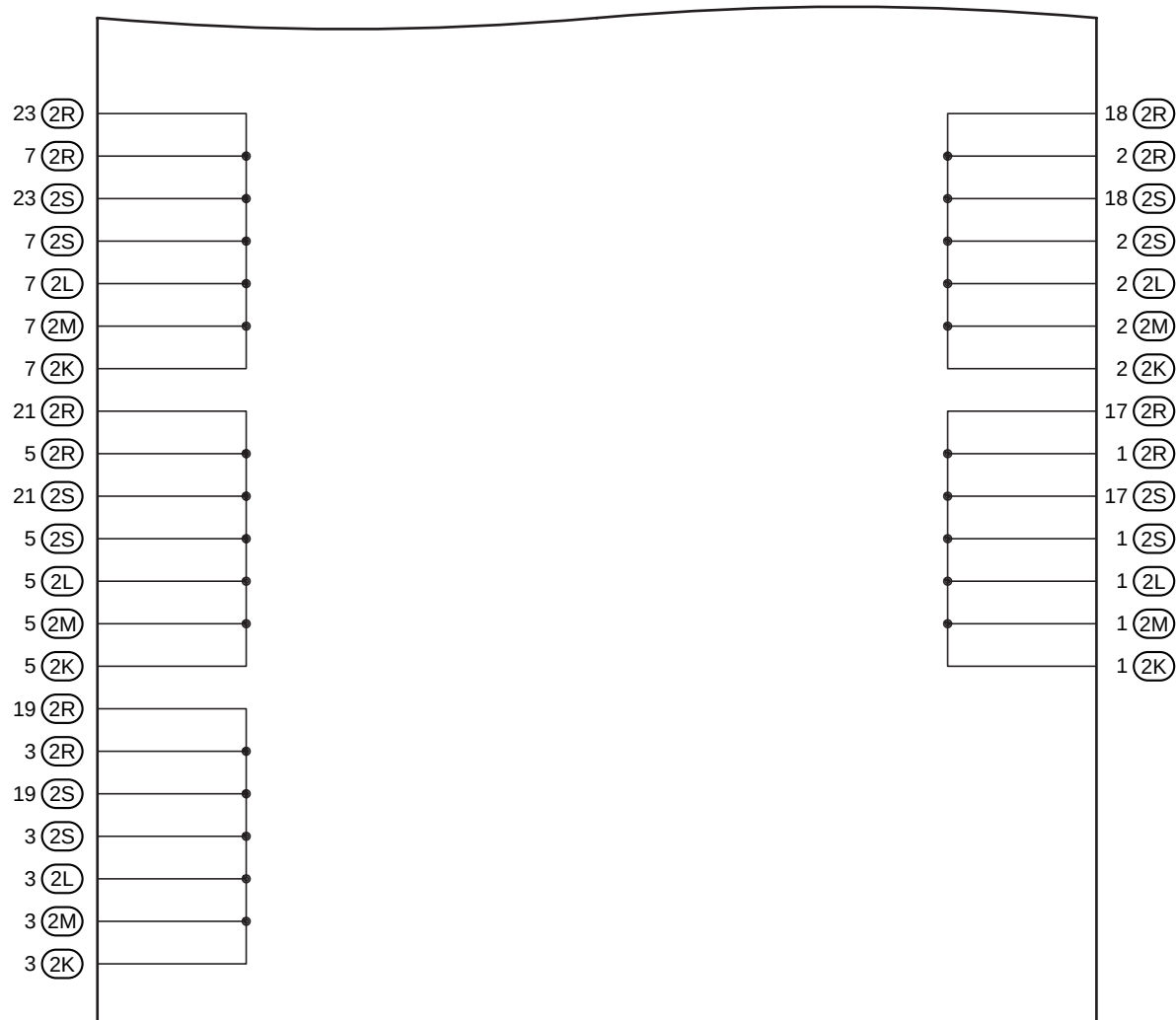


(Cont. Next Page)

F RELAY LOCATIONS

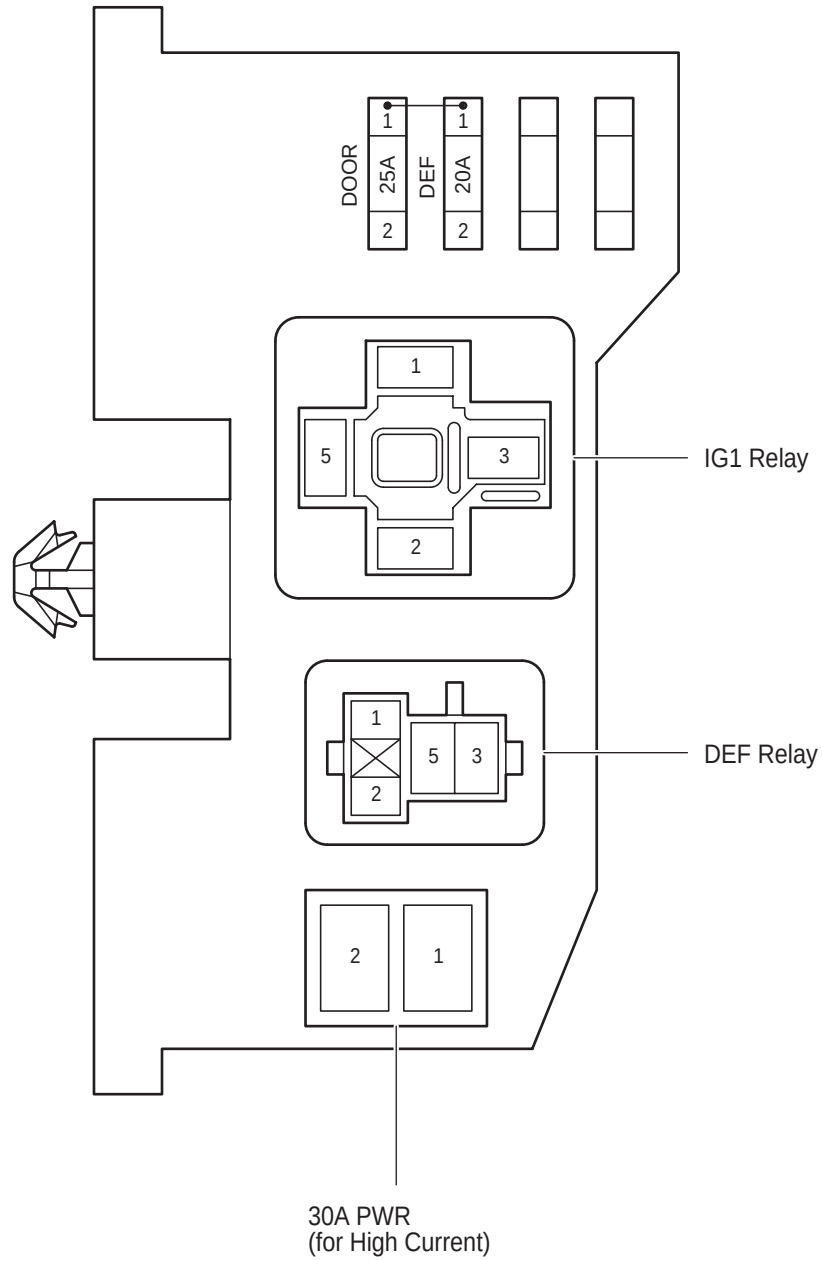
[Driver Side J/B Inner Circuit]

(Cont'd)



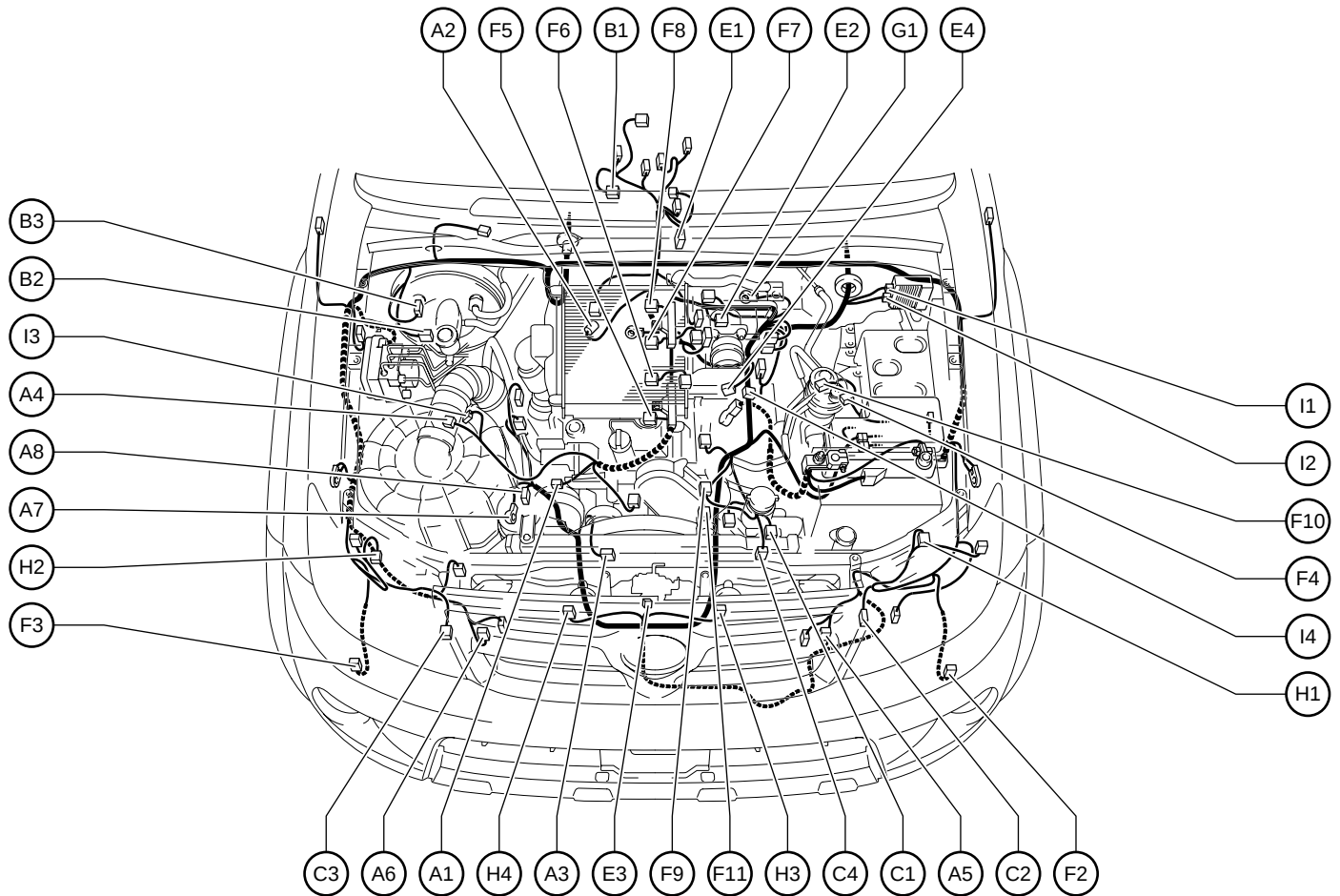
③ : R/B No.3

Cowl Side Panel LH (See Page 21)



G ELECTRICAL WIRING ROUTING

Position of Parts in Engine Compartment



A 1 A/C Compressor
 A 2 A/T Fluid Temp. Sensor
 A 3 ADD Actuator
 A 4 Air Flow Meter
 A 5 Airbag Sensor (Front LH)
 A 6 Airbag Sensor (Front RH)
 A 7 Alternator
 A 8 Alternator

B 1 Back-Up Lamp SW
 B 2 Brake Fluid Level Warning SW
 B 3 Brake Vacuum Warning SW

C 1 Camshaft Position Sensor
 C 2 Clearance Lamp (Front LH)
 C 3 Clearance Lamp (Front RH)
 C 4 Crankshaft Position Sensor

E 1 ECT Solenoid
 E 2 EGR Valve Position Sensor
 E 3 Engine Hood Courtesy SW
 E 4 Engine Oil Pressure Sensor

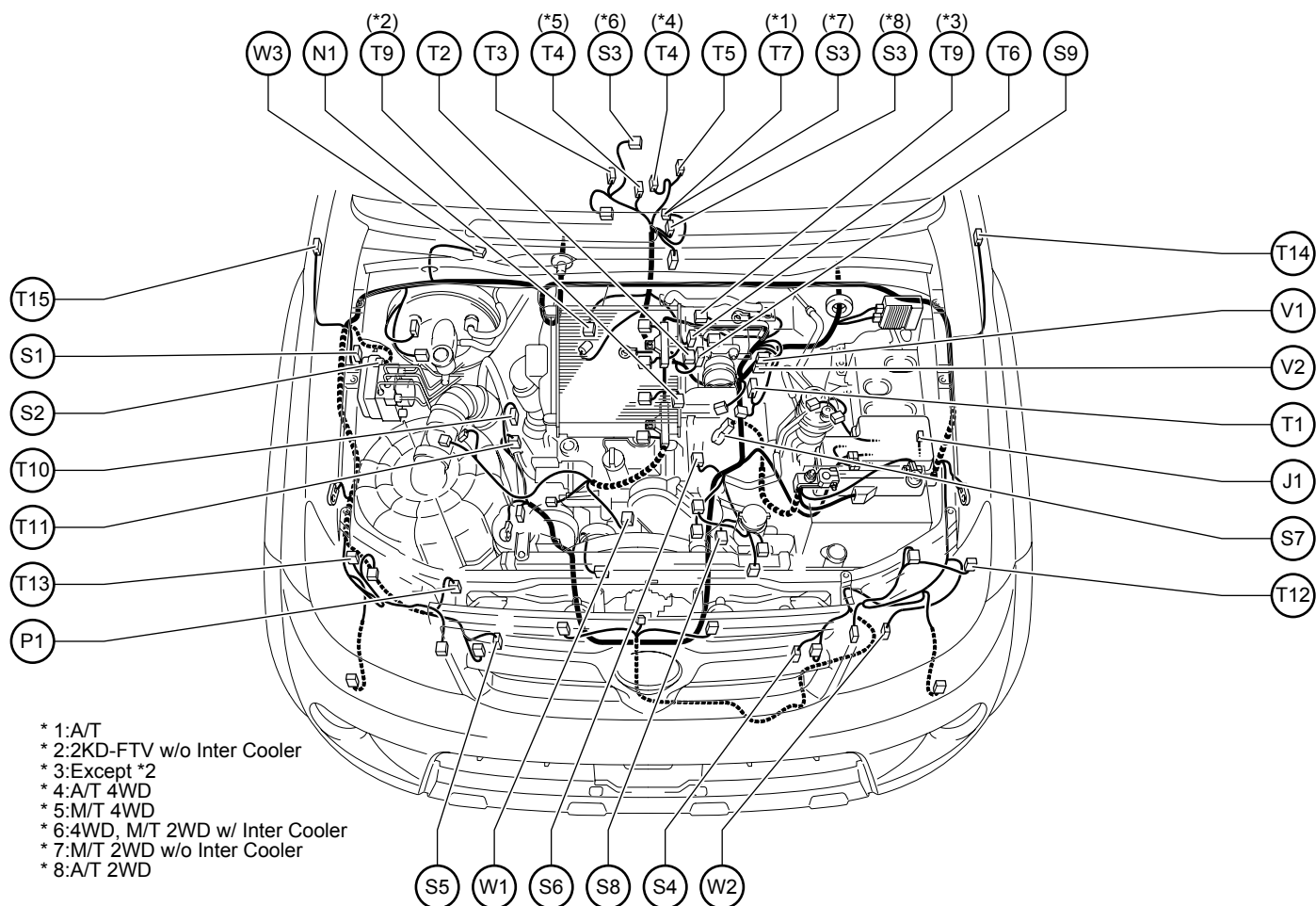
F 2 Fog Lamp (Front LH)
 F 3 Fog Lamp (Front RH)
 F 4 Fuel Filter Warning SW
 F 5 Fuel Injector (No.1)
 F 6 Fuel Injector (No.2)
 F 7 Fuel Injector (No.3)
 F 8 Fuel Injector (No.4)
 F 9 Fuel Pressure Sensor
 F 10 Fuel Sedimenter Level Warning SW
 F 11 Fuel Temp. Sensor

G 1 Glow Plug

H 1 Headlamp (LH)
 H 2 Headlamp (RH)
 H 3 Horn (High)
 H 4 Horn (Low)

I 1 Injector Driver (EDU)
 I 2 Injector Driver (EDU)
 I 3 Inlet Air Temp. Sensor (Air Cleaner)
 I 4 Inlet Air Temp. Sensor (Inter Cooler)

Position of Parts in Engine Compartment



J 1 Junction Connector

N 1 Neutral Start SW

P 1 Pressure SW

S 1 Security Horn

S 2 Skid Control ECU with Actuator

S 3 Speed Sensor

S 4 Speed Sensor (Front LH)

S 5 Speed Sensor (Front RH)

S 6 Starter

S 7 Starter

S 8 Suction Control Valve

S 9 Swirl Control Valve

T 1 Throttle Body Assembly

T 2 Throttle Body Assembly

T 3 Transfer Indicator SW (4WD Position)

T 4 Transfer Indicator SW (L4 Position)

T 5 Transfer Indicator SW (Neutral Position)

T 6 Transmission Revolution Sensor (Input)

T 7 Transmission Revolution Sensor (Output)

T 9 Turbo Pressure Sensor

T 10 Turbocharger Variable Nozzle Motor

T 11 Turbocharger Variable Nozzle Sensor

T 12 Turn Signal Lamp (Front LH)

T 13 Turn Signal Lamp (Front RH)

T 14 Turn Signal Lamp (Front Side LH)

T 15 Turn Signal Lamp (Front Side RH)

V 1 VRV (EGR)

V 2 VSV (EGR Cut Valve)

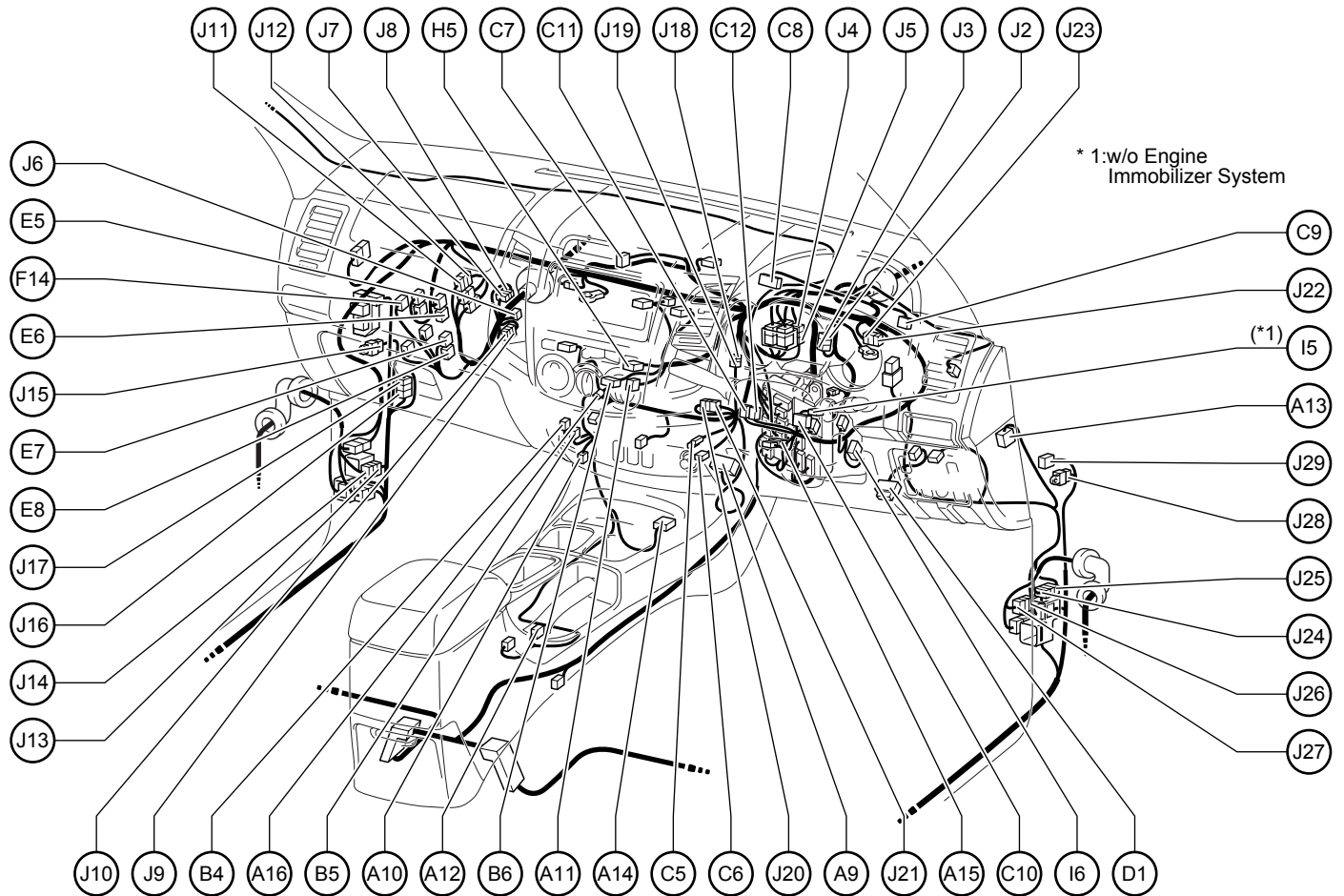
W 1 Water Temp. Sensor

W 2 Windshield Washer Motor

W 3 Windshield Wiper Motor

G ELECTRICAL WIRING ROUTING

Position of Parts in Instrument Panel



A 9 A/C Amplifier
 A10 A/C Thermistor
 A11 A/C Volume SW
 A12 A/T Shift Lever Illumination
 A13 Accelerator Position Sensor
 A14 Airbag Sensor Assembly Center
 A15 Airbag Squib (Steering Wheel Pad)
 A16 Ashtray Illumination

B 4 Blower Motor
 B 5 Blower Resistor
 B 6 Blower SW

C 5 Cigarette Lighter
 C 6 Cigarette Lighter Illumination
 C 7 Clock
 C 8 Combination Meter
 C 9 Combination Meter
 C10 Combination SW
 C11 Combination SW
 C12 Combination SW

D 1 DLC3

E 5 Engine ECU
 E 6 Engine ECU
 E 7 Engine ECU
 E 8 Engine ECU

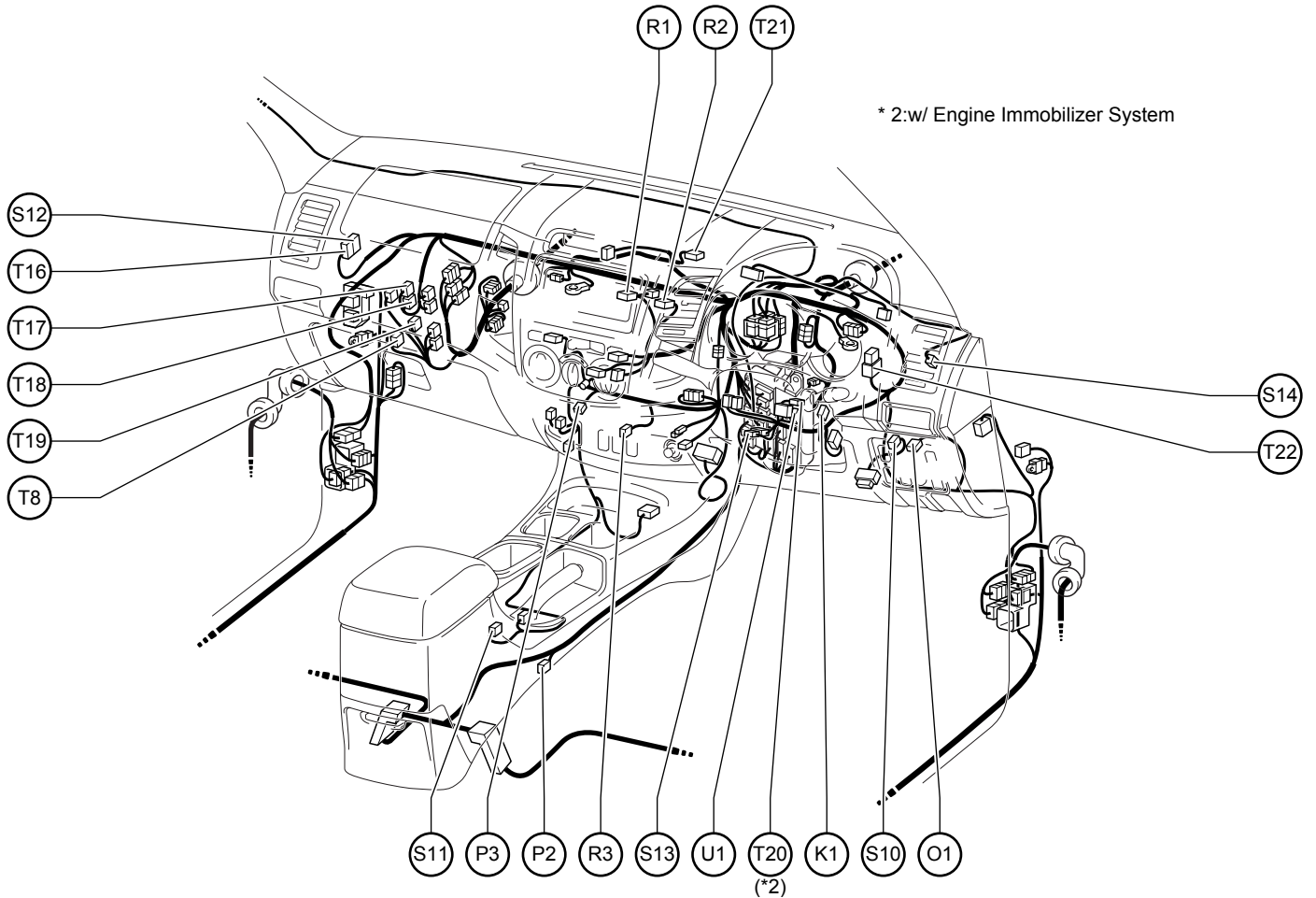
F 14 4WD Control ECU

H 5 Hazard Warning Signal SW

I 5 Ignition Key Cylinder Lamp
 I 6 Ignition SW

J 2 Junction Connector
 J 3 Junction Connector
 J 4 Junction Connector
 J 5 Junction Connector
 J 6 Junction Connector
 J 7 Junction Connector
 J 8 Junction Connector
 J 9 Junction Connector
 J10 Junction Connector
 J11 Junction Connector
 J12 Junction Connector
 J13 Junction Connector
 J14 Junction Connector
 J15 Junction Connector
 J16 Junction Connector
 J17 Junction Connector
 J18 Junction Connector
 J19 Junction Connector
 J20 Junction Connector
 J21 Junction Connector
 J22 Junction Connector
 J23 Junction Connector
 J24 Junction Connector
 J25 Junction Connector
 J26 Junction Connector
 J27 Junction Connector
 J28 Junction Connector
 J29 Junction Connector

Position of Parts in Instrument Panel



K 1 Key Interlock Solenoid

O 1 Outer Mirror SW

P 2 Parking Brake SW

P 3 Power Point Socket

R 1 Radio Receiver Assembly

R 2 Radio Receiver Assembly

R 3 Rear Window Defogger SW

S10 Security Indicator Lamp

S11 Shift Lock Control ECU

S12 Short Pin (Theft Warning ECU)

S13 Spiral Cable

S14 Stop Lamp SW

T 8 Turbo Motor Driver

T16 Theft Warning ECU

T17 Transmission Control ECU

T18 Transmission Control ECU

T19 Transmission Control ECU

T20 Transponder Key Amplifier

T21 Transponder Key ECU

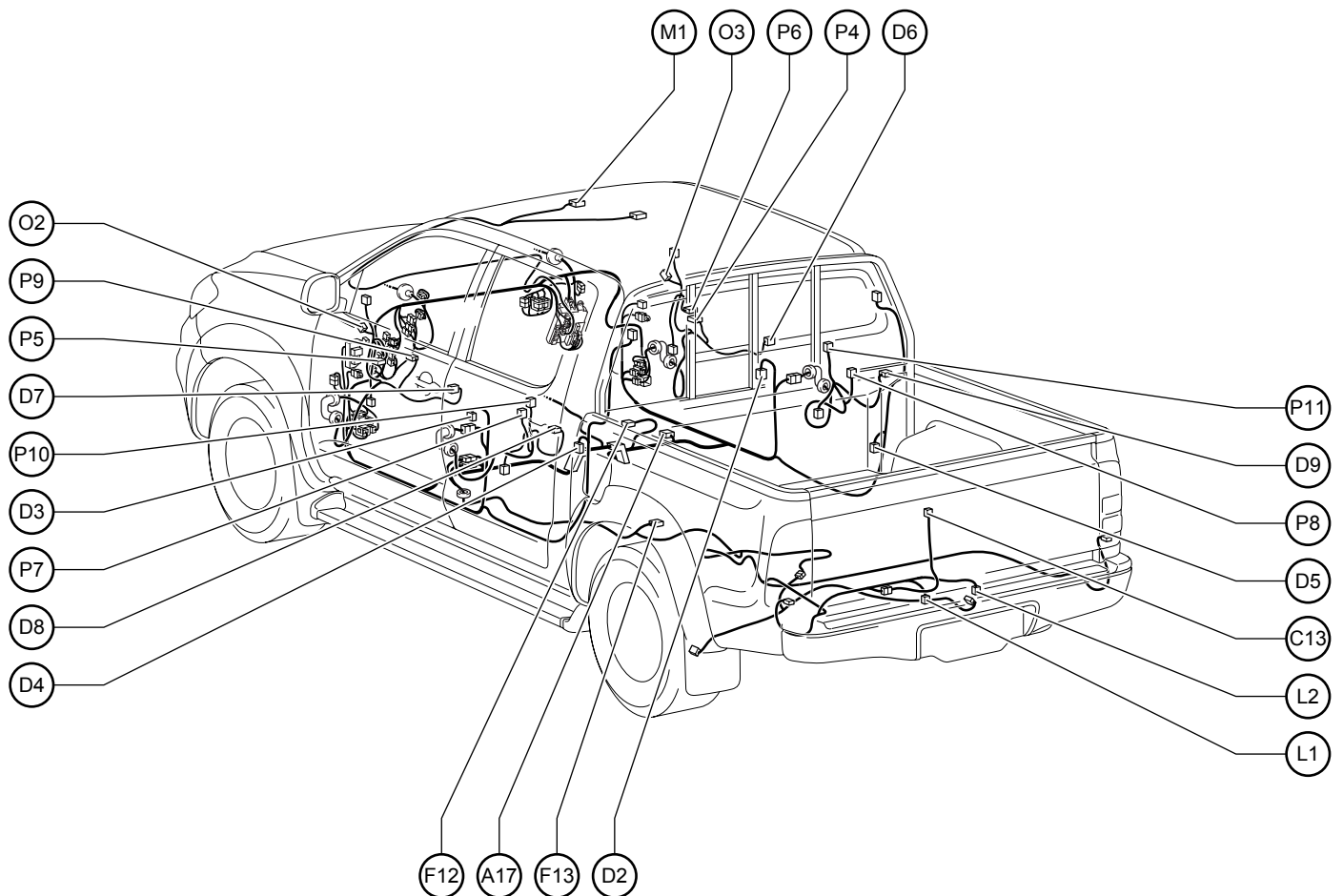
T22 Turn Signal Flasher

U 1 Unlock Warning SW

G ELECTRICAL WIRING ROUTING

Position of Parts in Body

[Double Cab]



A17 ABS Deceleration Sensor

C13 Center Stop Lamp

D 2 Door Courtesy SW (Driver's Side)
 D 3 Door Courtesy SW (Front Passenger's Side)
 D 4 Door Courtesy SW (Rear LH)
 D 5 Door Courtesy SW (Rear RH)
 D 6 Door Lock Assembly (Driver's Side)
 D 7 Door Lock Assembly (Front Passenger's Side)
 D 8 Door Lock Assembly (Rear LH)
 D 9 Door Lock Assembly (Rear RH)

F 12 Front Seat Inner Belt (Driver's Side)

F 13 Fuel Sender Gage

L 1 License Plate Lamp LH

L 2 License Plate Lamp RH

M 1 Map Lamp

O 2 Outer Rear View Mirror (LH)

O 3 Outer Rear View Mirror (RH)

P 4 Power Window Master SW

P 5 Power Window Regulator Motor (Front LH)

P 6 Power Window Regulator Motor (Front RH)

P 7 Power Window Regulator Motor (Rear LH)

P 8 Power Window Regulator Motor (Rear RH)

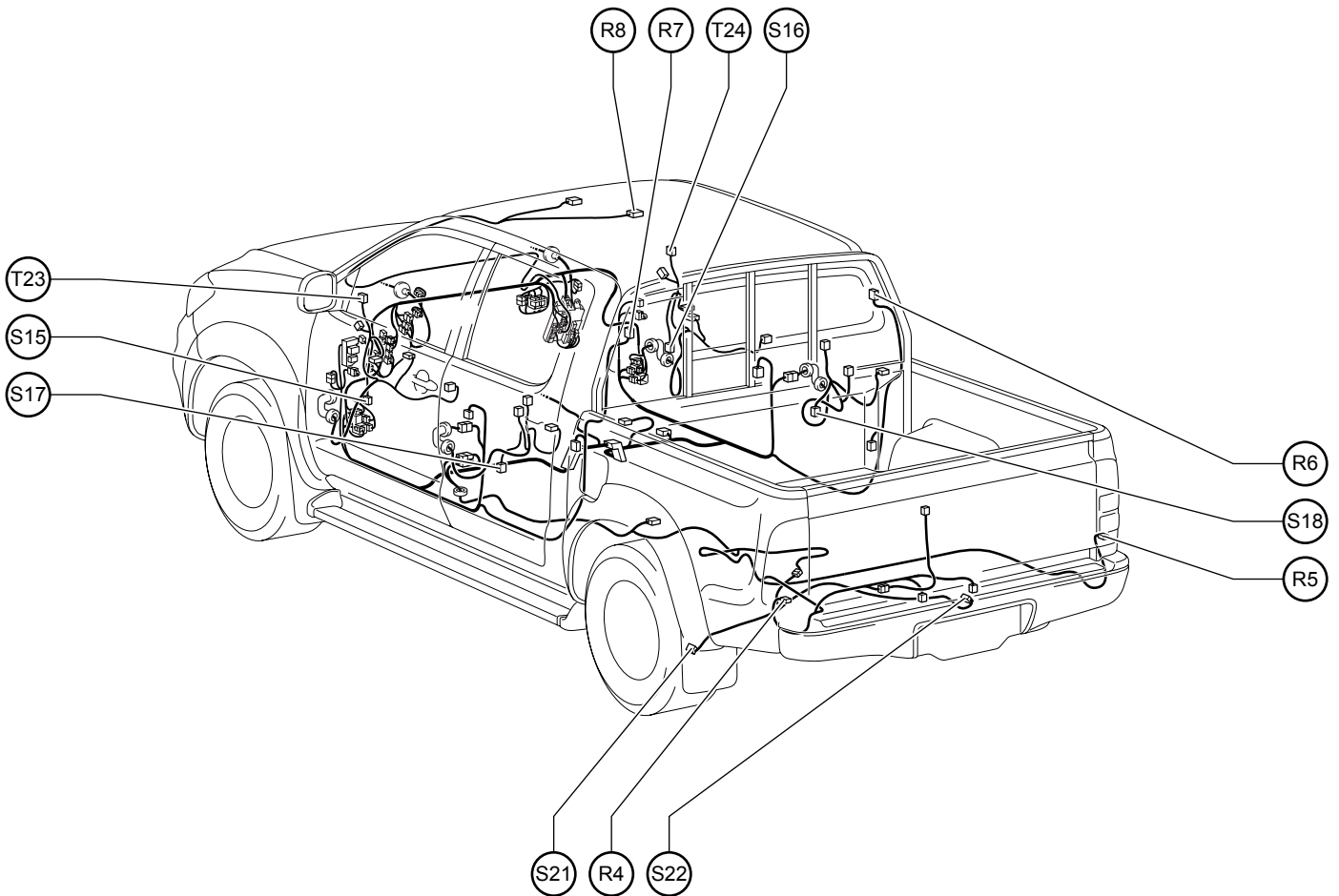
P 9 Power Window SW (Front Passenger's Side)

P10 Power Window SW (Rear LH)

P11 Power Window SW (Rear RH)

Position of Parts in Body

[Double Cab]



R 4 Rear Combination Lamp (LH)
 R 5 Rear Combination Lamp (RH)
 R 6 Rear Window Defogger
 R 7 Rear Window Defogger
 R 8 Room Lamp

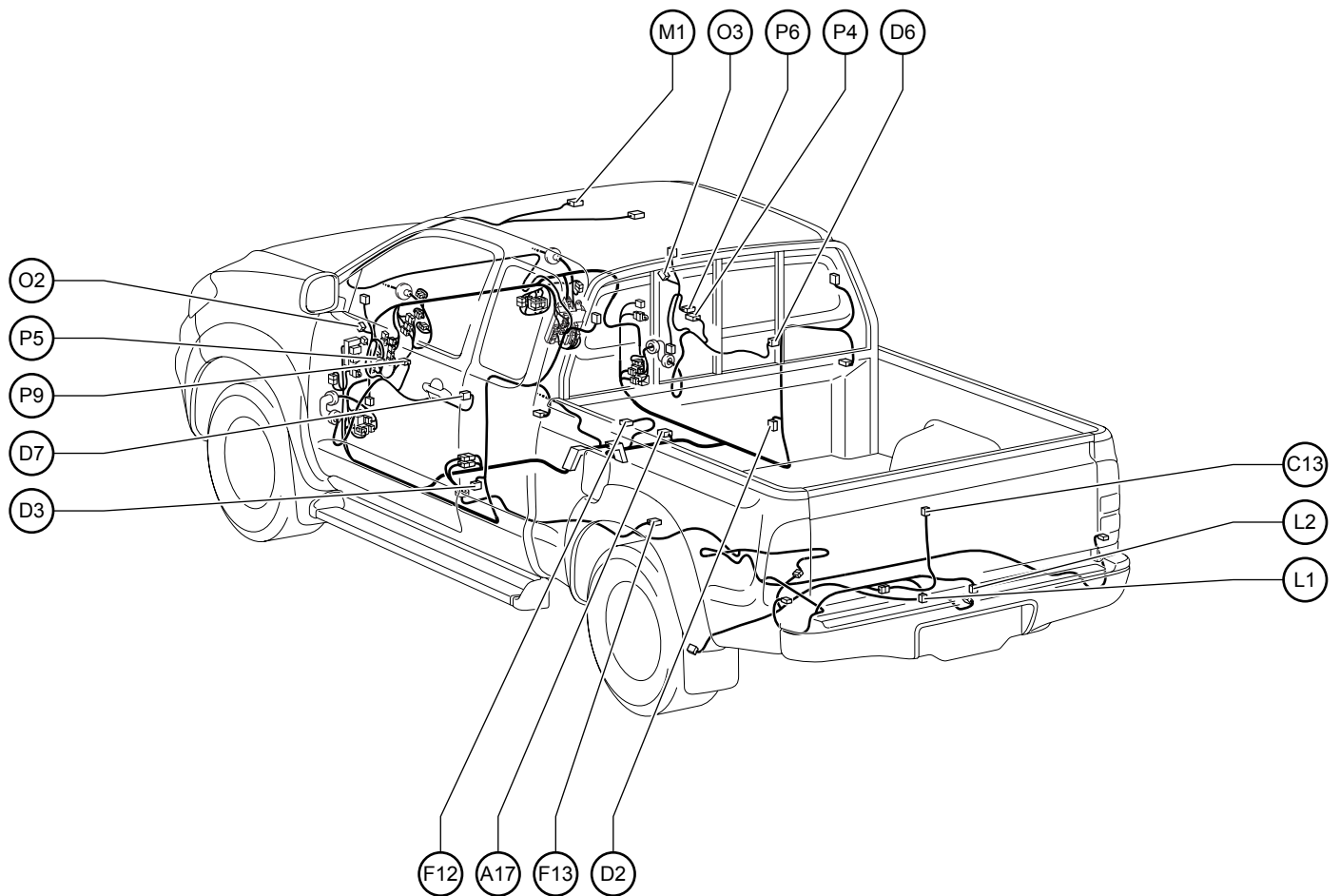
T 23 Tweeter (LH)
 T 24 Tweeter (RH)

S 15 Speaker (Front Door LH)
 S 16 Speaker (Front Door RH)
 S 17 Speaker (Rear Door LH)
 S 18 Speaker (Rear Door RH)
 S 21 Speed Sensor (Rear LH)
 S 22 Speed Sensor (Rear RH)

G ELECTRICAL WIRING ROUTING

Position of Parts in Body

[Extra Cab]



A17 ABS Deceleration Sensor

C13 Center Stop Lamp

D 2 Door Courtesy SW (Driver's Side)

D 3 Door Courtesy SW (Front Passenger's Side)

D 6 Door Lock Assembly (Driver's Side)

D 7 Door Lock Assembly (Front Passenger's Side)

F 12 Front Seat Inner Belt (Driver's Side)

F 13 Fuel Sender Gage

L 1 License Plate Lamp LH

L 2 License Plate Lamp RH

M 1 Map Lamp

O 2 Outer Rear View Mirror (LH)

O 3 Outer Rear View Mirror (RH)

P 4 Power Window Master SW

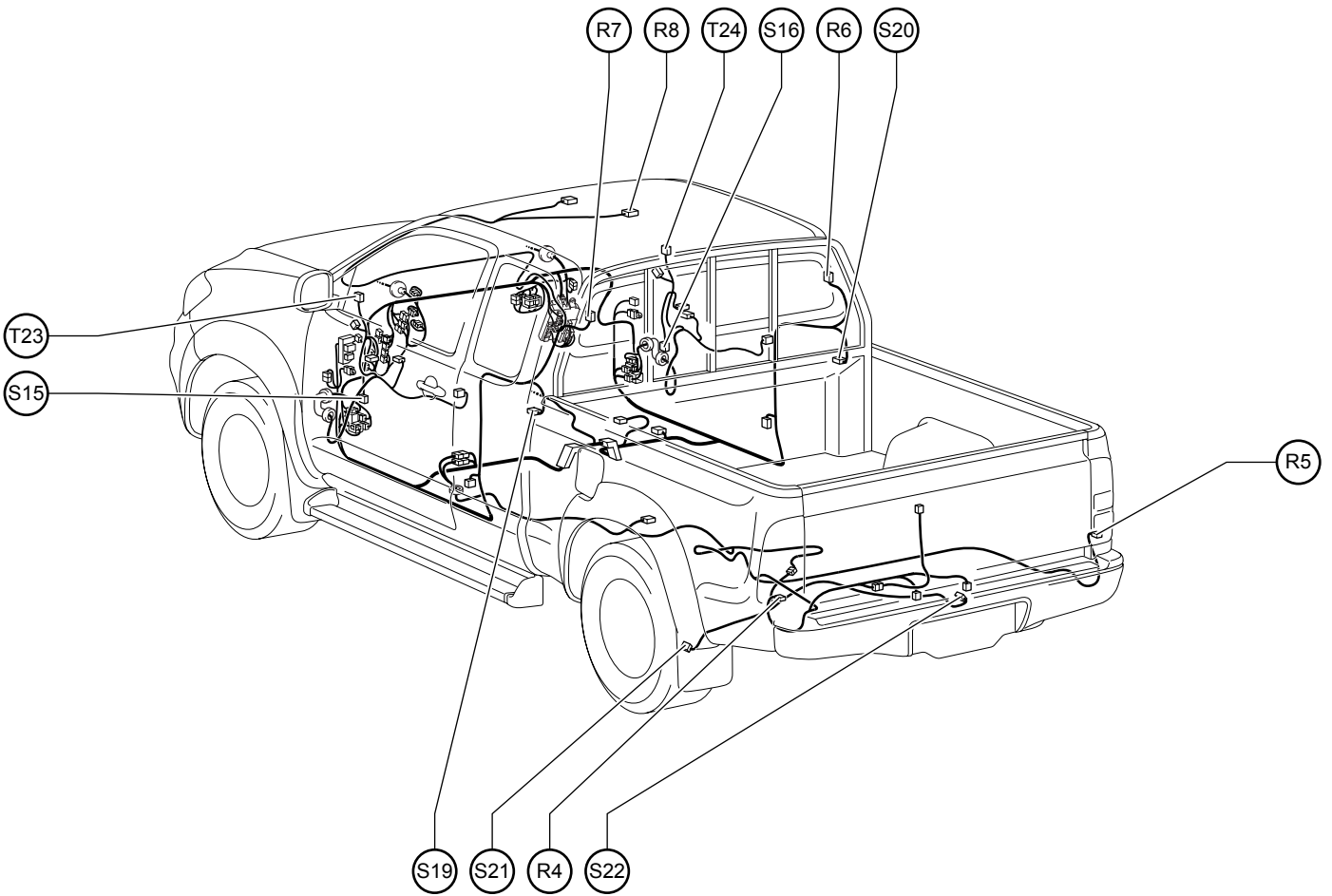
P 5 Power Window Regulator Motor (Front LH)

P 6 Power Window Regulator Motor (Front RH)

P 9 Power Window SW (Front Passenger's Side)

Position of Parts in Body

[Extra Cab]



R 4 Rear Combination Lamp (LH)
 R 5 Rear Combination Lamp (RH)
 R 6 Rear Window Defogger
 R 7 Rear Window Defogger
 R 8 Room Lamp

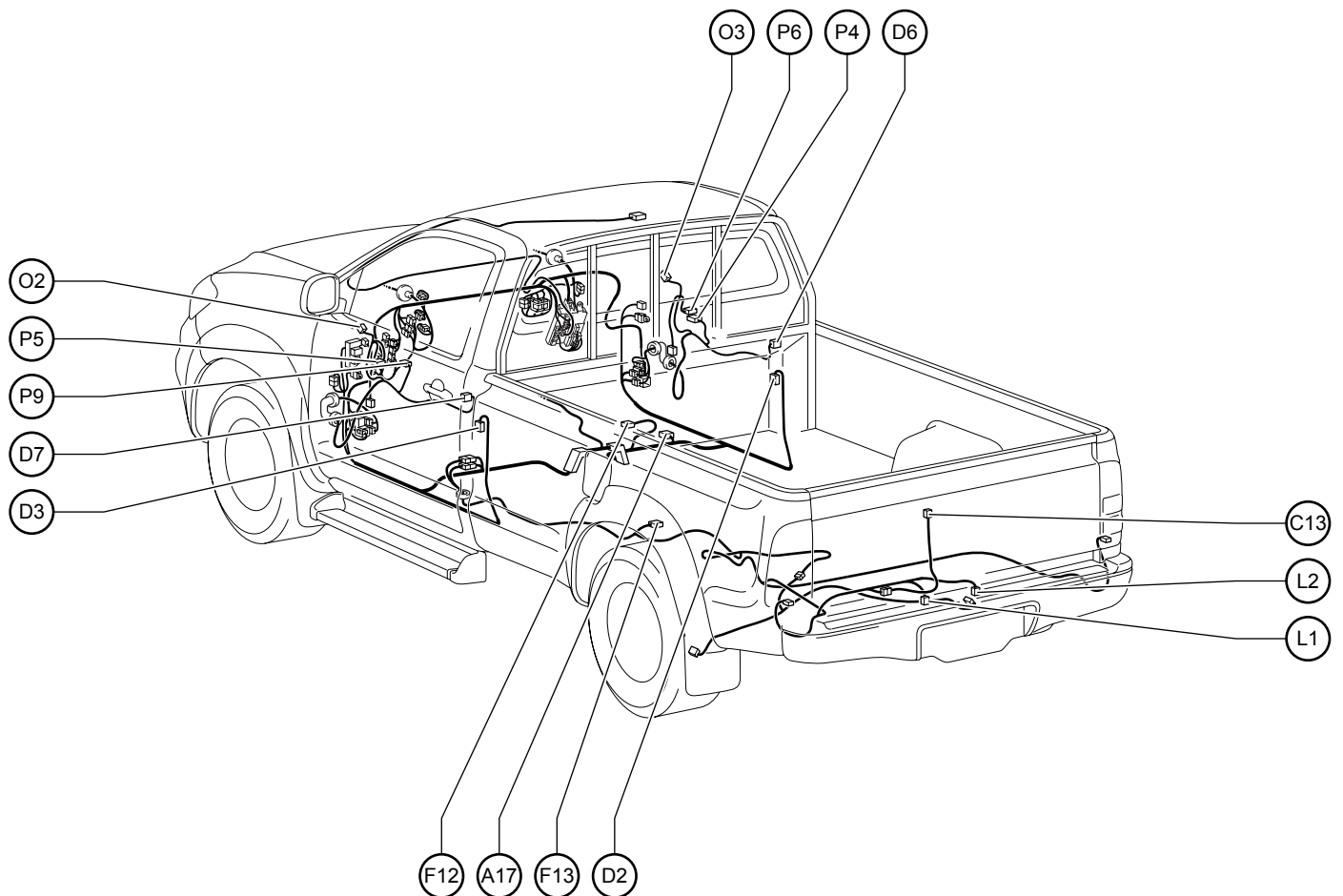
T 23 Tweeter (LH)
 T 24 Tweeter (RH)

S 15 Speaker (Front Door LH)
 S 16 Speaker (Front Door RH)
 S 19 Speaker (Rear LH)
 S 20 Speaker (Rear RH)
 S 21 Speed Sensor (Rear LH)
 S 22 Speed Sensor (Rear RH)

G ELECTRICAL WIRING ROUTING

Position of Parts in Body

[Single Cab]



A17 ABS Deceleration Sensor

C13 Center Stop Lamp

D 2 Door Courtesy SW (Driver's Side)

D 3 Door Courtesy SW (Front Passenger's Side)

D 6 Door Lock Assembly (Driver's Side)

D 7 Door Lock Assembly (Front Passenger's Side)

F 12 Front Seat Inner Belt (Driver's Side)

F 13 Fuel Sender Gage

L 1 License Plate Lamp LH

L 2 License Plate Lamp RH

O 2 Outer Rear View Mirror (LH)

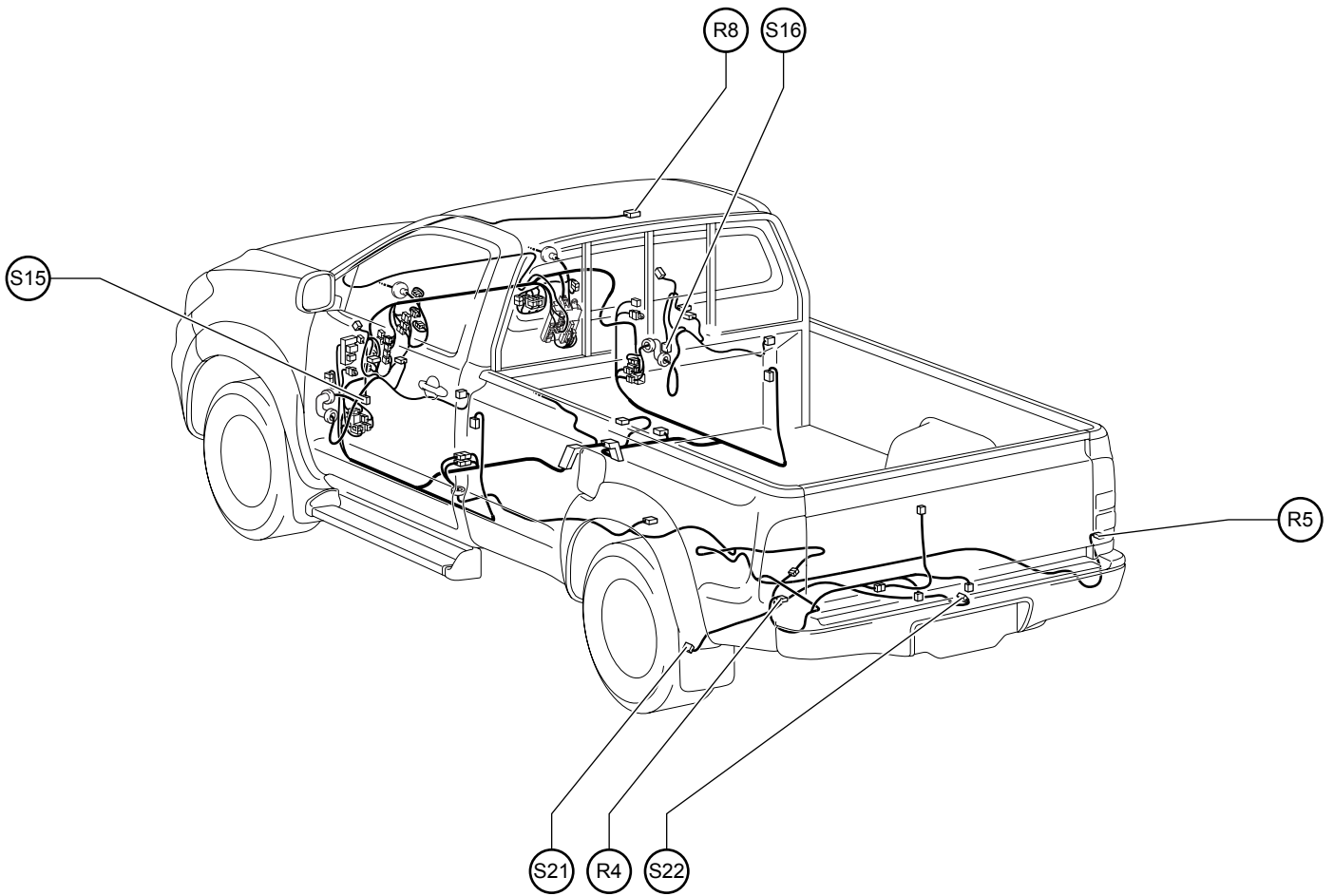
O 3 Outer Rear View Mirror (RH)

P 4 Power Window Master SW

P 5 Power Window Regulator Motor (Front LH)

P 6 Power Window Regulator Motor (Front RH)

P 9 Power Window SW (Front Passenger's Side)

Position of Parts in Body**[Single Cab]**

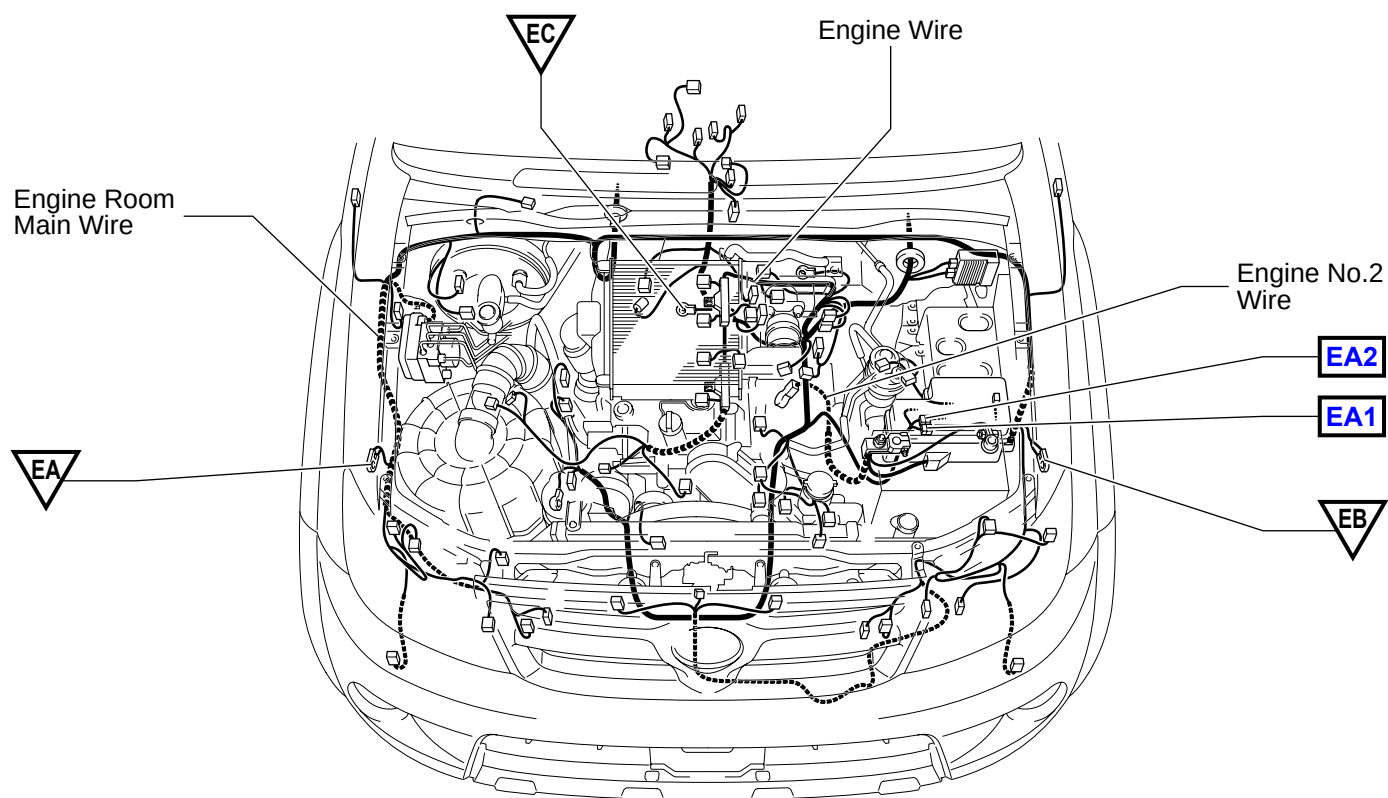
R 4 Rear Combination Lamp (LH)
R 5 Rear Combination Lamp (RH)
R 8 Room Lamp

S15 Speaker (Front Door LH)
S16 Speaker (Front Door RH)
S21 Speed Sensor (Rear LH)
S22 Speed Sensor (Rear RH)

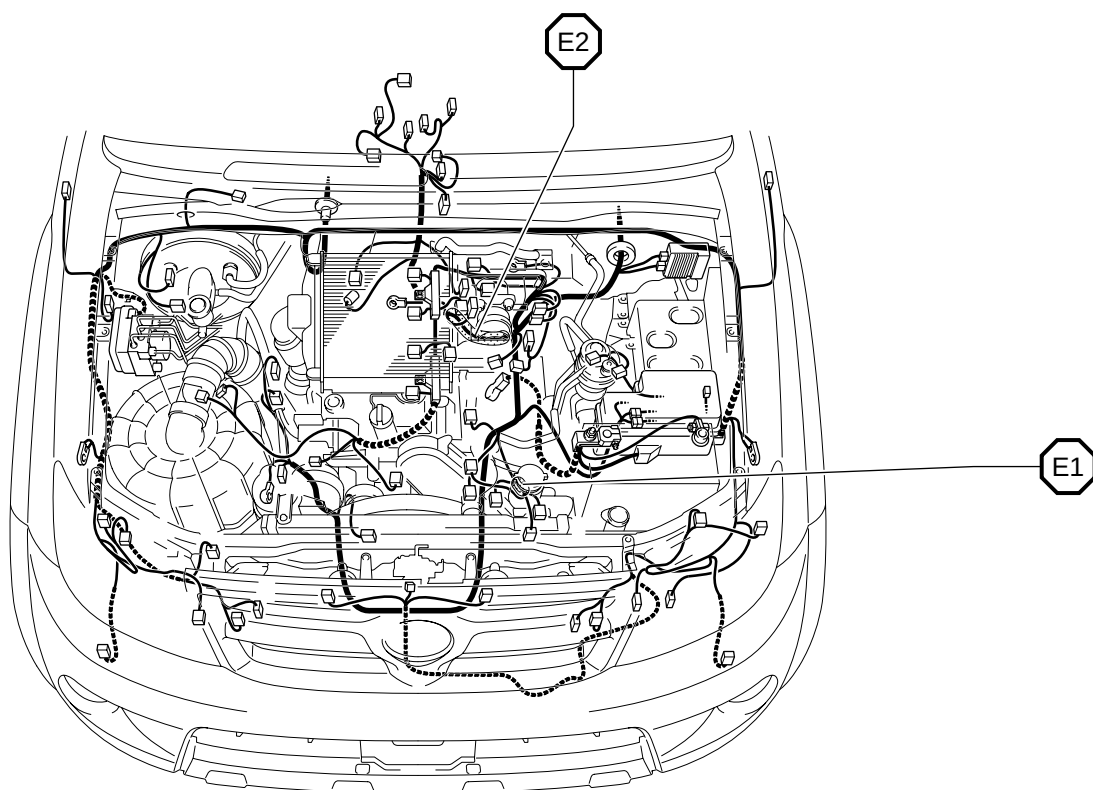
G ELECTRICAL WIRING ROUTING

□ : Location of Connector Joining Wire Harness and Wire Harness

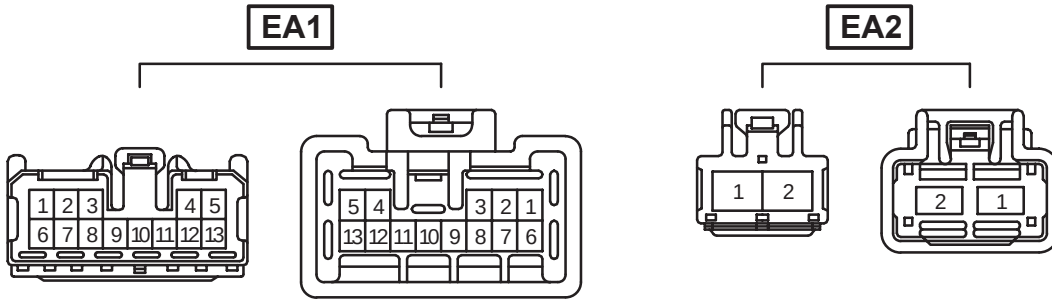
▽ : Location of Ground Points



○ : Location of Splice Points



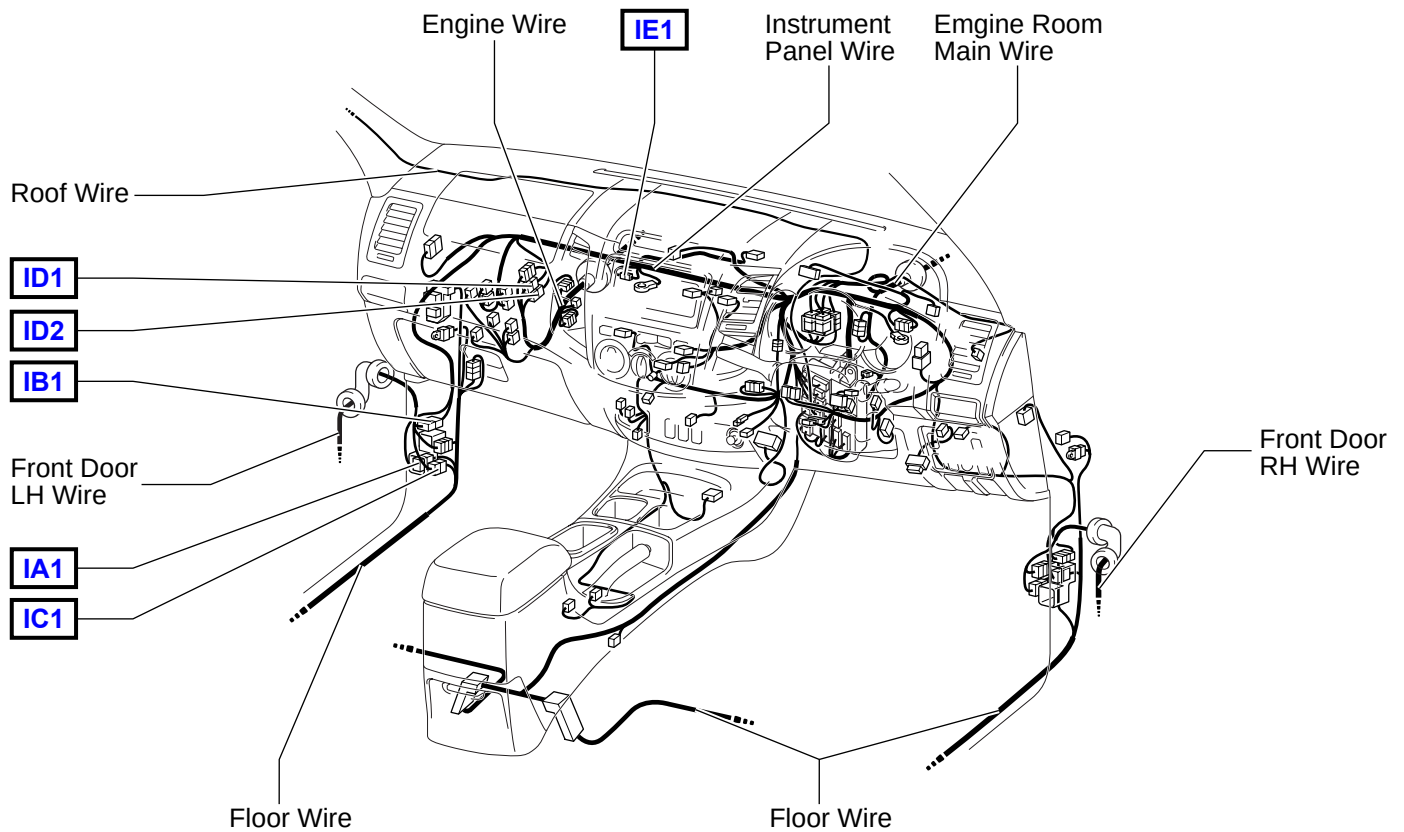
Connector Joining Wire Harness and Wire Harness



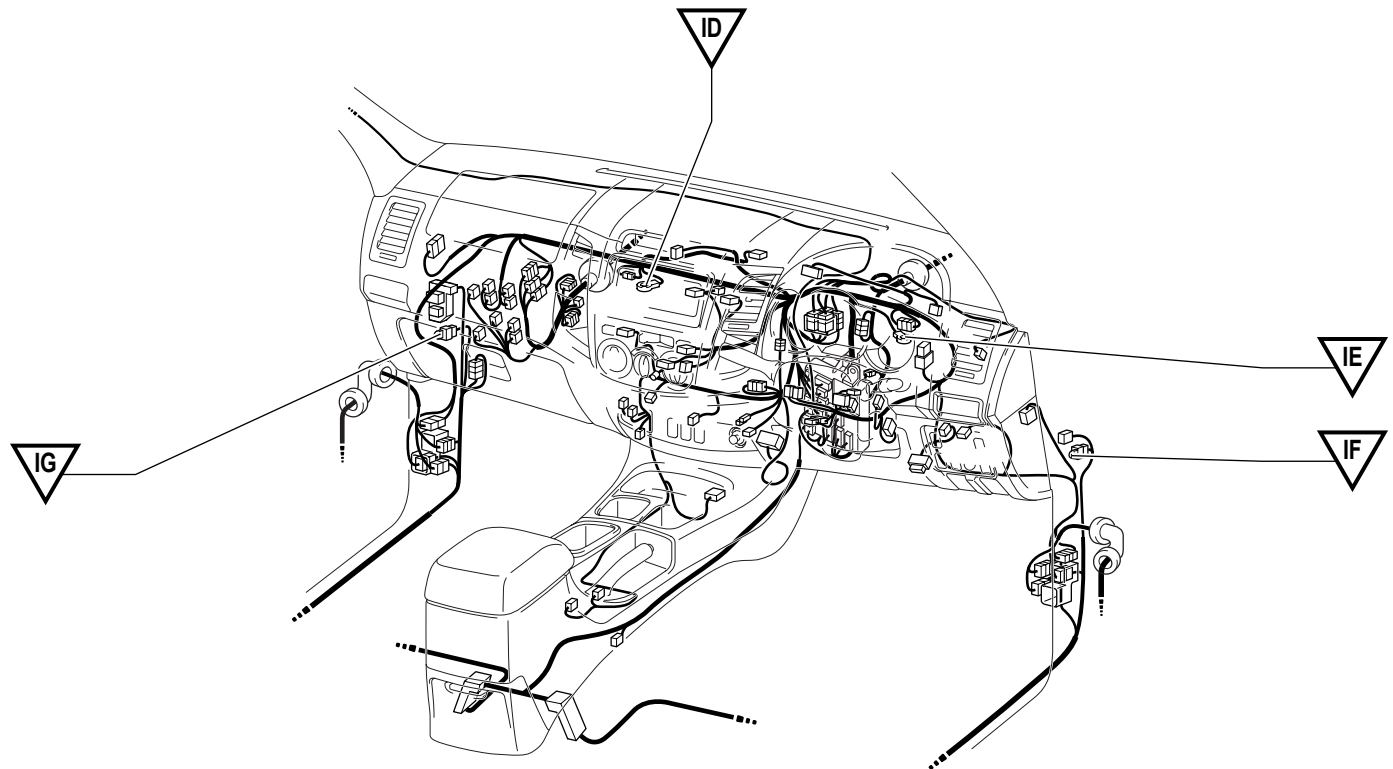
Code	Joining Wire Harness and Wire Harness (Connector Location)
EA1	Engine Wire and Engine Room Main Wire (Inside of Engine Room R/B and J/B)
EA2	

G ELECTRICAL WIRING ROUTING

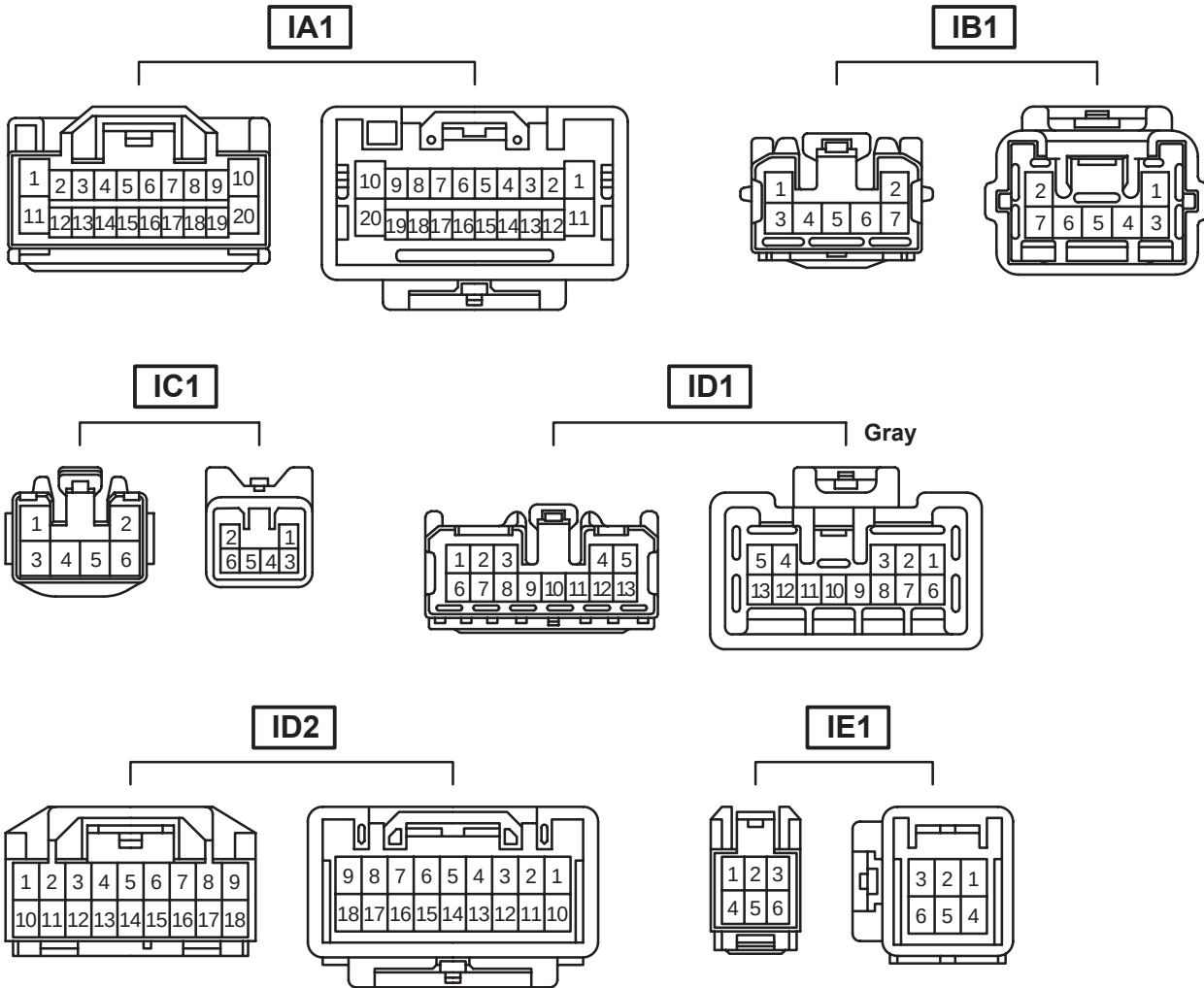
□ : Location of Connector Joining Wire Harness and Wire Harness



▽ : Location of Ground Points



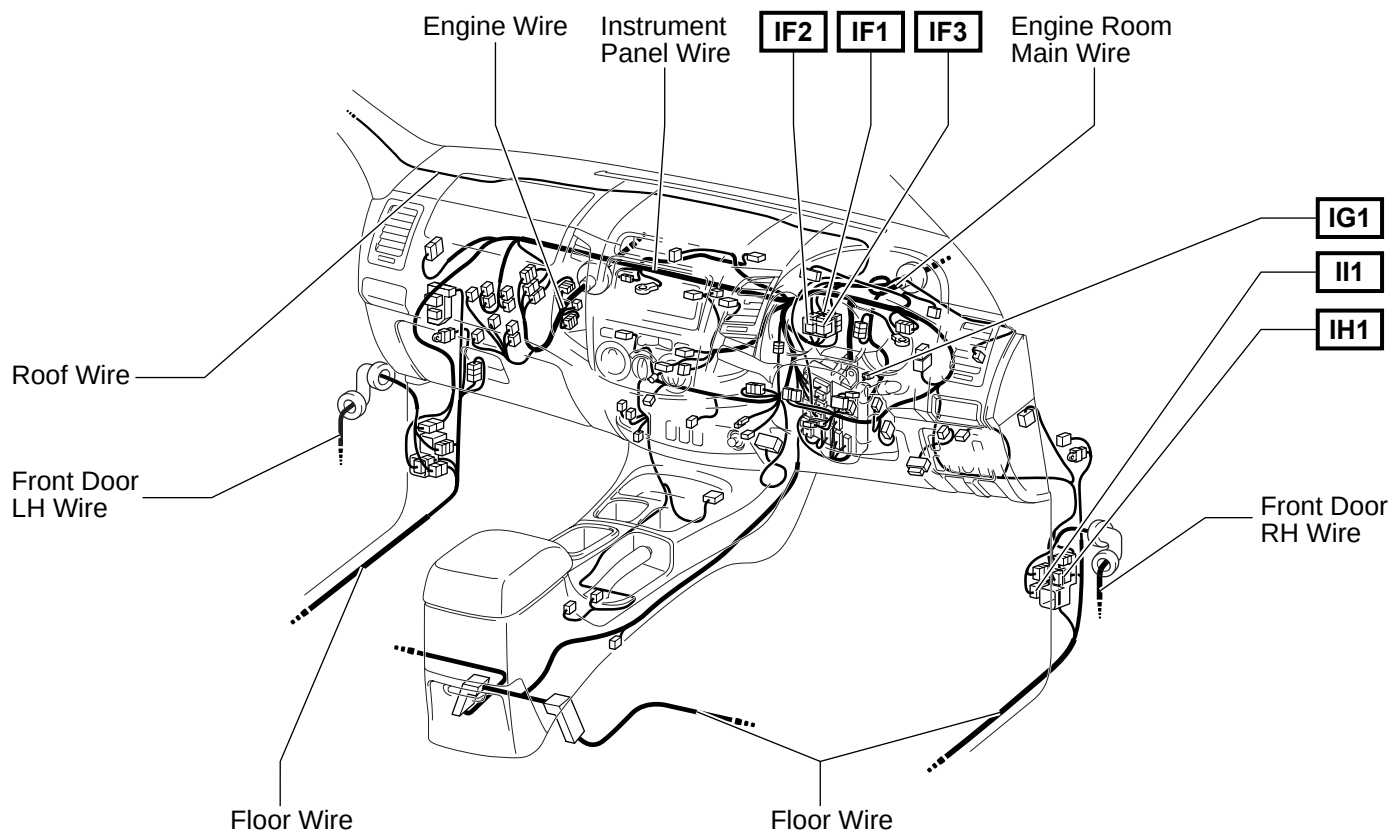
Connector Joining Wire Harness and Wire Harness



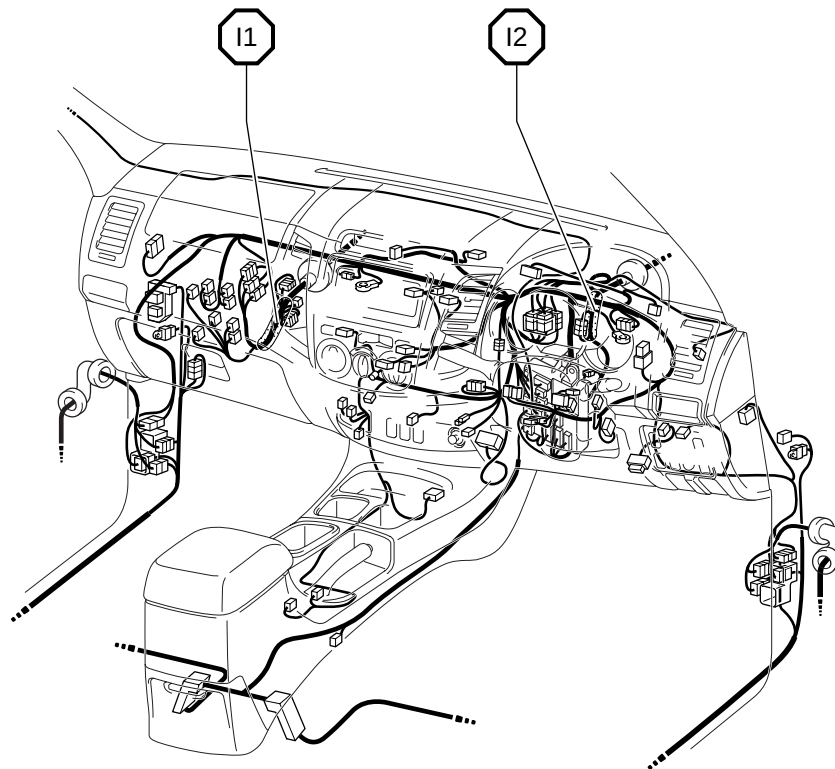
Code	Joining Wire Harness and Wire Harness (Connector Location)
IA1	Instrument Panel Wire and Floor Wire (Left Kick Panel)
IB1	Front Door LH Wire and Instrument Panel Wire (Left Kick Panel)
IC1	Front Door LH Wire and Floor Wire (Left Kick Panel)
ID1	Engine Wire and Instrument Panel Wire (Behind the Glove Box)
ID2	
IE1	Instrument Panel Wire and Instrument Panel Wire (Instrument Panel Reinforcement LH)

G ELECTRICAL WIRING ROUTING

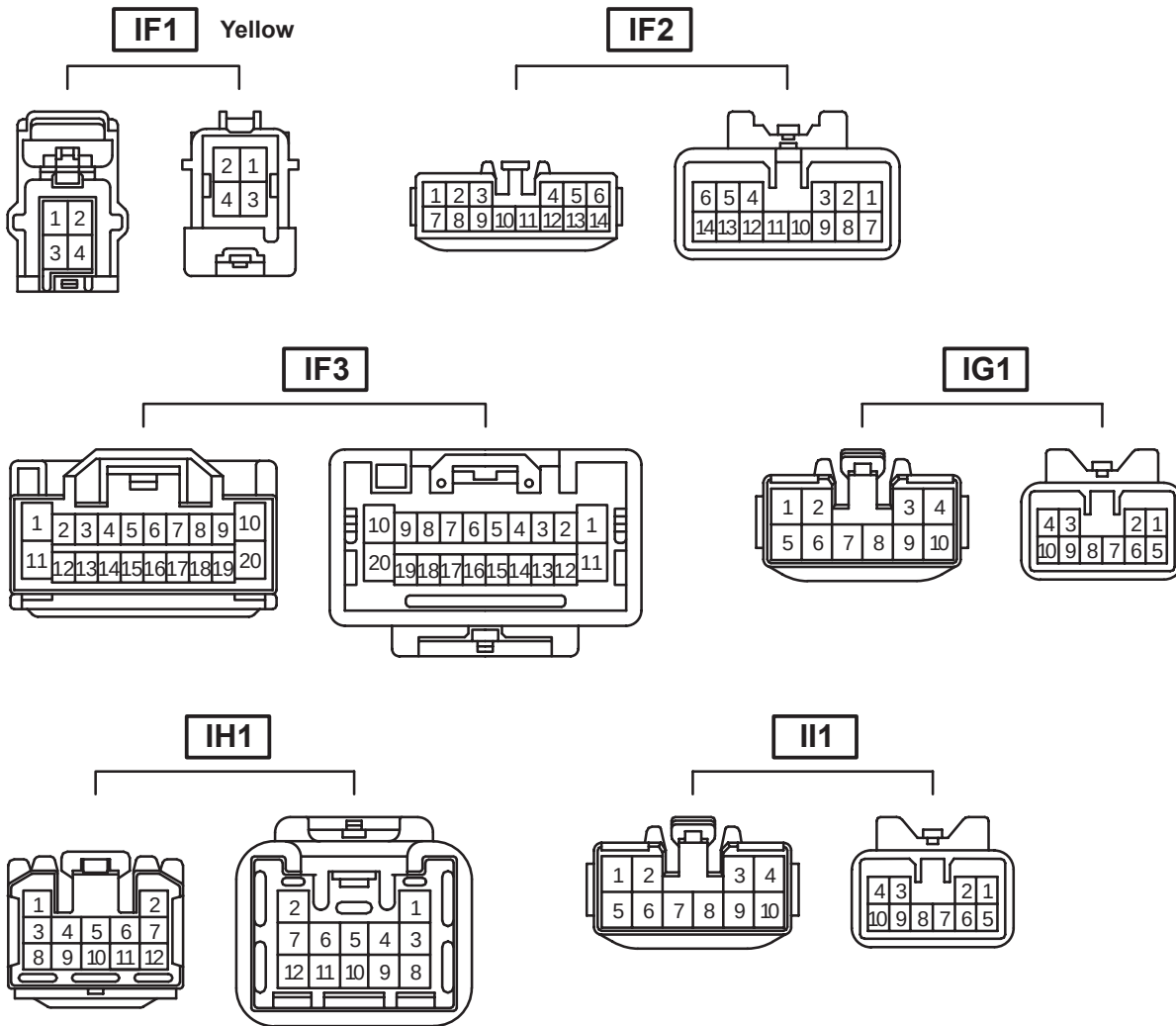
□ : Location of Connector Joining Wire Harness and Wire Harness



○ : Location of Splice Points



Connector Joining Wire Harness and Wire Harness

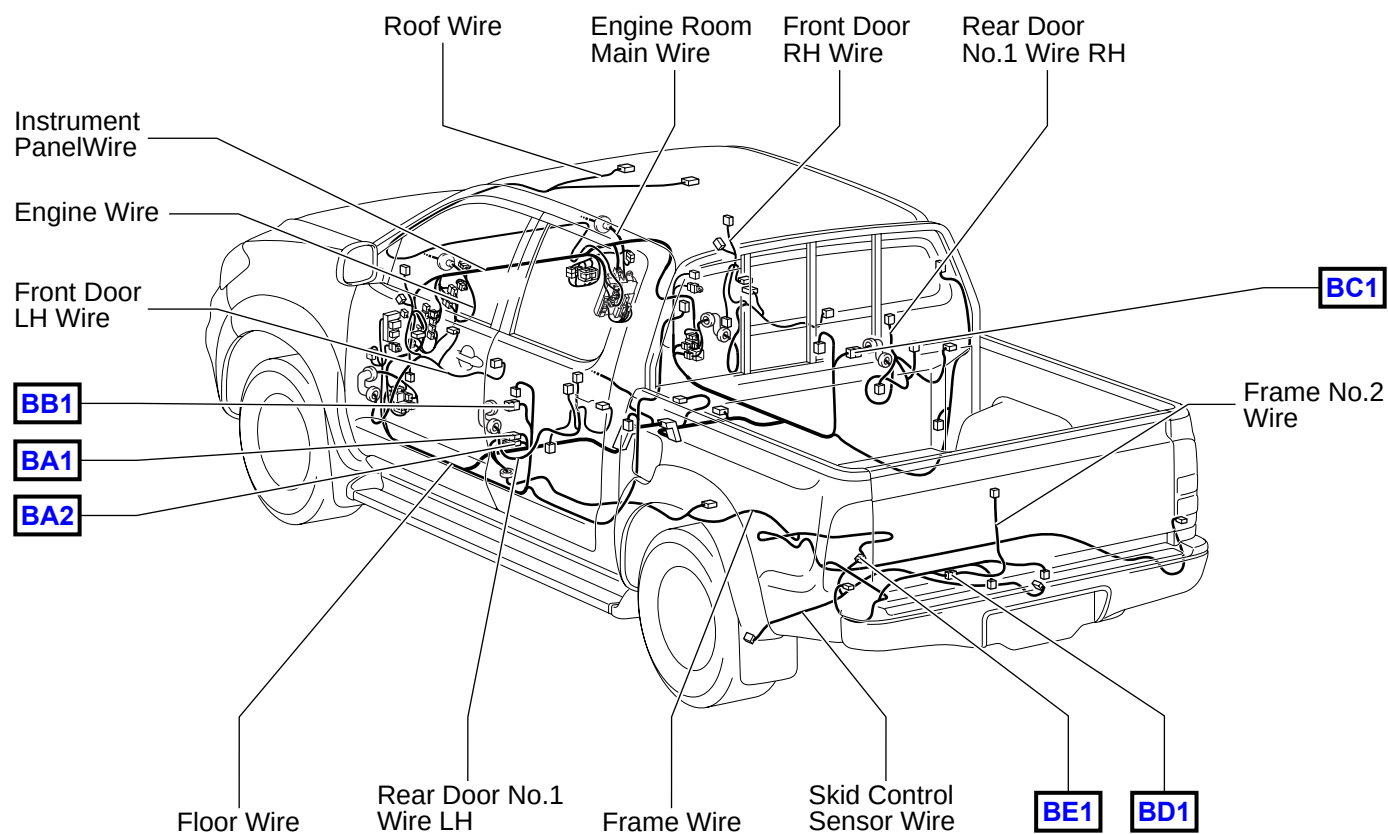


Code	Joining Wire Harness and Wire Harness (Connector Location)
IF1	Engine Room Main Wire and Instrument Panel Wire (Behind the Driver Side J/B)
IF2	
IF3	
IG1	Engine Room Main Wire and Floor Wire (Near the Driver Side J/B)
IH1	Instrument Panel Wire and Floor Wire (Right Kick Panel)
II1	Front Door RH Wire and Floor Wire (Right Kick Panel)

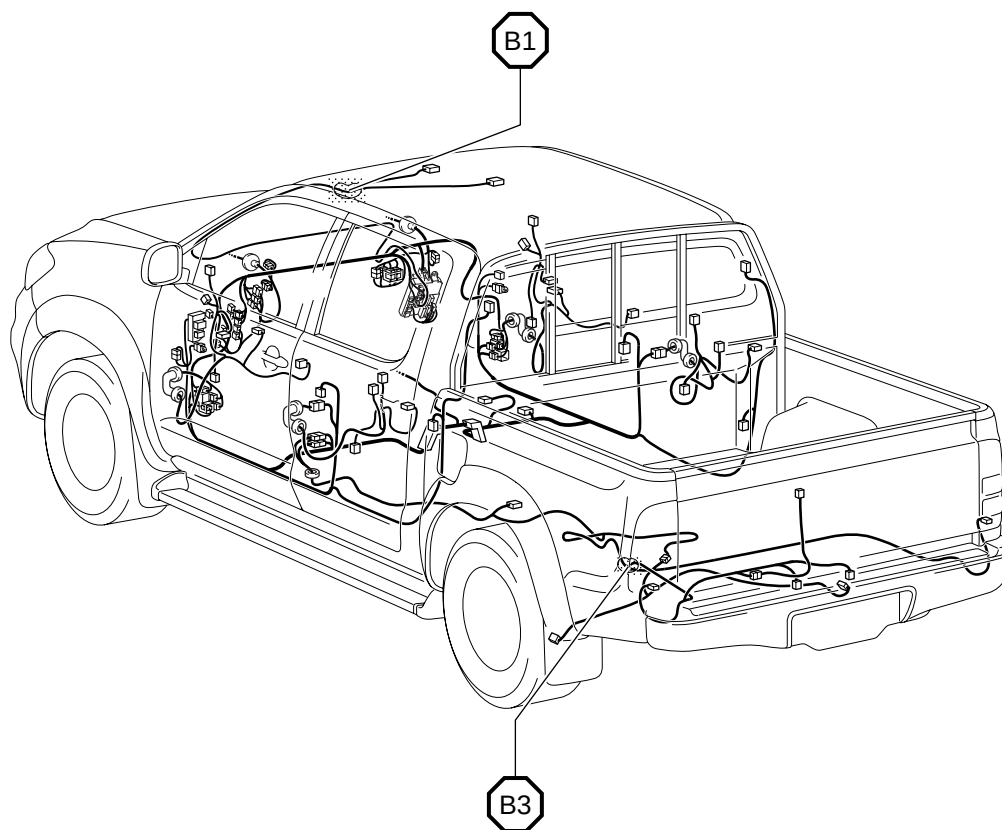
G ELECTRICAL WIRING ROUTING

□ : Location of Connector Joining Wire Harness and Wire Harness

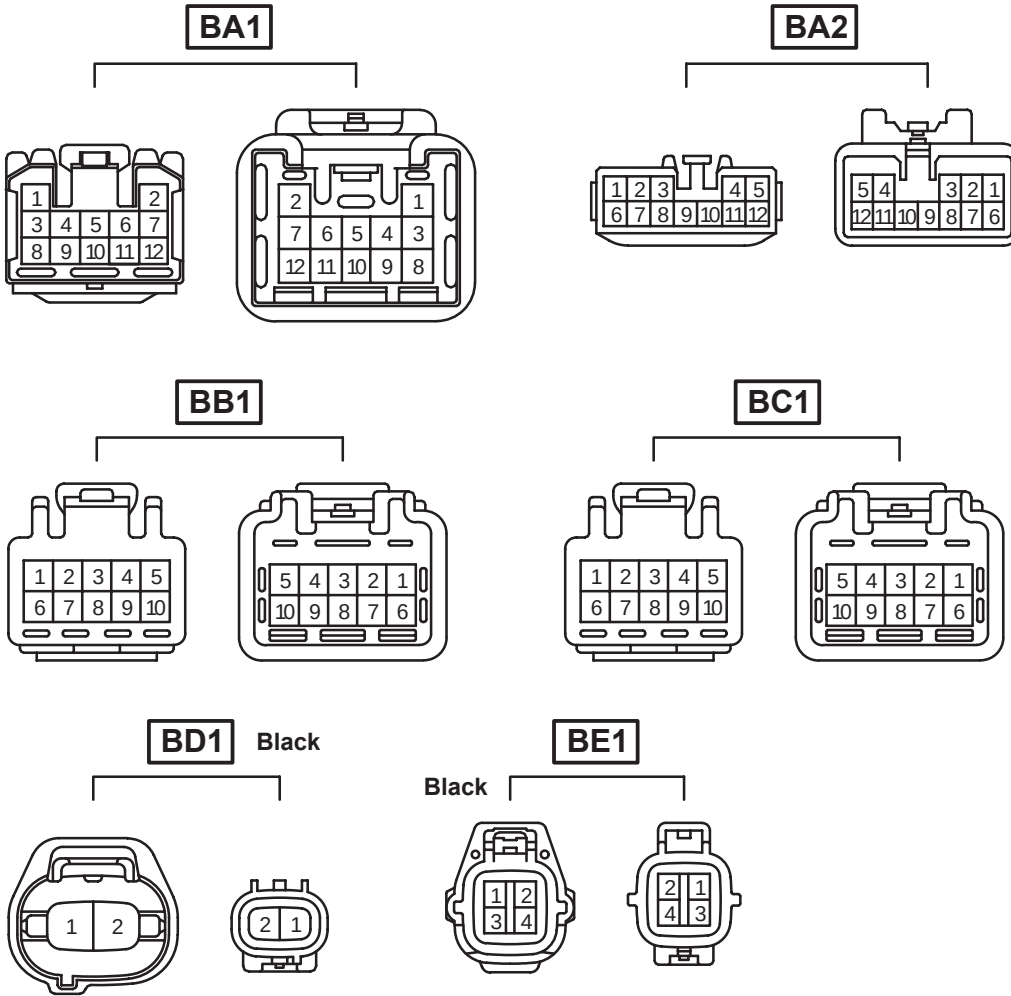
[Double Cab]



○ : Location of Splice Points



Connector Joining Wire Harness and Wire Harness

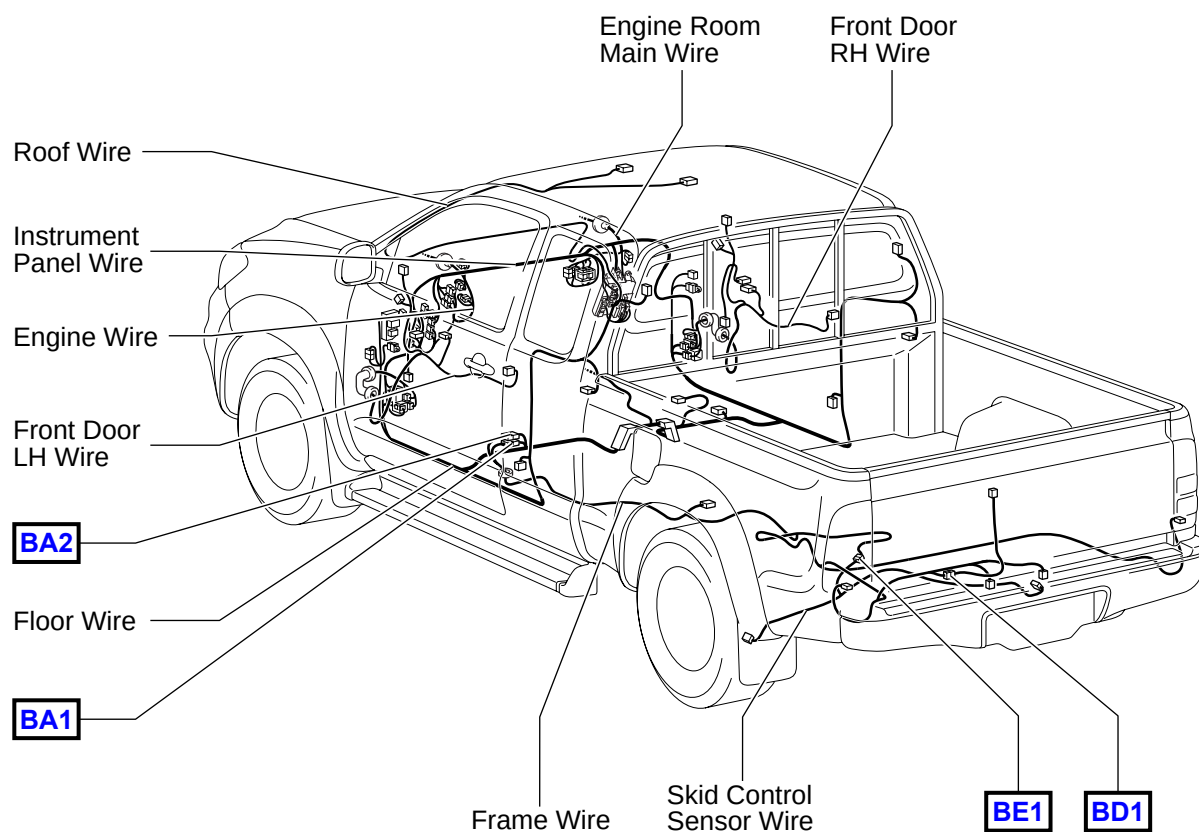


Code	Joining Wire Harness and Wire Harness (Connector Location)
BA1	Frame Wire and Floor Wire (Under the Front Seat LH)
BA2	Frame Wire and Floor Wire (Under the Front Seat LH)
BB1	Rear Door No.1 Wire LH and Floor Wire (Center Pillar LH)
BC1	Rear Door No.1 Wire RH and Floor Wire (Center Pillar RH)
BD1	Frame No.2 Wire and Frame Wire (Under the Back Panel LH)
BE1	Frame Wire and Skid Control Sensor Wire (Rear of the Rear Differential)

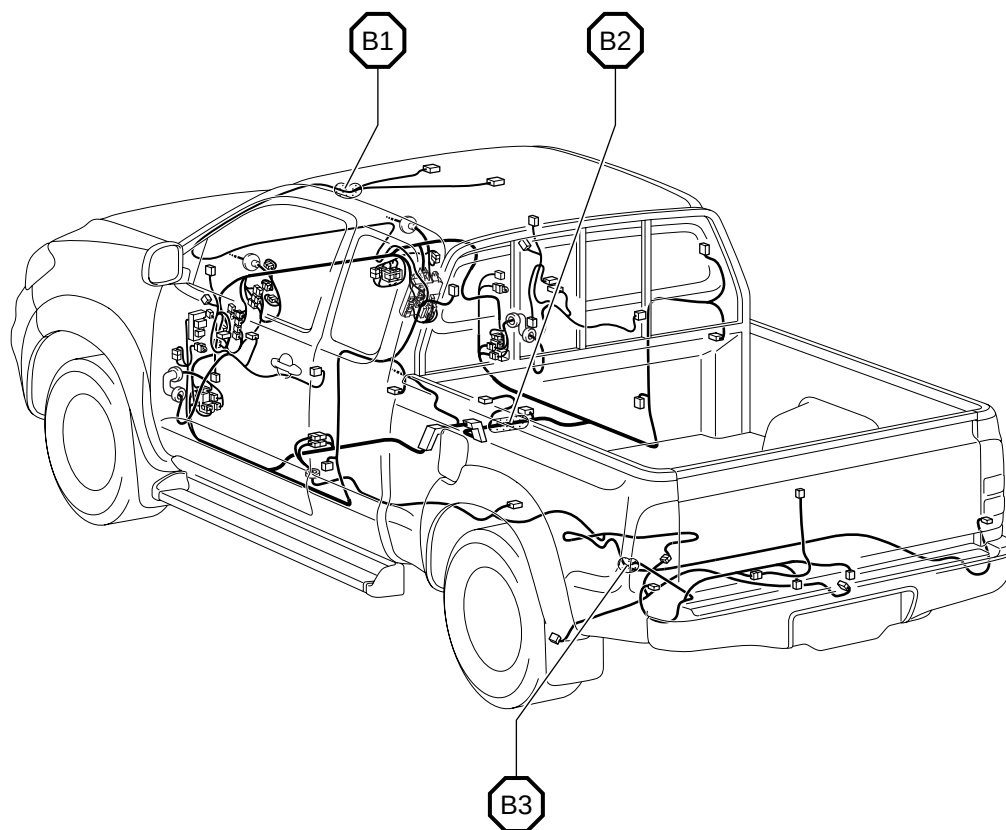
G ELECTRICAL WIRING ROUTING

□ : Location of Connector Joining Wire Harness and Wire Harness

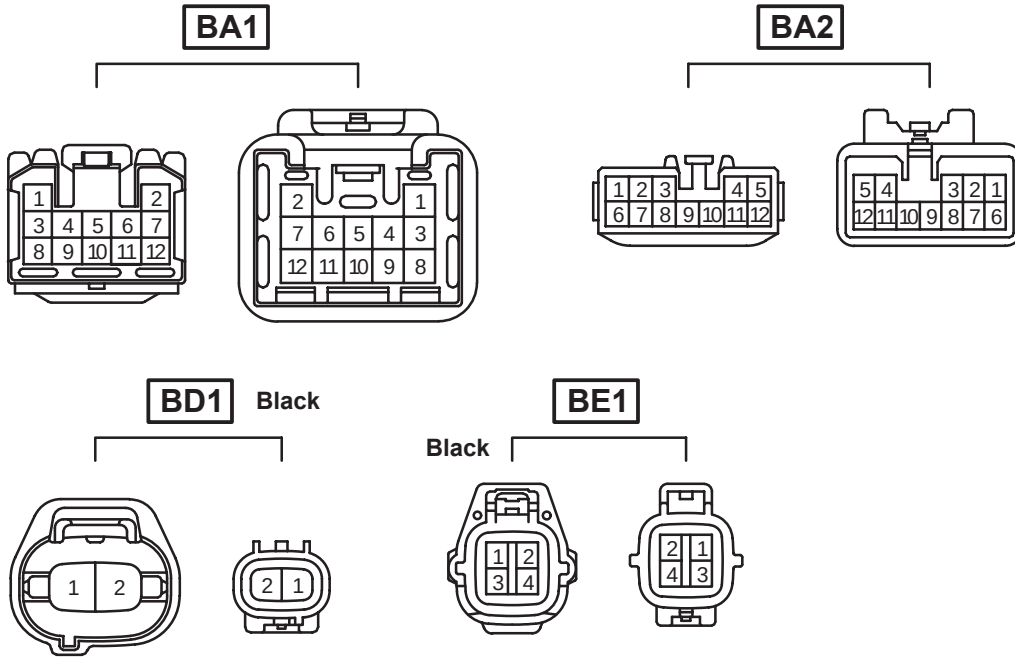
[Extra Cab]



○ : Location of Splice Points



Connector Joining Wire Harness and Wire Harness

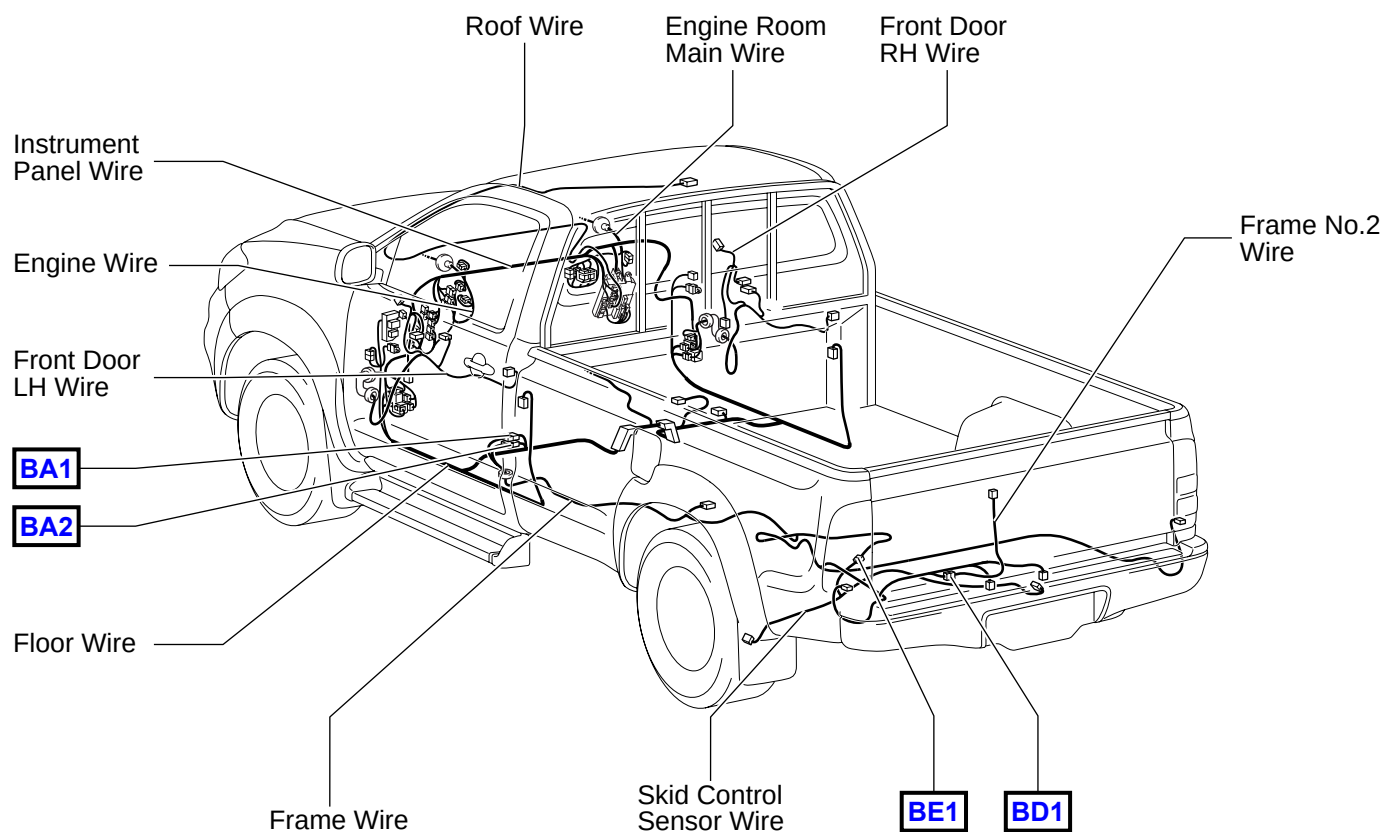


Code	Joining Wire Harness and Wire Harness (Connector Location)
BA1	Frame Wire and Floor Wire (Under the Front Seat LH)
BA2	
BD1	Frame No.2 Wire and Frame Wire (Under the Back Panel LH)
BE1	Frame Wire and Skid Control Sensor Wire (Rear of the Rear Differential)

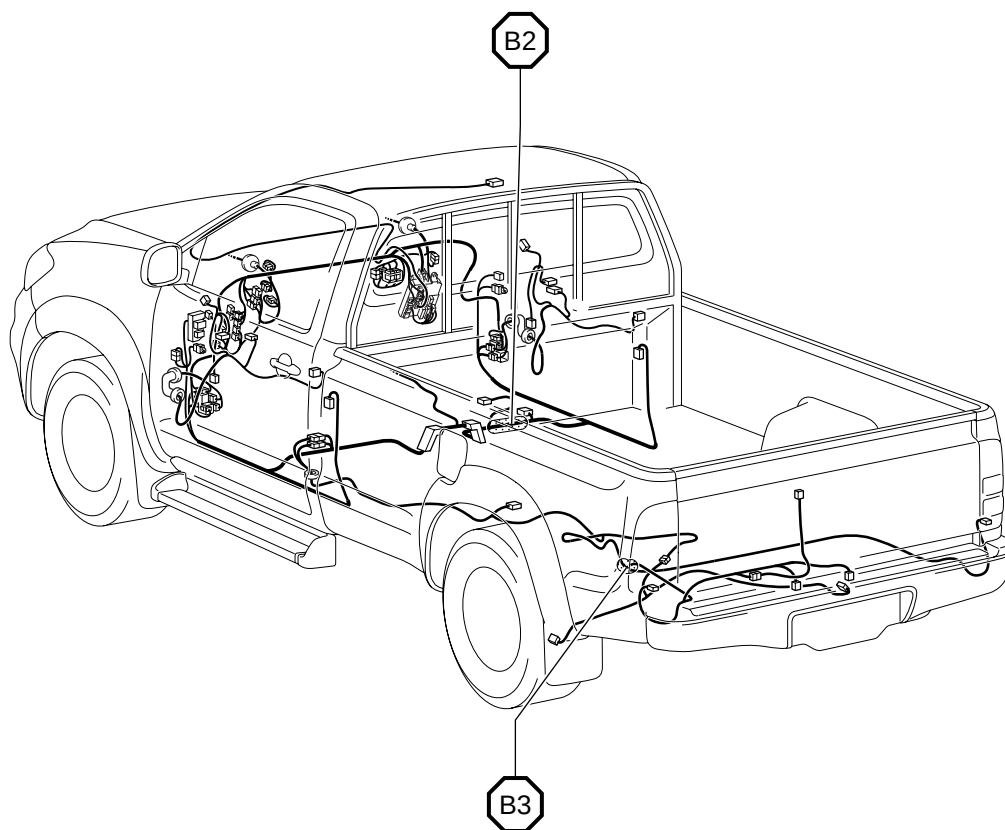
G ELECTRICAL WIRING ROUTING

□ : Location of Connector Joining Wire Harness and Wire Harness

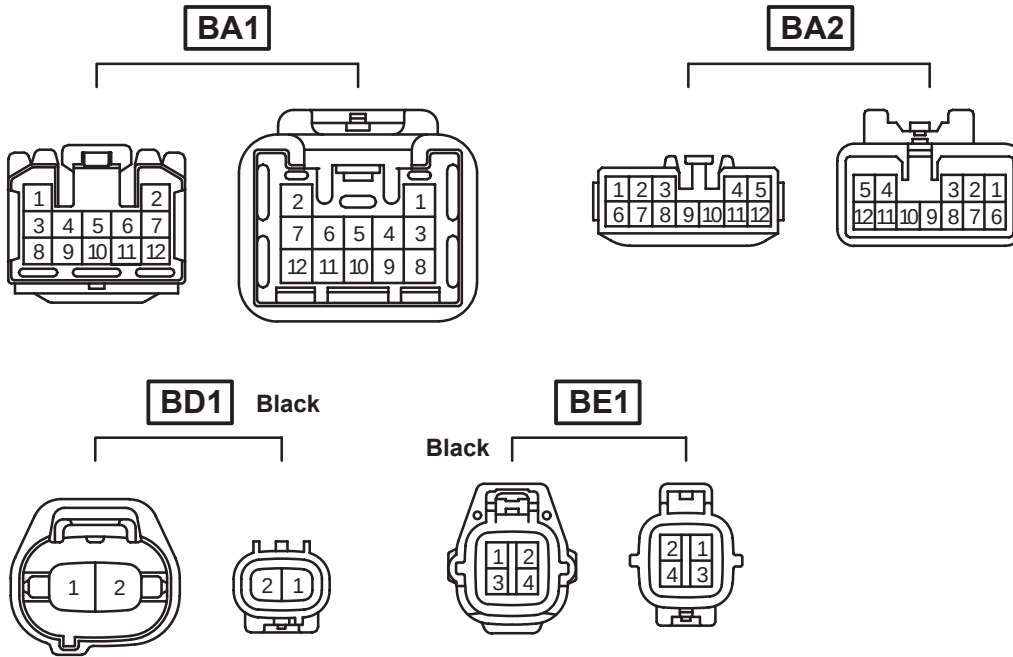
[Single Cab]



○ : Location of Splice Points



Connector Joining Wire Harness and Wire Harness



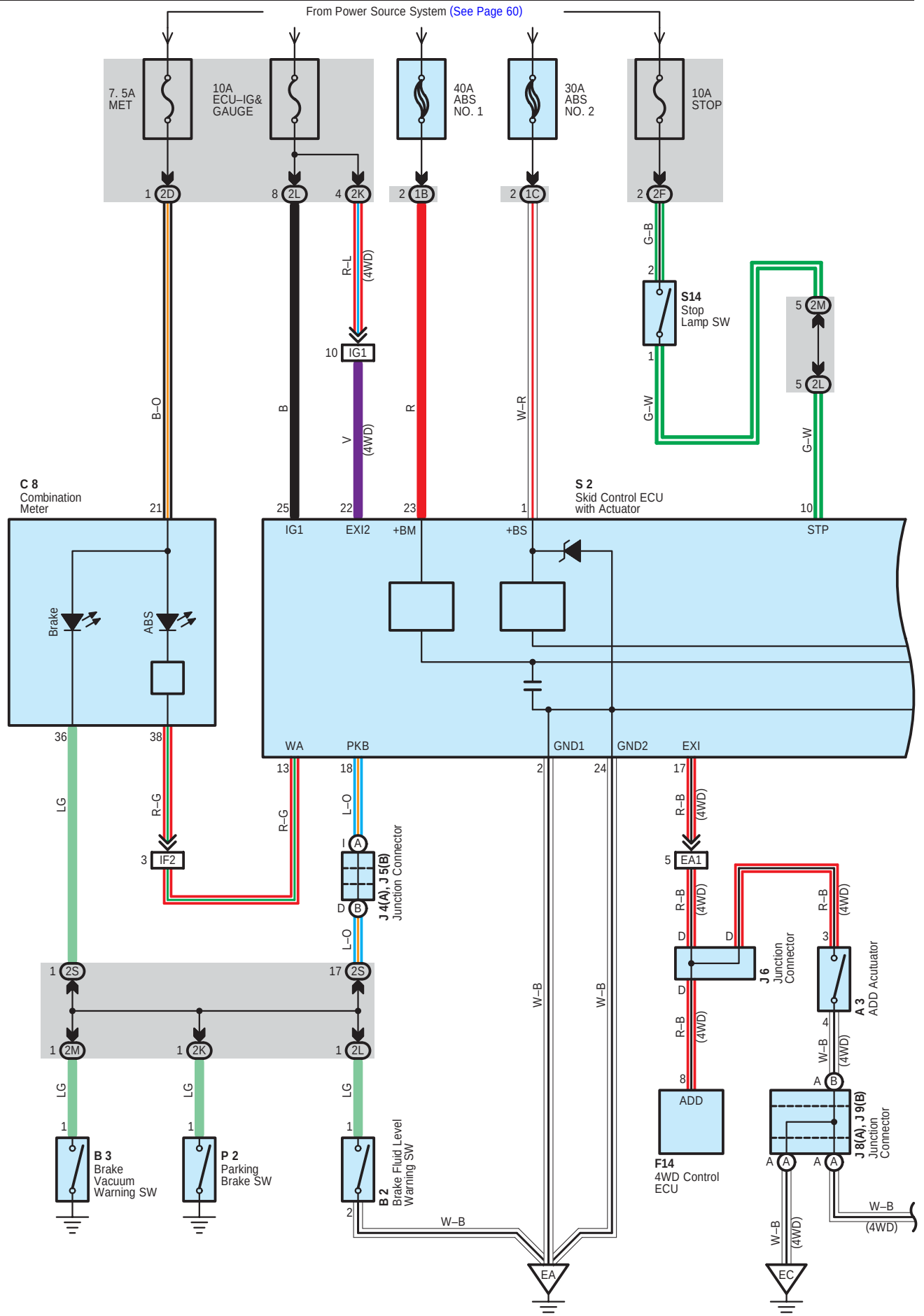
Code	Joining Wire Harness and Wire Harness (Connector Location)
BA1	Frame Wire and Floor Wire (Under the Front Seat LH)
BA2	
BD1	Frame No.2 Wire and Frame Wire (Under the Back Panel LH)
BE1	Frame Wire and Skid Control Sensor Wire (Rear of the Rear Differential)

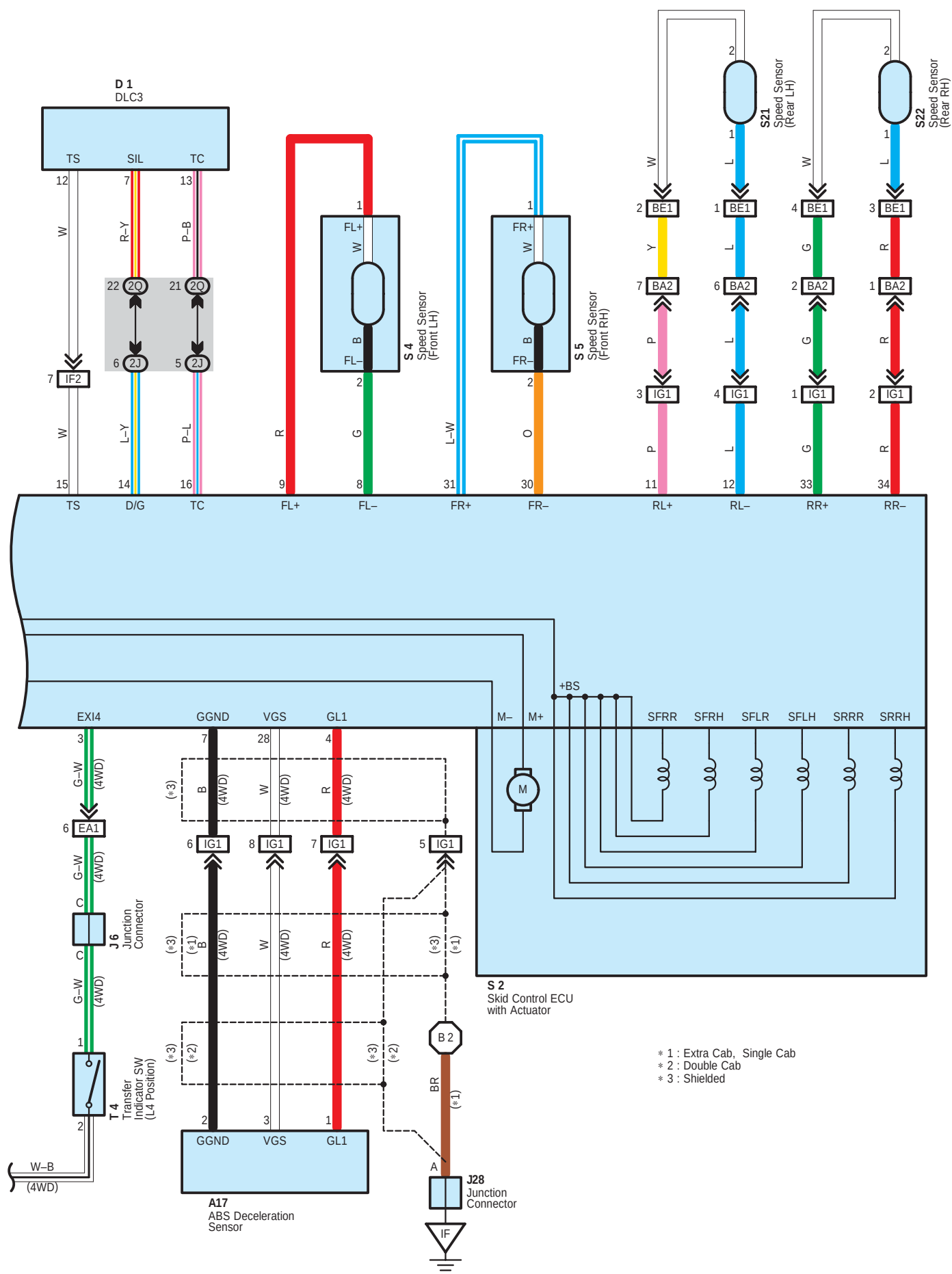
HILUX

ELECTRICAL WIRING DIAGRAM

SYSTEM CIRCUITS

	Page
ABS	116
Air Conditioner	178
Audio System	168
Back-Up Light	98
Charging	66
Cigarette Lighter	158
Clock	162
Combination Meter	172
Door Lock Control	130
ECT	108
Engine Control	68
Engine Immobilizer System	80
Front Fog Light	86
Front Wiper and Washer	152
Headlight	84
Horn	156
Illumination	88
Interior Light	100
Key Reminder	104
Light Reminder	104
Multiplex Communication System (CAN)	82
Power Outlet	160
Power Source	60
Power Window	148
Rear Window Defogger	164
Remote Control Mirror	166
Seat Belt Warning	106
Shift Lock	128
SRS	121
Starting	64
Stop Light	96
Taillight	88
Theft Deterrent	142
Turn Signal and Hazard Warning Light	92
Wireless Door Lock Control	136
4WD	124





System Outline

This system controls the respective brake fluid pressures acting on the brake cylinders of the right front wheel, left front wheel, and rear wheels when the brakes are applied in a panic stop so that the wheels do not lock. This results in improved directional stability and steerability during panic braking.

1. Input Signal

(1) Speed sensor signal

The speed of the wheels is detected and input to TERMINALS FL+, FR+, RL+ and RR+ of the skid control ECU with actuator.

(2) Stop lamp SW signal

A signal is input to TERMINAL STP of the skid control ECU with actuator when the brake pedal is depressed.

2. System Operation

During sudden braking, the skid control ECU with actuator which has signals input from each sensor lets the hydraulic pressure acting on each wheel cylinder escape to the reservoir.

The pump inside the skid control ECU with actuator is also operating at this time and it returns the brake fluid from the reservoir to the master cylinder, thus preventing locking of vehicle wheels.

If the skid control ECU with actuator judges that the hydraulic pressure acting on the wheel cylinder is insufficient, the current acting on the solenoid is controlled and the hydraulic pressure is increased.

Holding of the hydraulic pressure is also controlled by the ECU, by the same method as above, by repeated pressure reduction. Holding and increase are repeated to maintain vehicle stability and to improve steerability during sudden braking.

Service Hints

S4, S5 Speed Sensor (Front LH), (Front RH)

1-2 : 0.9-1.3 kΩ (25°C, 77°F)

S2 Skid Control ECU with Actuator

25-Ground : Approx. 12 volts with the ignition SW at ON position

10-Ground : Approx. 12 volts with the stop lamp SW on (Brake pedal depressed)

2, 24-Ground : Always continuity

1, 23-Ground : Always approx. 12 volts

S14 Stop Lamp SW

2-1 : Closed with the brake pedal depressed

○ : Parts Location

Code	See Page	Code	See Page	Code	See Page
A3	36	J4	A 38	S5	37
A17	40 (Double Cab)	J5	B 38	S14	39
	42 (Extra Cab)	J6 38		S21	41 (Double Cab)
	44 (Single Cab)	J8	A 38		43 (Extra Cab)
B2	36	J9	B 38		45 (Single Cab)
B3	36	J28 38		S22	41 (Double Cab)
C8	38	P2	39		43 (Extra Cab)
D1	38	S2	37		45 (Single Cab)
F14	39	S4	37	T4	37

○ : Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
1B	23	Engine Room Main Wire and Engine Room J/B (Engine Compartment Left)
1C		
2D	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2F	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2J		
2K	28	Floor Wire and Driver Side J/B (Instrument Panel Brace RH)
2L	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2M		
2Q	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2S		

 : **Connector Joining Wire Harness and Wire Harness**

Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
EA1	46	Engine Wire and Engine Room Main Wire (Inside of Engine Room R/B and Engine Room J/B)
IF2	50	Engine Room Main Wire and Instrument Panel Wire (Behind the Driver Side J/B)
IG1	50	Engine Room Main Wire and Floor Wire (Near the Driver Side J/B)
BA2	52 (Double Cab)	Frame Wire and Floor Wire (Under the Front Seat LH)
	54 (Extra Cab)	
	56 (Single Cab)	
BE1	52 (Double Cab)	Frame Wire and Skid Control Sensor Wire (Rear of the Rear Differential)
	54 (Extra Cab)	
	56 (Single Cab)	

 : **Ground Points**

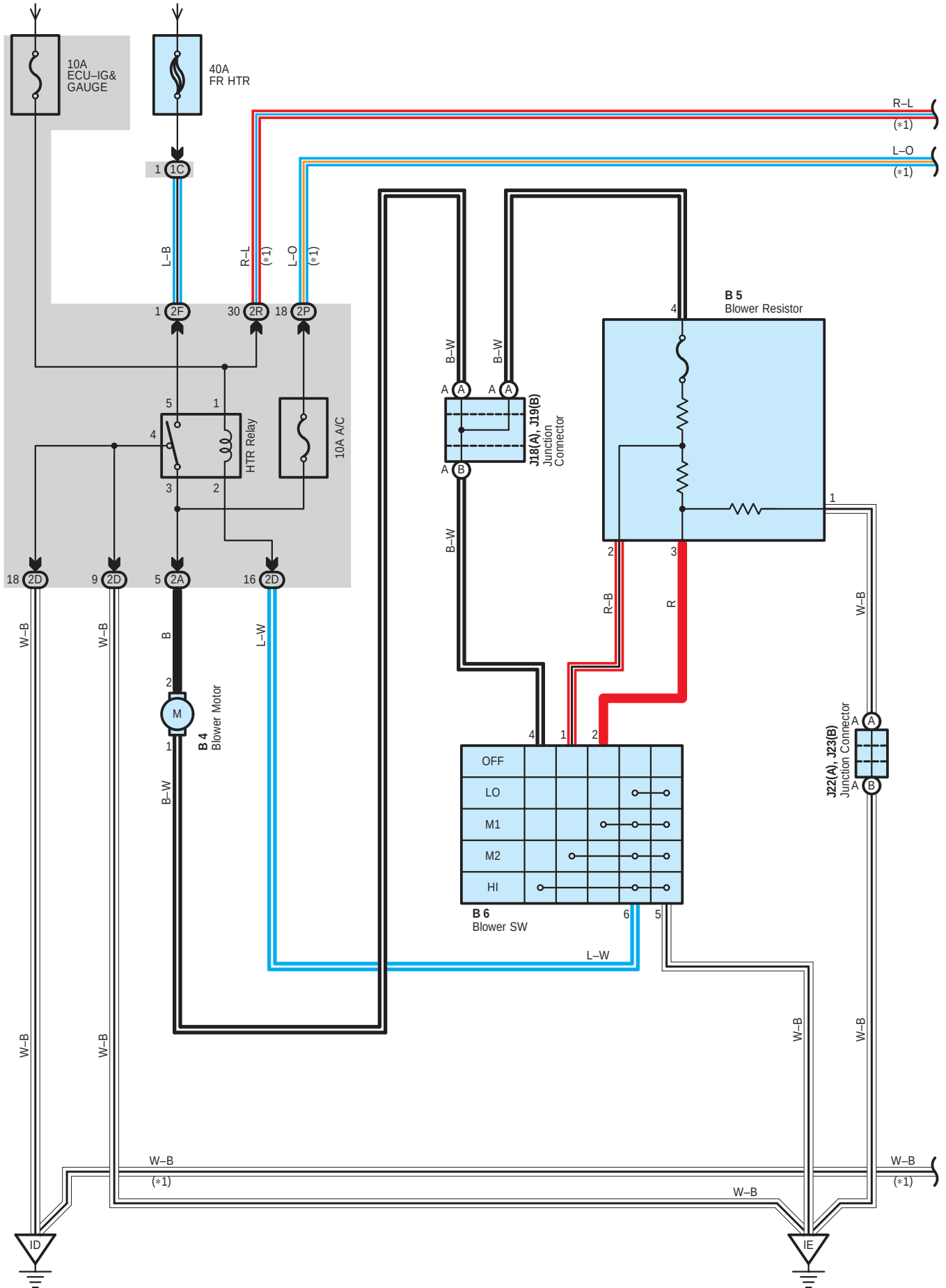
Code	See Page	Ground Points Location
EA	46	Right Side of the Fender Apron
EC	46	Rear Side of the Cylinder Block
IF	48	Cowl Side Panel RH

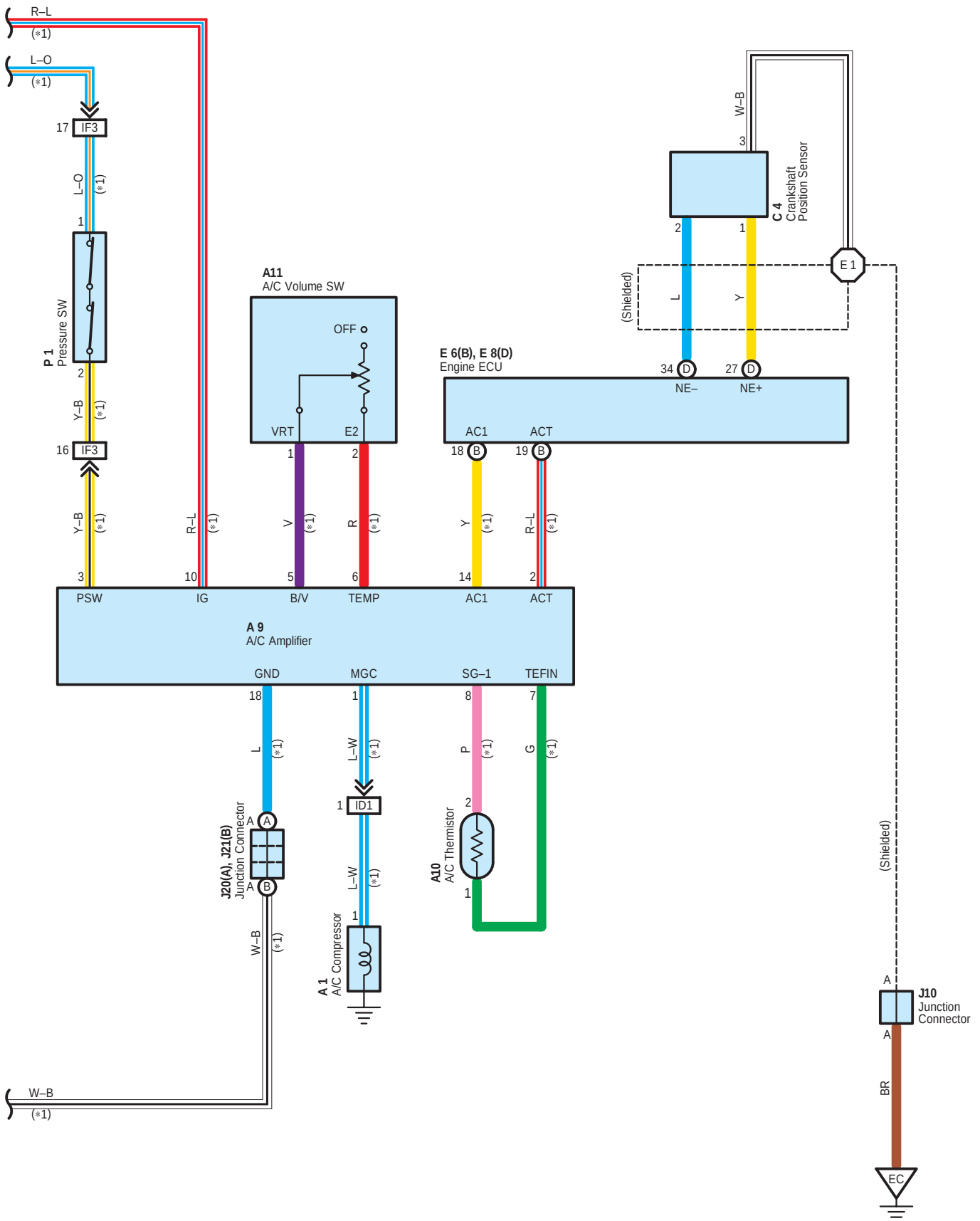
 : **Splice Points**

Code	See Page	Wire Harness with Splice Points	Code	See Page	Wire Harness with Splice Points
B2	54 (Extra Cab)	Frame Wire	B2	56 (Single Cab)	Frame Wire

Air Conditioner

From Power Source System (See Page 60)





Air Conditioner

System Outline

Current is applied at all times through the FR HTR fuse to TERMINAL 5 of the HTR relay.

When the ignition SW is turned on, the current flows through the ECU-IG&GAUGE fuse to TERMINAL 1 of the HTR relay to TERMINAL 2 to TERMINAL 6 of the blower SW.

Blower Motor Operation

* Low speed operation

When the blower SW is moved to LO position, the current flows to TERMINAL 6 of the blower SW to TERMINAL 5 to GROUND, causing the HTR relay to turn on. This causes the current flows from the FR HTR fuse to TERMINAL 5 of the HTR relay to TERMINAL 3 to TERMINAL 2 of the blower motor to TERMINAL 1 to TERMINAL 4 of the blower resistor to TERMINAL 1 to GROUND, rotating the blower motor at low speed.

* Medium speed operation (Operation at M1, M2)

When the blower SW is moved to M1 position, the current flows to TERMINAL 6 of the blower SW to TERMINAL 5 to GROUND, causing the HTR relay to turn on. This causes the current flows from the FR HTR fuse to TERMINAL 5 of the HTR relay to TERMINAL 3 to TERMINAL 2 of the blower motor to TERMINAL 1 to TERMINAL 4 of the blower resistor to TERMINAL 3 to TERMINAL 2 of the blower SW to TERMINAL 5 to GROUND. At this time, the blower resistance of the blower resistor is smaller than at low speed, so the blower motor rotates at medium low speed.

When the blower SW is moved to M2 position, the current flows through the HTR relay to TERMINAL 2 of the blower motor to TERMINAL 1 to TERMINAL 4 of the blower resistor to TERMINAL 2 to TERMINAL 1 of the blower SW to TERMINAL 5 to GROUND. At this time, resistance of the blower resistor is smaller than at M1 position, so the blower motor rotates at medium high speed.

* High speed operation

When the blower SW is moved to HI position, the current flows to TERMINAL 6 of the blower SW to TERMINAL 5 to GROUND, causing the HTR relay to turn on.

This causes the current flows from the FR HTR fuse to TERMINAL 5 of the HTR relay to TERMINAL 3 to TERMINAL 2 of the blower motor to TERMINAL 1 to TERMINAL 4 of the blower SW to TERMINAL 5 to GROUND, rotating the blower motor at high speed.

Service Hints

HTR Relay

5-3 : Closed with the ignition SW at ON position and the blower SW on

P1 Pressure SW

1-2 : Open with the refrigerant pressure at less than approx. 2.0 kgf/cm² (28.4 psi, 196.1 kpa) or more than approx. 32.0 kgf/cm² (455 psi, 3138.1 kpa)

A9 A/C Amplifier

10-Ground : Approx. 12 volts with the ignition SW at ON position

18-Ground : Always continuity

○ : Parts Location

Code	See Page	Code	See Page	Code	See Page
A1	36	B6	38	J19	B 38
A9	38	C4	36	J20	A 38
A10	38	E6	B 38	J21	B 38
A11	38	E8	D 38	J22	A 38
B4	38	J10	38	J23	B 38
B5	38	J18	A 38	P1	37

○ : Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
1C	23	Engine Room Main Wire and Engine Room J/B (Engine Compartment Left)
2A	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2D		
2F	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2P	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2R		

 : **Connector Joining Wire Harness and Wire Harness**

Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
ID1	48	Engine Wire and Instrument Panel Wire (Behind the Glove Box)
IF3	50	Engine Room Main Wire and Instrument Panel Wire (Behind the Driver Side J/B)

 : **Ground Points**

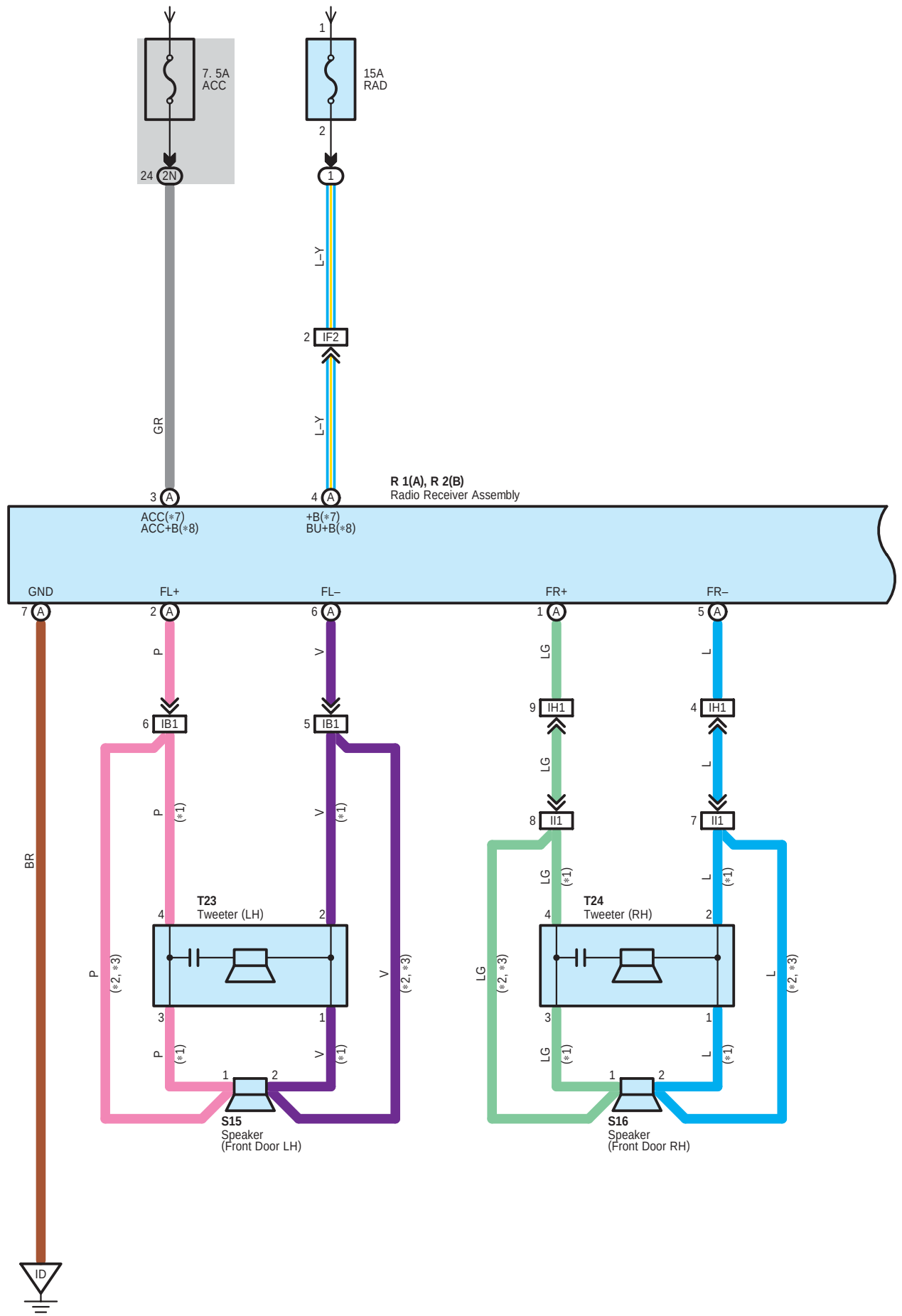
Code	See Page	Ground Points Location
EC	46	Rear Side of the Cylinder Block
ID	48	Instrument Panel Reinforcement Center
IE	48	Instrument Panel Reinforcement RH

 : **Splice Points**

Code	See Page	Wire Harness with Splice Points	Code	See Page	Wire Harness with Splice Points
E1	46	Engine Wire			

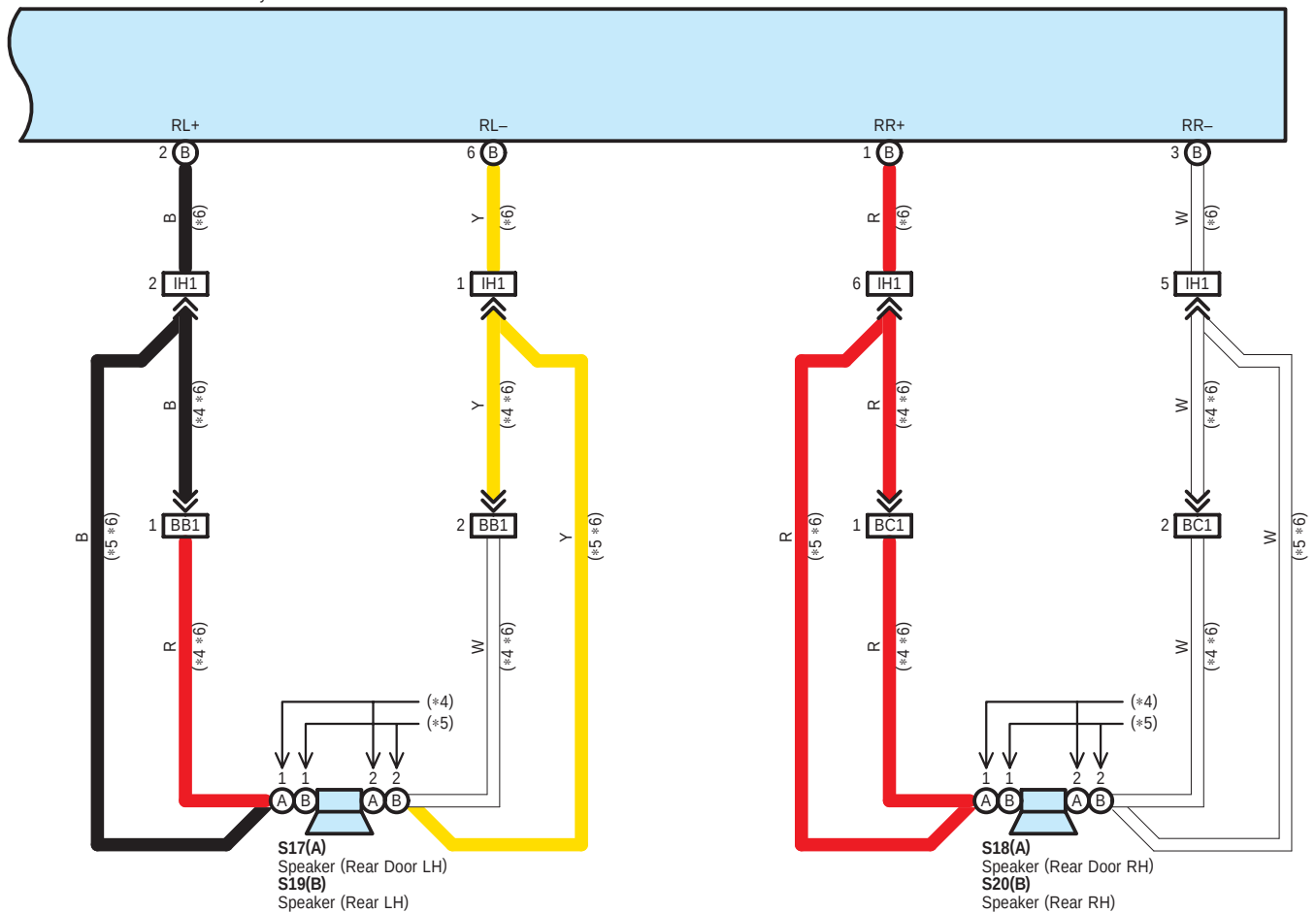
Audio System

From Power Source System (See Page 60)



- * 1 : 6 Speaker
- * 2 : 4 Speaker
- * 3 : 2 Speaker
- * 4 : Double Cab
- * 5 : Extra Cab
- * 6 : 6 Speaker, 4 Speaker
- * 7 : w/ CD Player
- * 8 : w/o CD Player

R 1(A), R 2(B)
Radio Receiver Assembly



Audio System

Service Hints

R1 (A) Radio Receiver Assembly

- (A) 4–Ground : Always approx. 12 volts
- (A) 3–Ground : Approx. 12 volts with the ignition SW at ACC or ON position
- (A) 7–Ground : Always continuity

○ : Parts Location

Code		See Page	Code		See Page	Code		See Page
R1	A	39	S16		43 (Extra Cab)	T23		41 (Double Cab)
R2	B	39			45 (Single Cab)			43 (Extra Cab)
S15		41 (Double Cab)	S17	A	41 (Double Cab)	T24		41 (Double Cab)
		43 (Extra Cab)	S18	A	41 (Double Cab)			43 (Extra Cab)
		45 (Single Cab)	S19	B	43 (Extra Cab)			
S16		41 (Double Cab)	S20	B	43 (Extra Cab)			

○ : Relay Blocks

Code	See Page	Relay Blocks (Relay Block Location)
1	22	Engine Room R/B (Engine Compartment Left)

○ : Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
2N	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)

□ : Connector Joining Wire Harness and Wire Harness

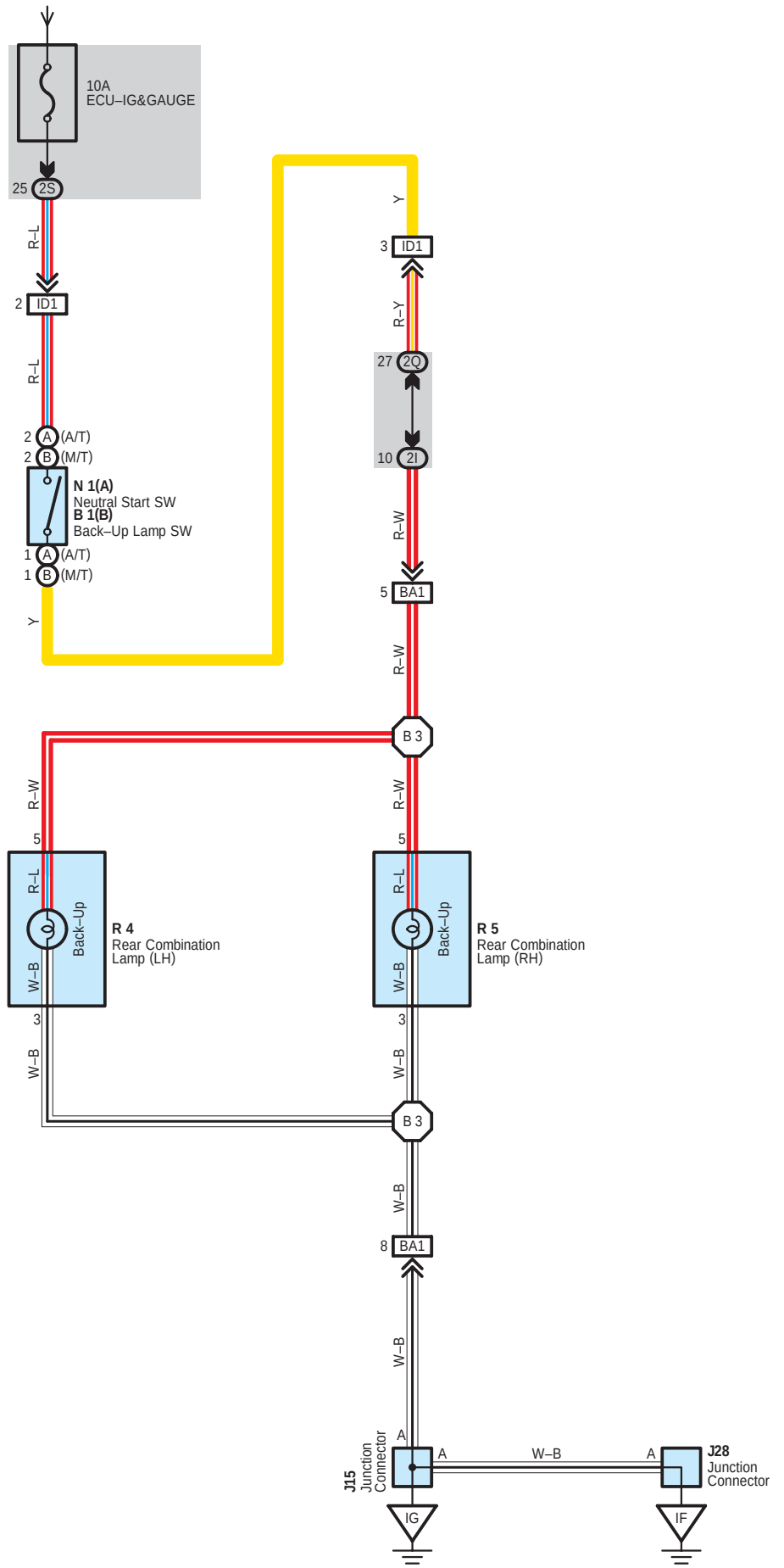
Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
IB1	48	Front Door LH Wire and Instrument Panel Wire (Left Kick Panel)
IF2	50	Engine Room Main Wire and Instrument Panel Wire (Behind the Driver Side J/B)
IH1	50	Instrument Panel Wire and Floor Wire (Right Kick Panel)
II1	50	Front Door RH Wire and Floor Wire (Right Kick Panel)
BB1	52 (Double Cab)	Rear Door No.1 Wire LH and Floor Wire (Center Pillar LH)
BC1	52 (Double Cab)	Rear Door No.1 Wire RH and Floor Wire (Center Pillar RH)

▽ : Ground Points

Code	See Page	Ground Points Location
ID	48	Instrument Panel Reinforcement Center

Back-Up Light

From Power Source System (See Page 60)



Service Hints

B1 (B) Back-Up Light SW (M/T)

(B) 2-(B) 1 : Closed with the shift lever in R position

N1 (A) Neutral Start SW (A/T)

(A) 2-(A) 1 : Closed with the shift lever in R position

○ : Parts Location

Code	See Page	Code	See Page	Code	See Page
B1	B	36	41 (Double Cab)	R5	43 (Extra Cab)
J15	38	R4	43 (Extra Cab)		45 (Single Cab)
J28	38		45 (Single Cab)		
N1	A	37	R5	41 (Double Cab)	

○ : Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
2I	28	Floor Wire and Driver Side J/B (Instrument Panel Brace RH)
2Q	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2S		

□ : Connector Joining Wire Harness and Wire Harness

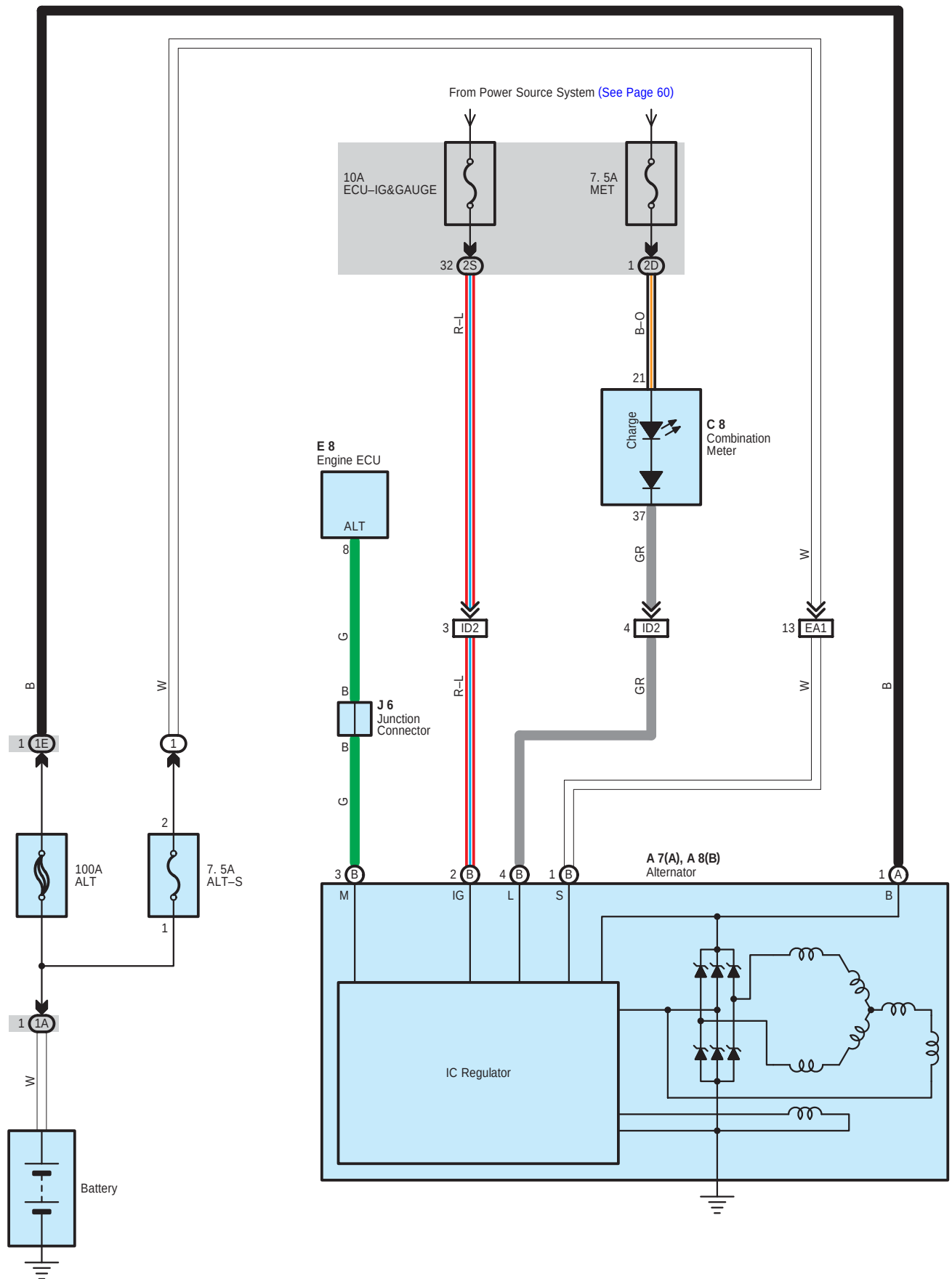
Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
ID1	48	Engine Wire and Instrument Panel Wire (Behind the Glove Box)
BA1	52 (Double Cab) 54 (Extra Cab) 56 (Single Cab)	Frame Wire and Floor Wire (Under the Front Seat LH)

▽ : Ground Points

Code	See Page	Ground Points Location
IF	48	Cowl Side Panel RH
IG	48	Cowl Side Panel LH

○ : Splice Points

Code	See Page	Wire Harness with Splice Points	Code	See Page	Wire Harness with Splice Points
B3	52 (Double Cab) 54 (Extra Cab)	Frame Wire	B3	56 (Single Cab)	Frame Wire



Service Hints

A7 (A) Alternator

(A) 1–Ground : 13.2–14.0 volts with the engine running at 5000 rpm and 115°C (239°F)

: Parts Location

Code		See Page	Code		See Page	Code		See Page
A7	A	36	C8	38		J6	38	
A8	B	36	E8	38				

: Relay Blocks

Code	See Page	Relay Blocks (Relay Block Location)
1	22	Engine Room R/B (Engine Compartment Left)

: Junction Block and Wire Harness Connector

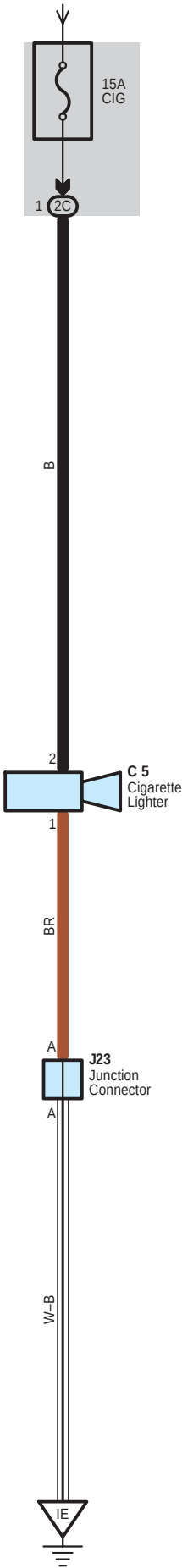
Code	See Page	Junction Block and Wire Harness (Connector Location)
1A	23	Engine Room Main Wire and Engine Room J/B (Engine Compartment Left)
1E	23	Engine Wire and Engine Room J/B (Engine Compartment Left)
2D	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2S	29	

: Connector Joining Wire Harness and Wire Harness

Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
EA1	46	Engine Wire and Engine Room Main Wire (Inside of Engine Room R/B and Engine Room J/B)
ID2	48	Engine Wire and Instrument Panel Wire (Behind the Glove Box)

Cigarette Lighter

From Power Source System (See Page 60)



Service Hints

C5 Cigarette Lighter

2–Ground : Approx. 12 volts with the ignition SW at ACC or ON position

1–Ground : Always continuity

: Parts Location

Code	See Page	Code	See Page	Code	See Page
C5	38	J23	38		

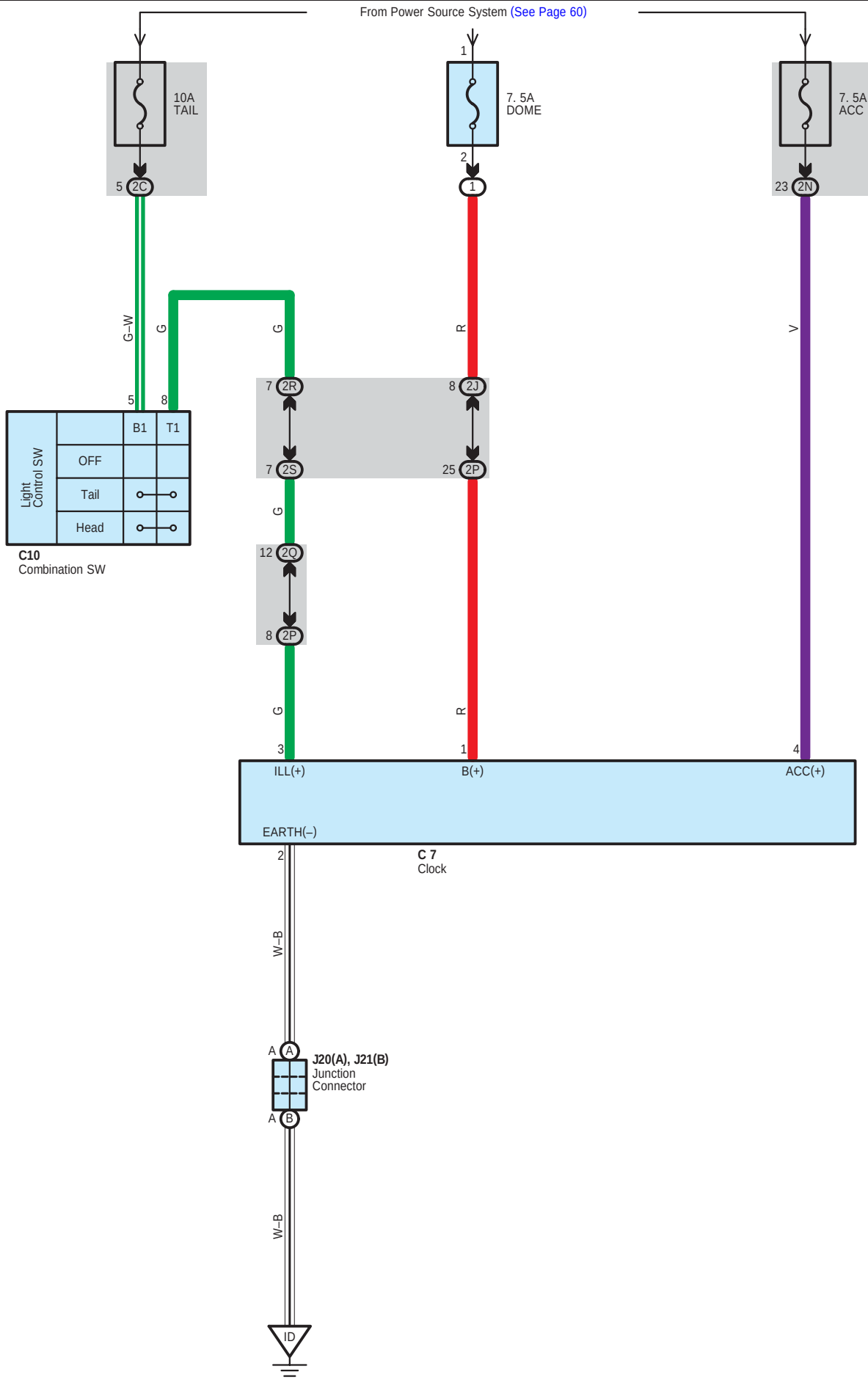
: Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
2C	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)

: Ground Points

Code	See Page	Ground Points Location
IE	48	Instrument Panel Reinforcement RH

Clock



Service Hints

C7 Clock

- 4—Ground : Approx. 12 volts with the ignition SW at ACC or ON position
- 1—Ground : Always approx. 12 volts
- 3—Ground : Approx. 12 volts with the light control SW at TAIL or HEAD position
- 2—Ground : Always continuity

: Parts Location

Code	See Page	Code	See Page	Code	See Page
C7	38	J20	A	38	
C10	38	J21	B	38	

: Relay Blocks

Code	See Page	Relay Blocks (Relay Block Location)
1	22	Engine Room R/B (Engine Compartment Left)

: Junction Block and Wire Harness Connector

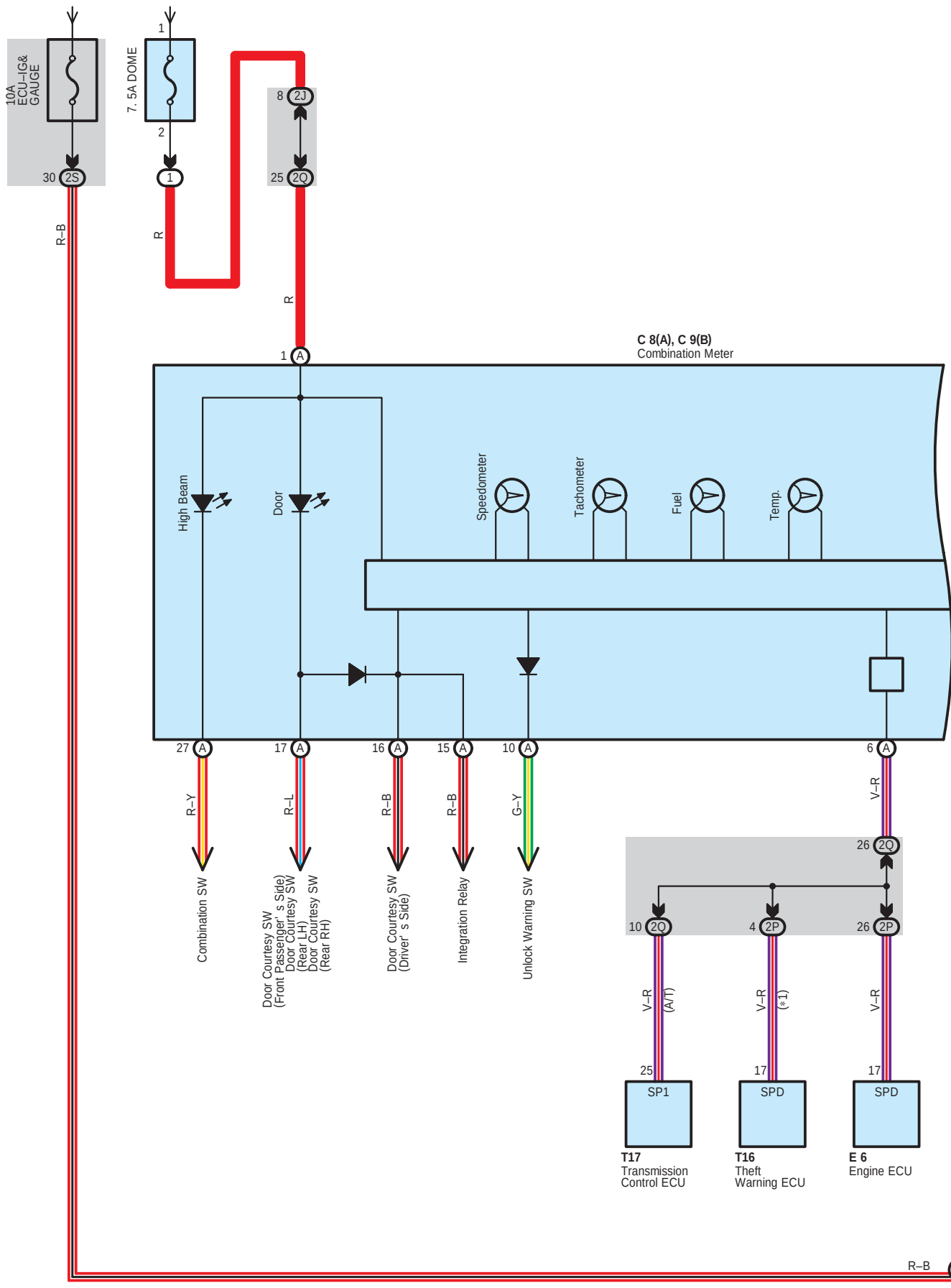
Code	See Page	Junction Block and Wire Harness (Connector Location)
2C	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2J	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2N	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2P		
2Q		
2R		
2S		

: Ground Points

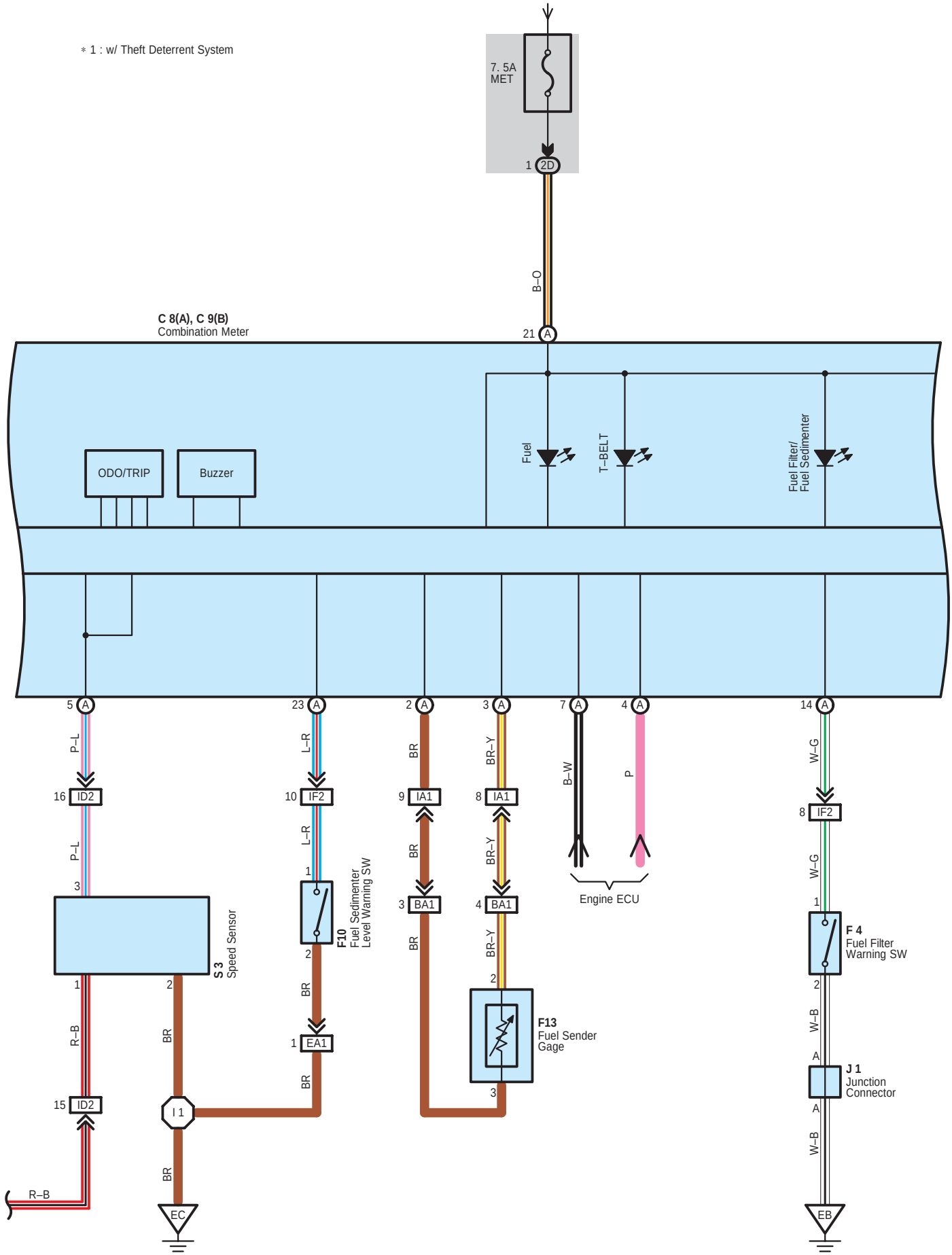
Code	See Page	Ground Points Location
ID	48	Instrument Panel Reinforcement Center

Combination Meter

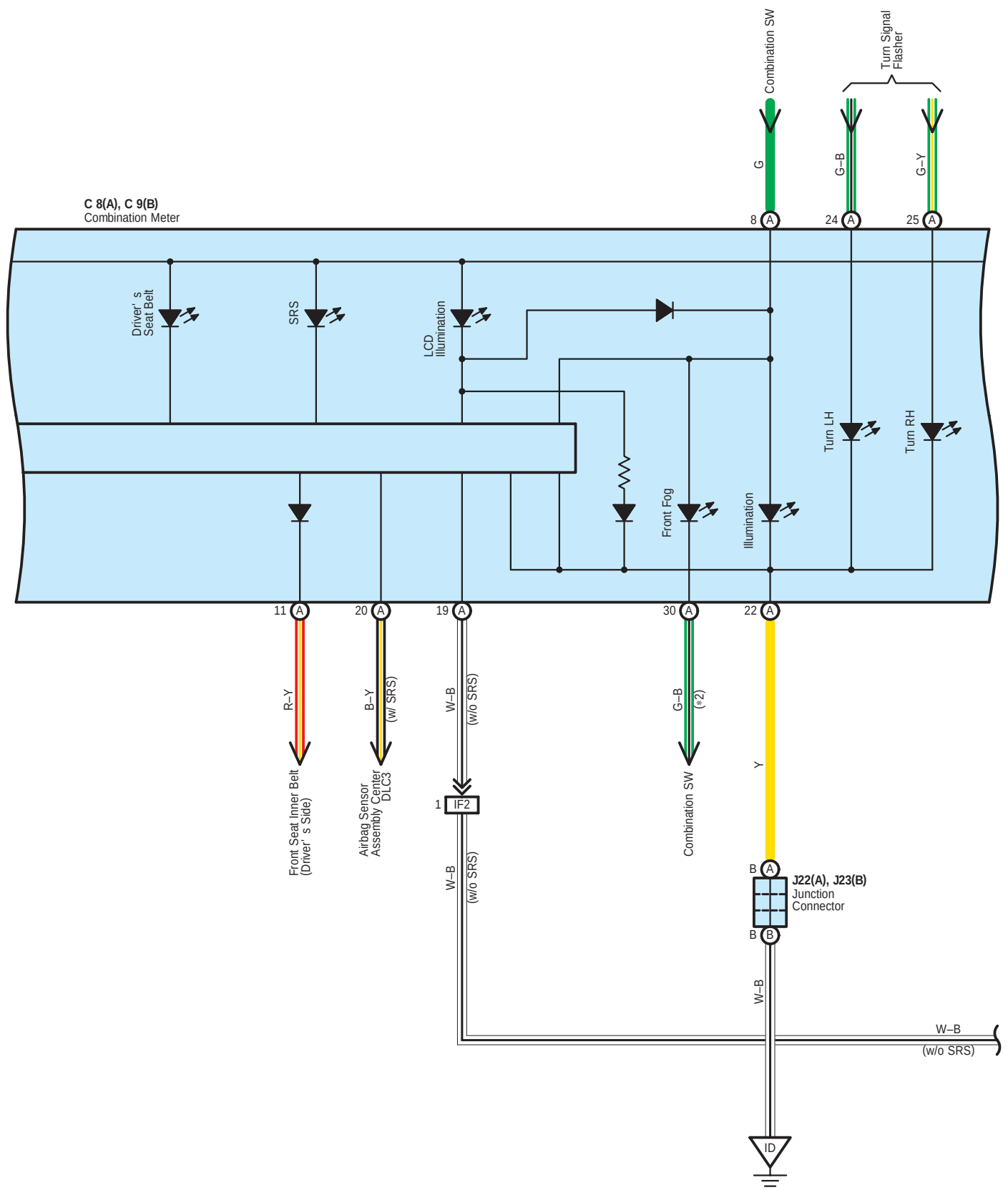
From Power Source System (See Page 60)



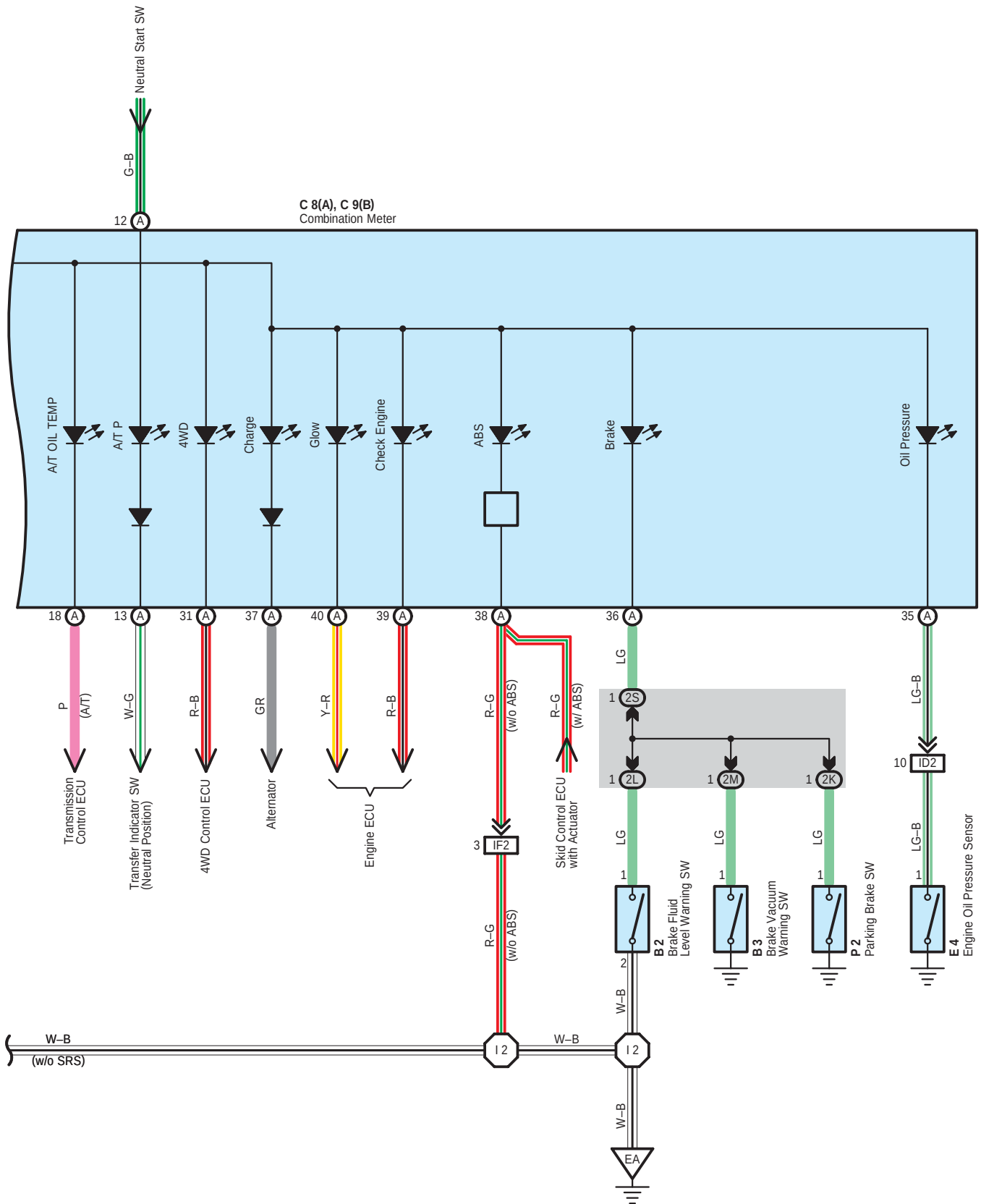
* 1 : w/ Theft Deterrent System



Combination Meter



* 2 : w/ Front Fog Light



Combination Meter

Service Hints

C8 (A) Combination Meter

(A) 21–Ground : Approx. 12 volts with the ignition SW at ON position

(A) 1–Ground : Always approx. 12 volts

(A) 22–Ground : Always continuity

F13 Fuel Sender Gage

2–3 : Approx. 4 Ω with the fuel full

Approx. 110 Ω with the fuel empty

○ : Parts Location

Code	See Page	Code	See Page	Code	See Page
B2	36	F4	36	J22	A 38
B3	36	F10	36	J23	B 38
C8	A 38	F13	40 (Double Cab)	P2	39
C9	B 38		42 (Extra Cab)	S3	37
E4	36		44 (Single Cab)	T16	39
E6	38	J1	37	T17	39

○ : Relay Blocks

Code	See Page	Relay Blocks (Relay Block Location)
1	22	Engine Room R/B (Engine Compartment Left)

○ : Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
2D	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2J	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2K	28	Floor Wire and Driver Side J/B (Instrument Panel Brace RH)
2L	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2M		
2P	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2Q		
2S		

□ : Connector Joining Wire Harness and Wire Harness

Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
EA1	46	Engine Wire and Engine Room Main Wire (Inside of Engine Room R/B and Engine Room J/B)
IA1	48	Instrument Panel Wire and Floor Wire (Left Kick Panel)
ID2	48	Engine Wire and Instrument Panel Wire (Behind the Glove Box)
IF2	50	Engine Room Main Wire and Instrument Panel Wire (Behind the Driver Side J/B)
BA1	52 (Double Cab)	Frame Wire and Floor Wire (Under the Front Seat LH)
	54 (Extra Cab)	
	56 (Single Cab)	

▽ : Ground Points

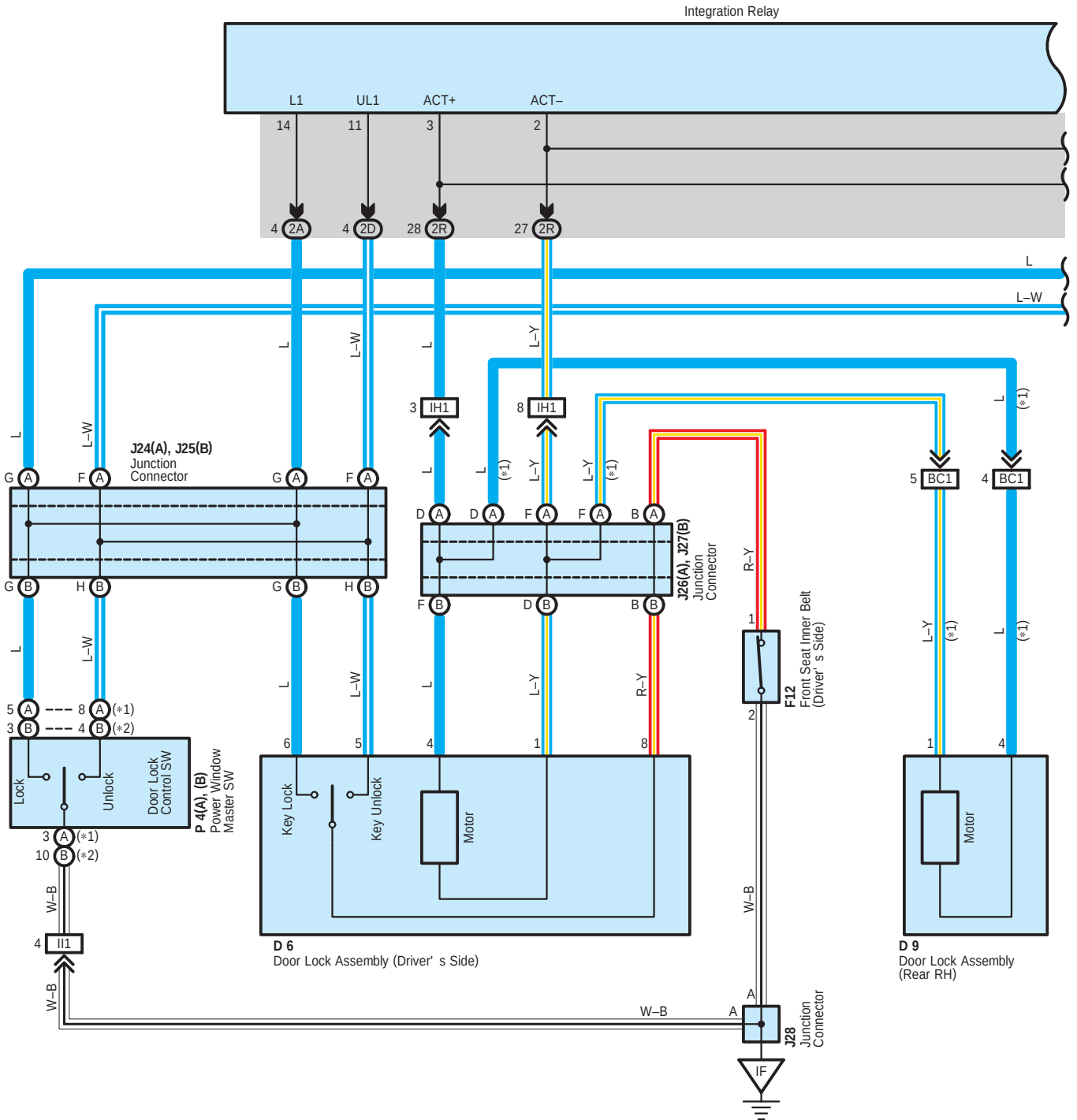
Code	See Page	Ground Points Location
EA	46	Right Side of the Fender Apron
EB	46	Left Side of the Fender Apron
EC	46	Rear Side of the Cylinder Block
ID	48	Instrument Panel Reinforcement Center

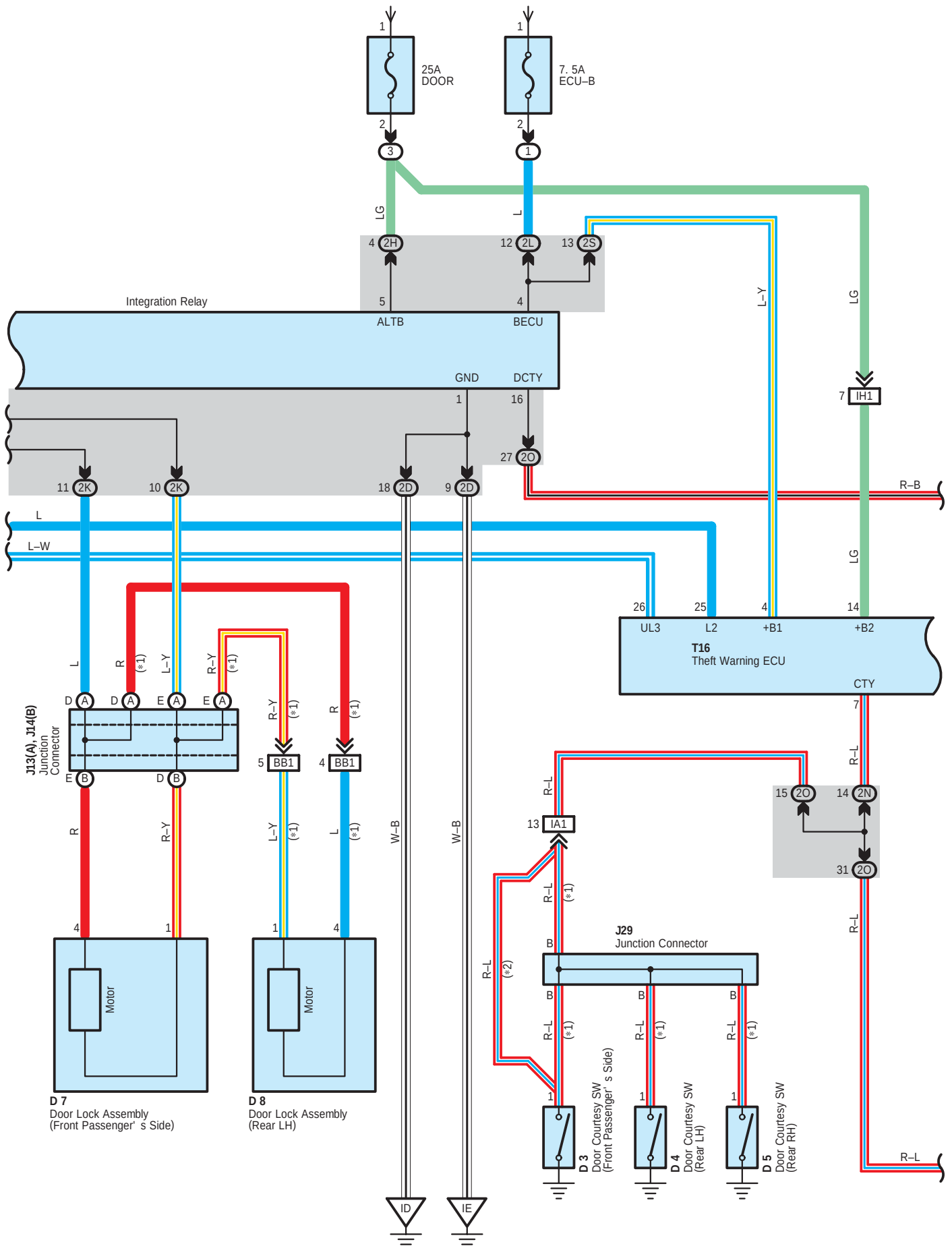
○ : Splice Points

Code	See Page	Wire Harness with Splice Points	Code	See Page	Wire Harness with Splice Points
I1	50	Engine Wire	I2	50	Engine Wire

Door Lock Control

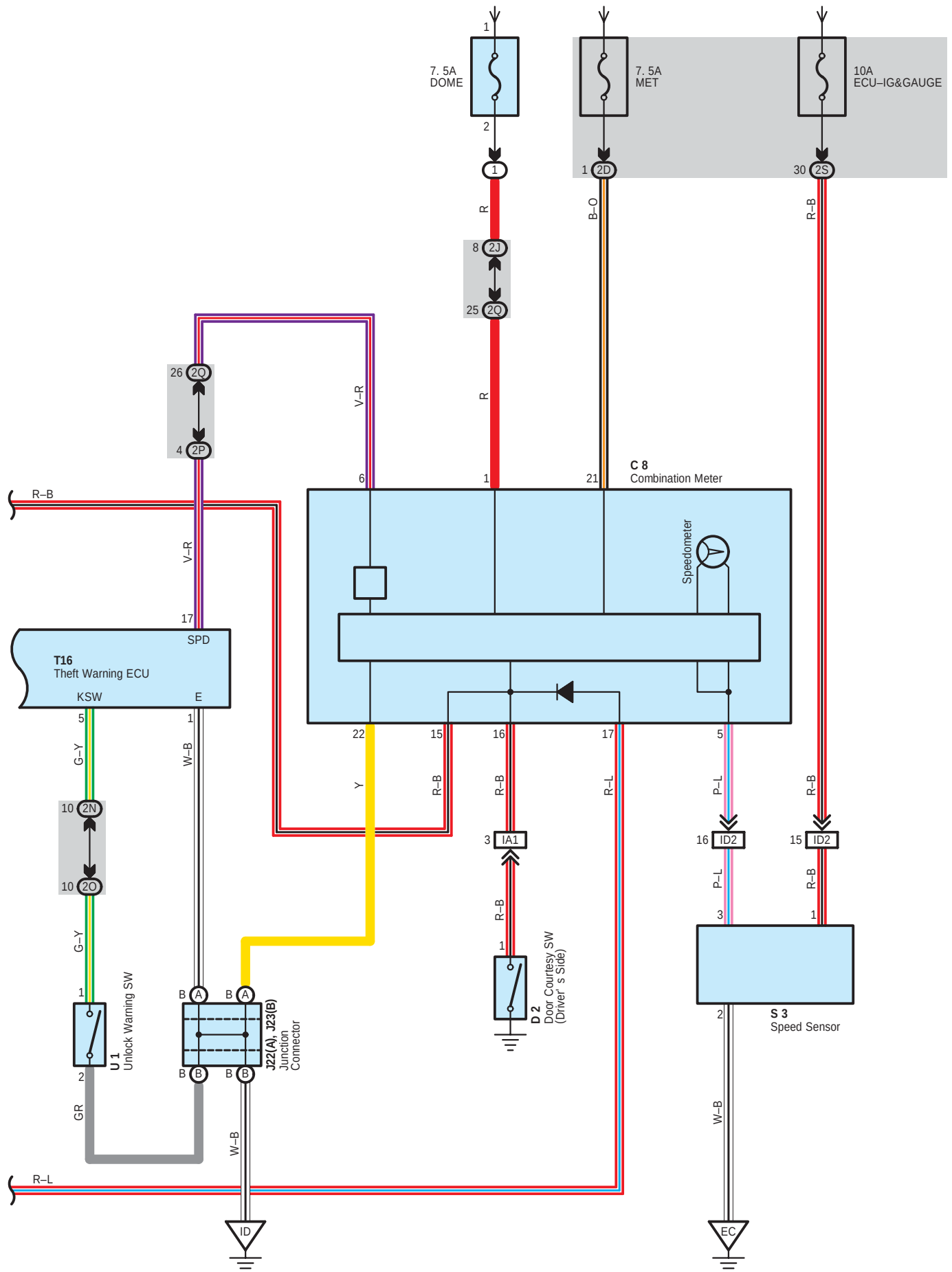
- * 1 : Double Cab
- * 2 : Extra Cab





Door Lock Control

From Power Source System (See Page 60)



System Outline

1. Manual Lock Operation

When the door lock control SW or driver's side door key lock and unlock SW (When driver's seat belt is unfastened) are operated to LOCK position, a lock signal is input to TERMINAL 14 of the integration relay and causes the integration relay to function. The current flows from TERMINAL 5 of the integration relay to TERMINAL 3 to the door lock motors to TERMINAL 2 of the integration relay to TERMINAL 1 to GROUND and the door lock motors locks the door.

2. Manual Unlock Operation

When the door lock control SW or driver's side door key lock and unlock SW (When driver's seat belt is unfastened) are operated to UNLOCK position, an unlock signal is input to TERMINAL 11 of the integration relay and causes the integration relay to function. The current flows from TERMINAL 5 of the integration relay to TERMINAL 2 to the door lock motors to TERMINAL 3 of the integration relay to TERMINAL 1 to GROUND and the door lock motors unlocks the door.

3. Speed-Sensitive Automatic Door Lock

Under the precondition that one of the doors is opened with the ignition SW OFF, the theft warning ECU outputs lock signal to the integration relay and the relay sends the lock signal to all the door lock motors when all the flowing conditions are satisfied.

- * All the door are shut.
- * The unlock warning SW is ON besides the ignition SW is ON.
- * The vehicle speed reaches about 25km/h.

4. Ignition SW Linked Automatic Door Unlock

Under the precondition that the unlock warning SW is ON as well as the ignition SW is ON after door locked by the speed-sensitive automatic door lock, the theft warning ECU outputs unlock signal to the integration relay and the relay sends the unlock signal to all the door lock motor when all the flowing conditions are satisfied.

- * The unlock warning SW is ON.
- * The ignition SW is OFF.

Service Hints

Integration Relay

- 1-Ground : Always continuity
- 4, 5-Ground : Always approx. 12 volts
- 14-Ground : Continuity with the door lock control SW or door key lock and unlock SW (When driver's seat belt is unfastened) locked
- 11-Ground : Continuity with the door lock control SW or door key lock and unlock SW (When driver's seat belt is unfastened) unlocked

D6 Door Lock Assembly (Driver's Side)

- 5-8 : Continuity with the door lock cylinder unlocked with the key
- 6-8 : Continuity with the door lock cylinder locked with the key

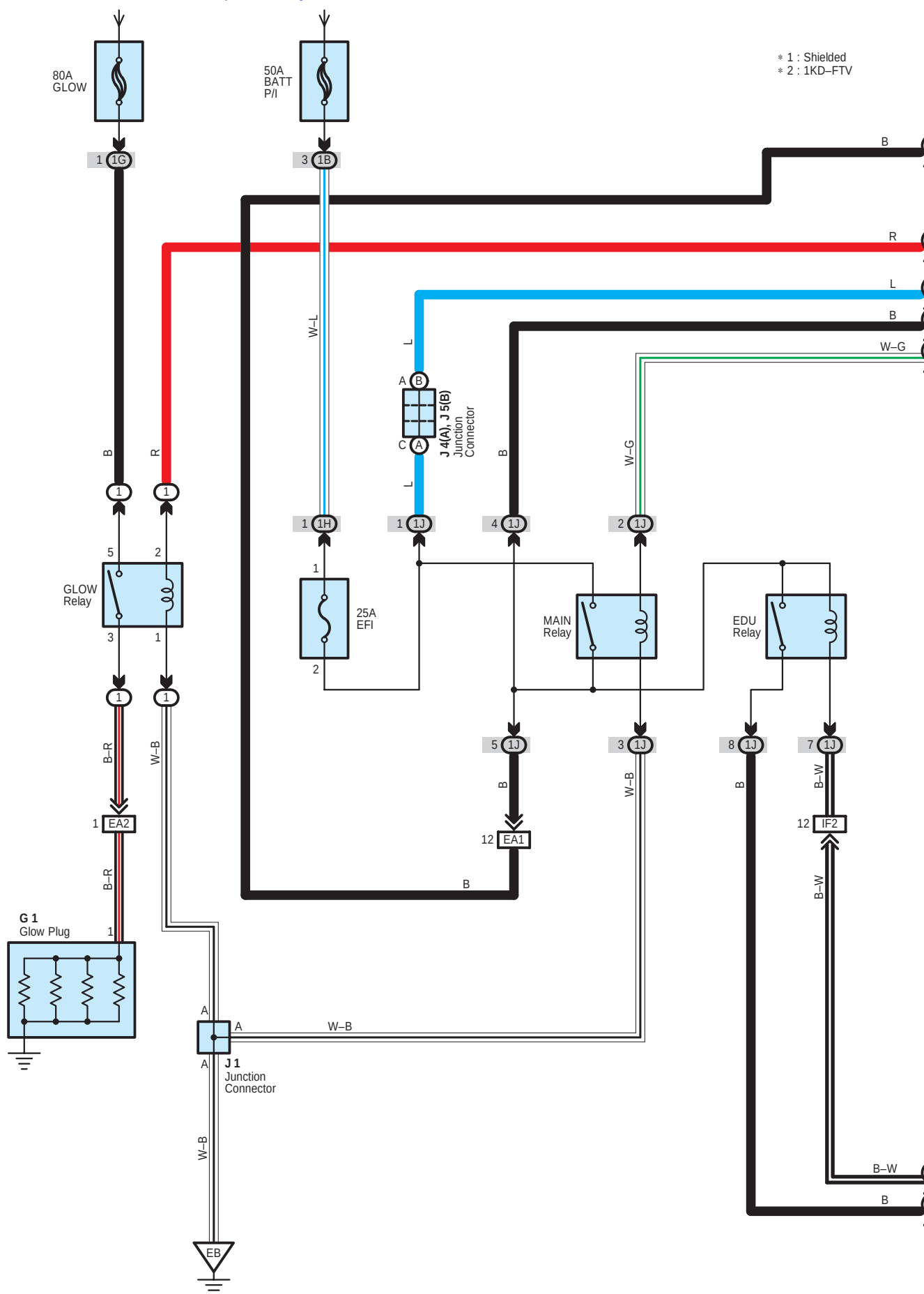
○ : Parts Location

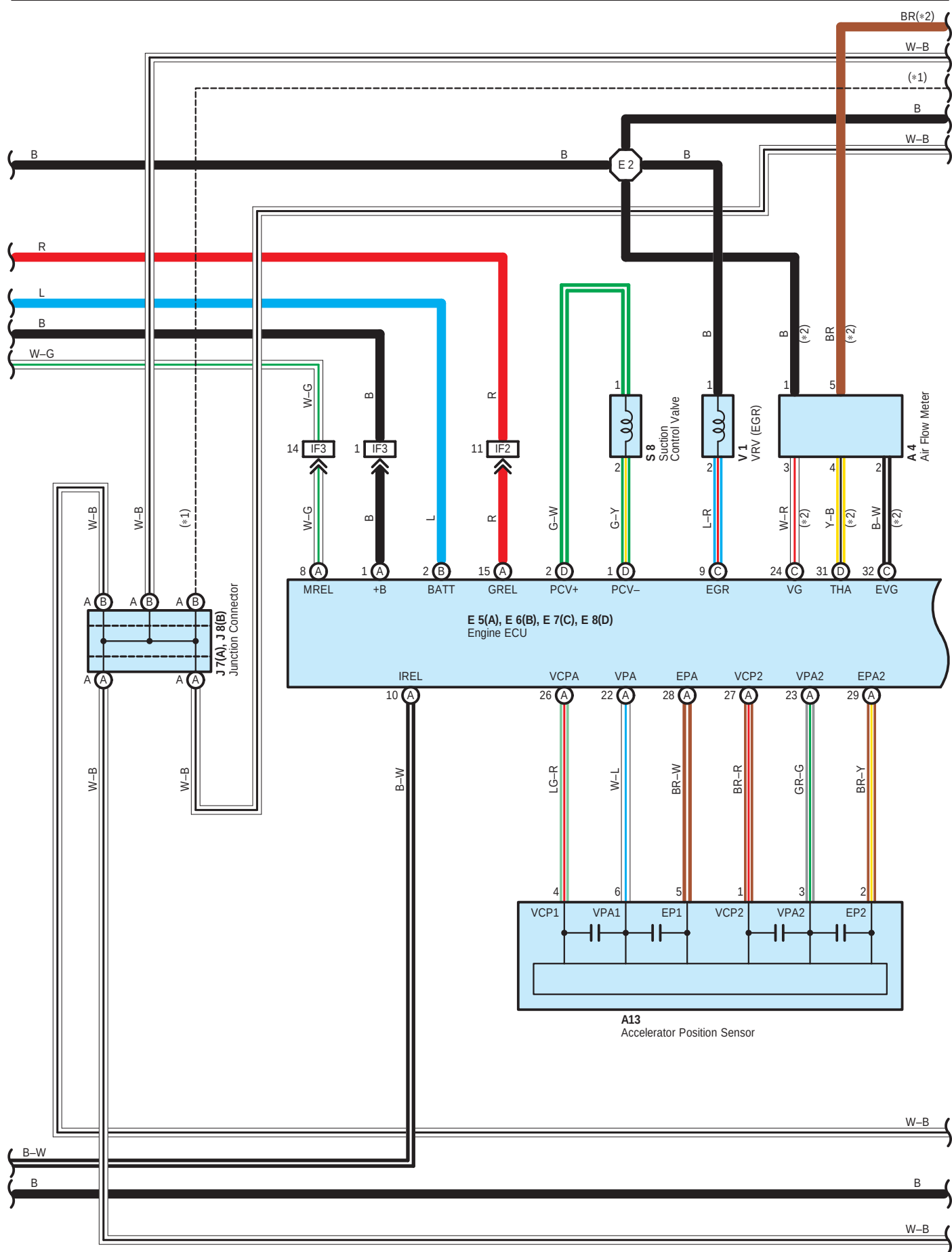
Code	See Page	Code	See Page	Code	See Page
C8	38	D7	40 (Double Cab)	J24	A 38
D2	40 (Double Cab)		42 (Extra Cab)	J25	B 38
	42 (Extra Cab)		44 (Single Cab)	J26	A 38
	44 (Single Cab)	D8	40 (Double Cab)	J27	B 38
D3	40 (Double Cab)	D9	40 (Double Cab)	J28	38
	42 (Extra Cab)		40 (Double Cab)	J29	38
	44 (Single Cab)		42 (Extra Cab)	P4	40 (Double Cab)
D4	40 (Double Cab)	F12	44 (Single Cab)		42 (Extra Cab)
D5	40 (Double Cab)				44 (Single Cab)
D6	40 (Double Cab)	J13	A 38		
	42 (Extra Cab)	J14	B 38	S3	37
	44 (Single Cab)	J22	A 38	T16	39
		J23	B 38	U1	39

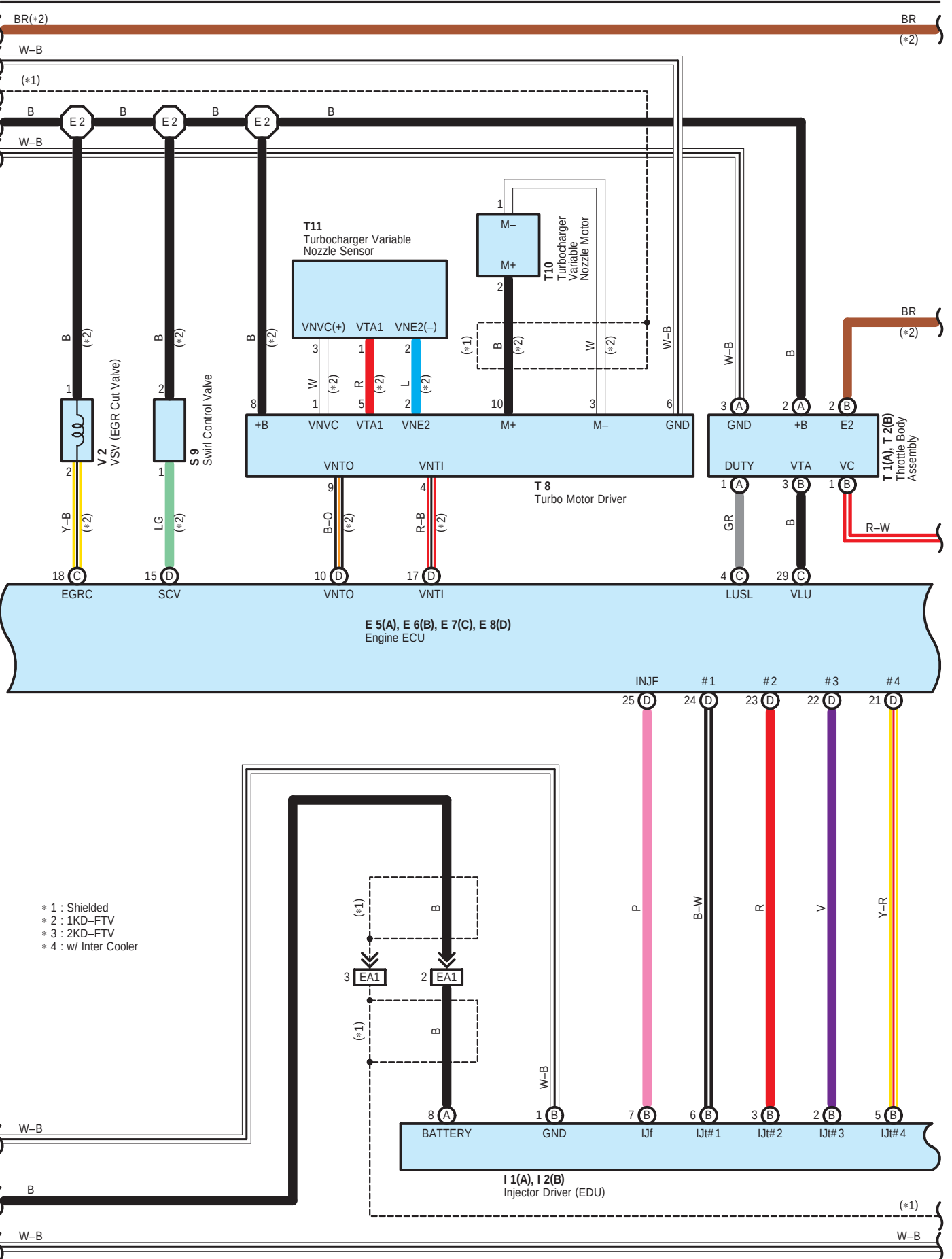
○ : Relay Blocks

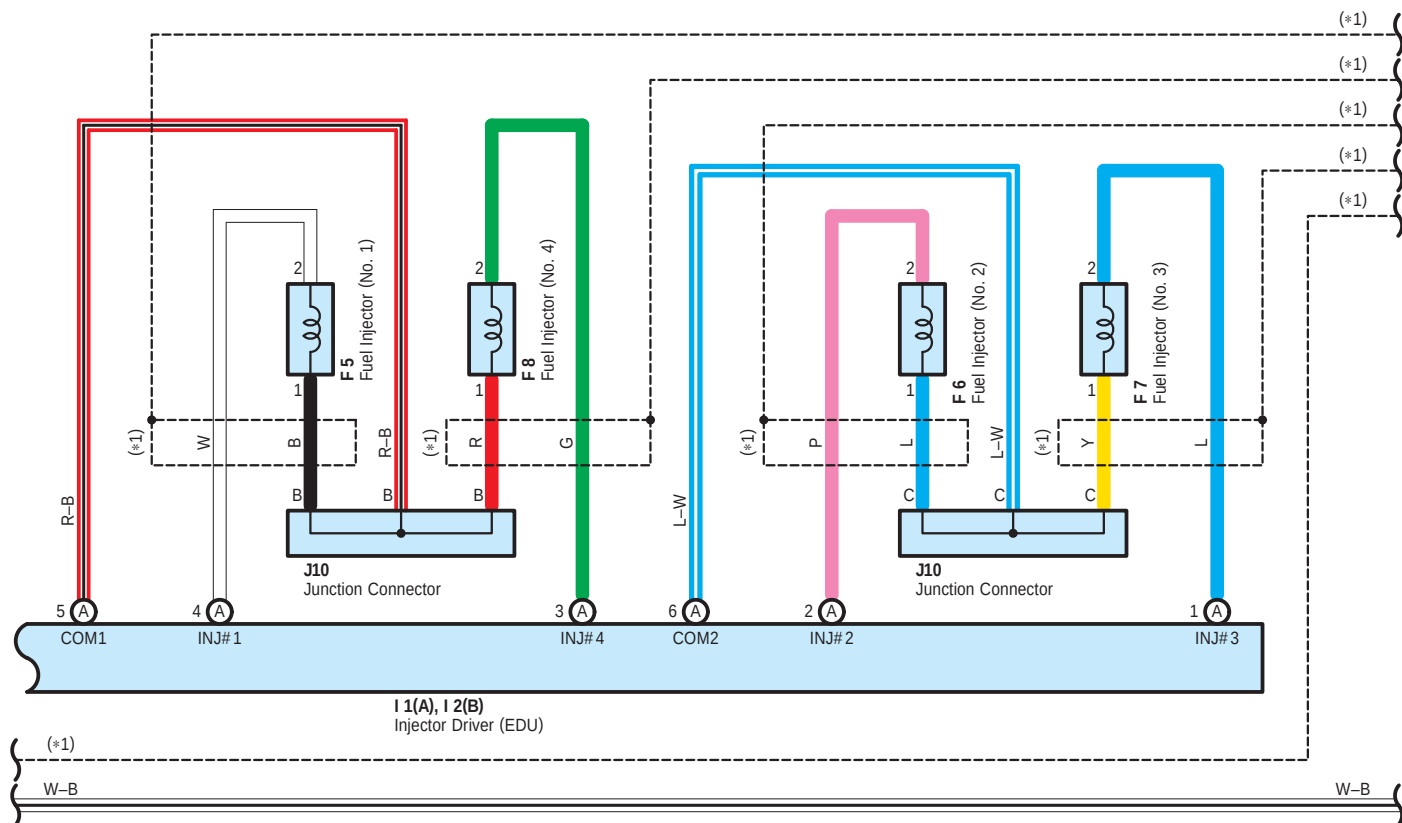
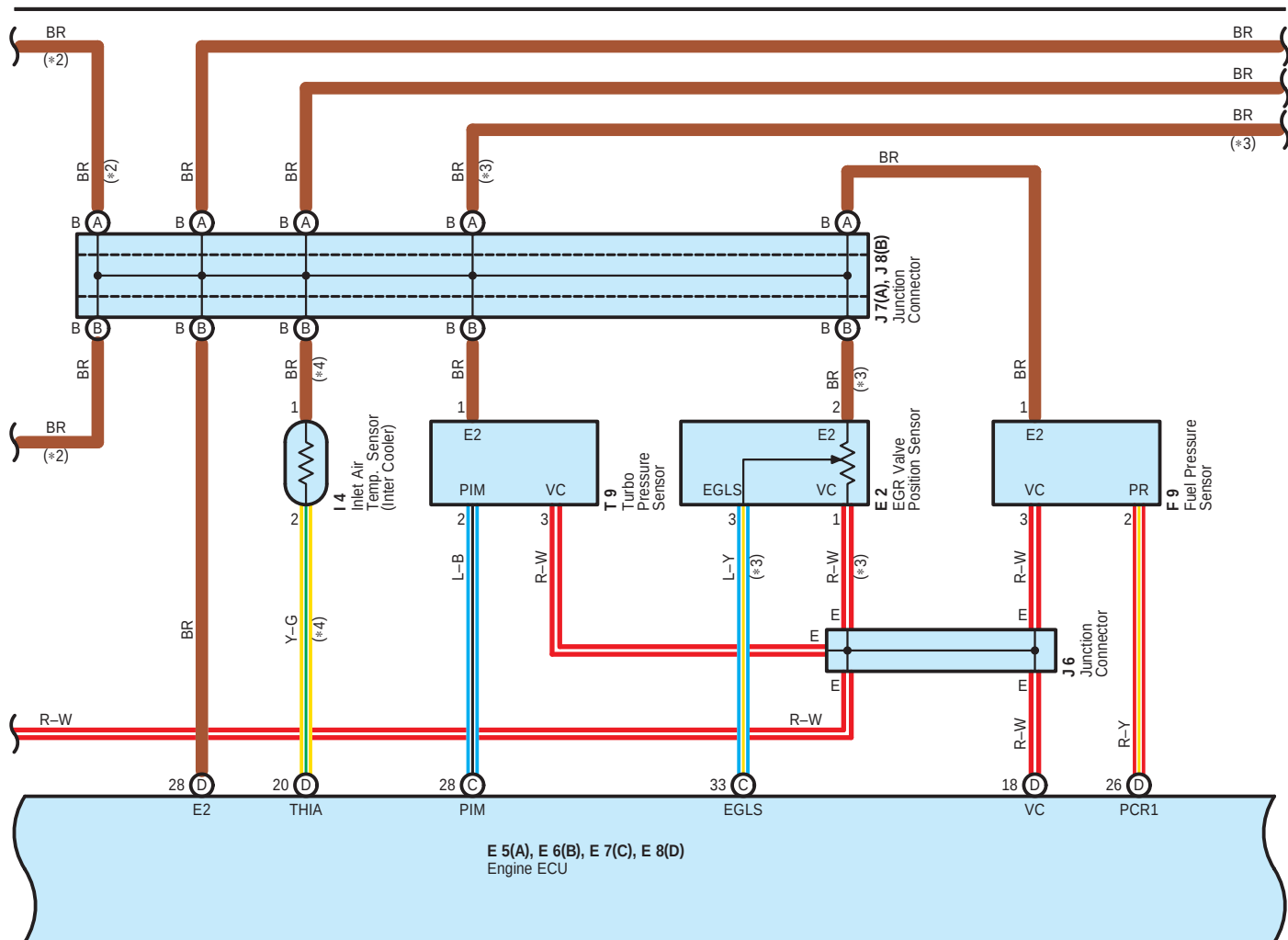
Code	See Page	Relay Blocks (Relay Block Location)
1	22	Engine Room R/B (Engine Compartment Left)
3	35	R/B No.3 (Cowl Side Panel LH)

From Power Source System (See Page 60)

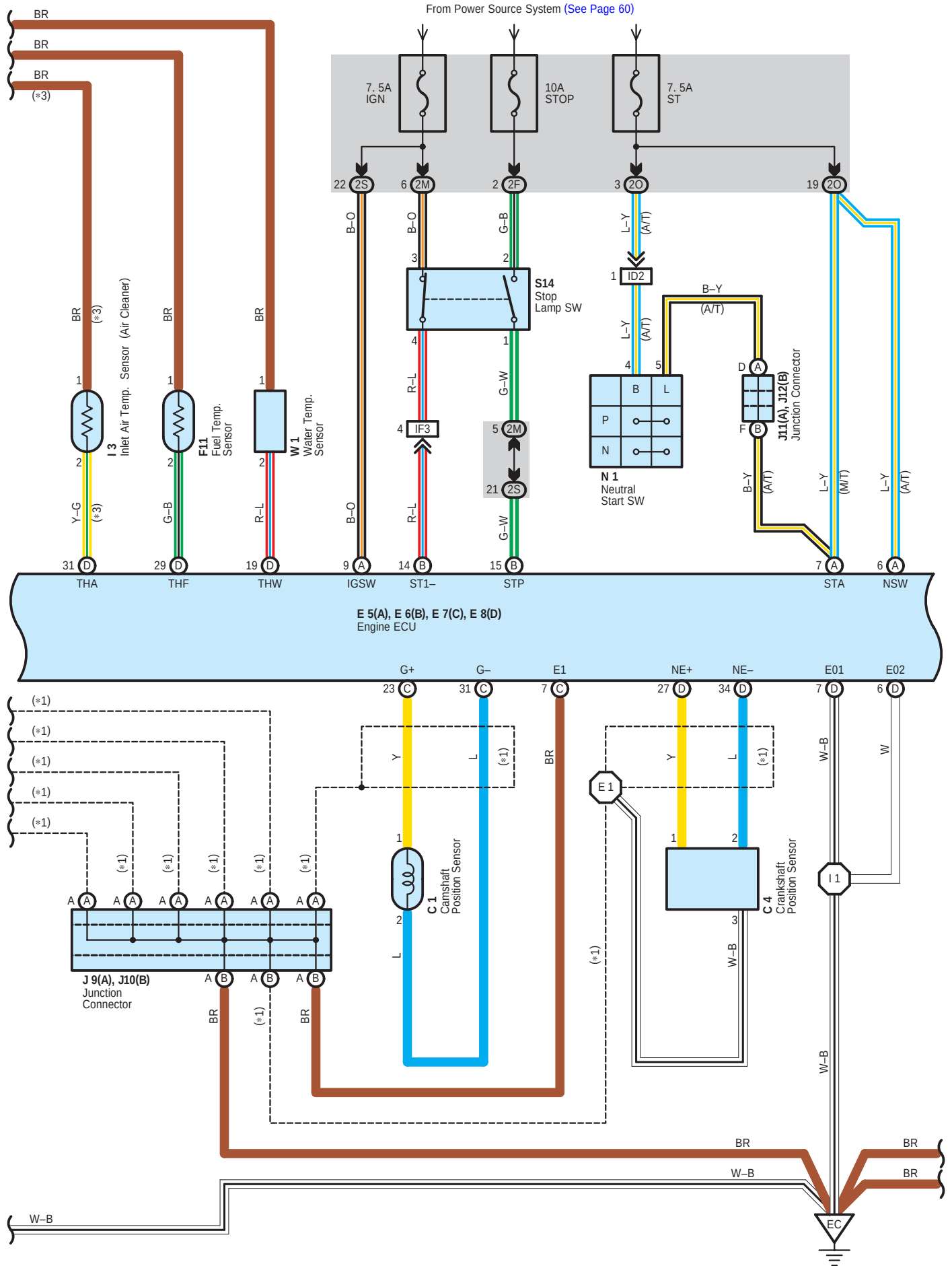


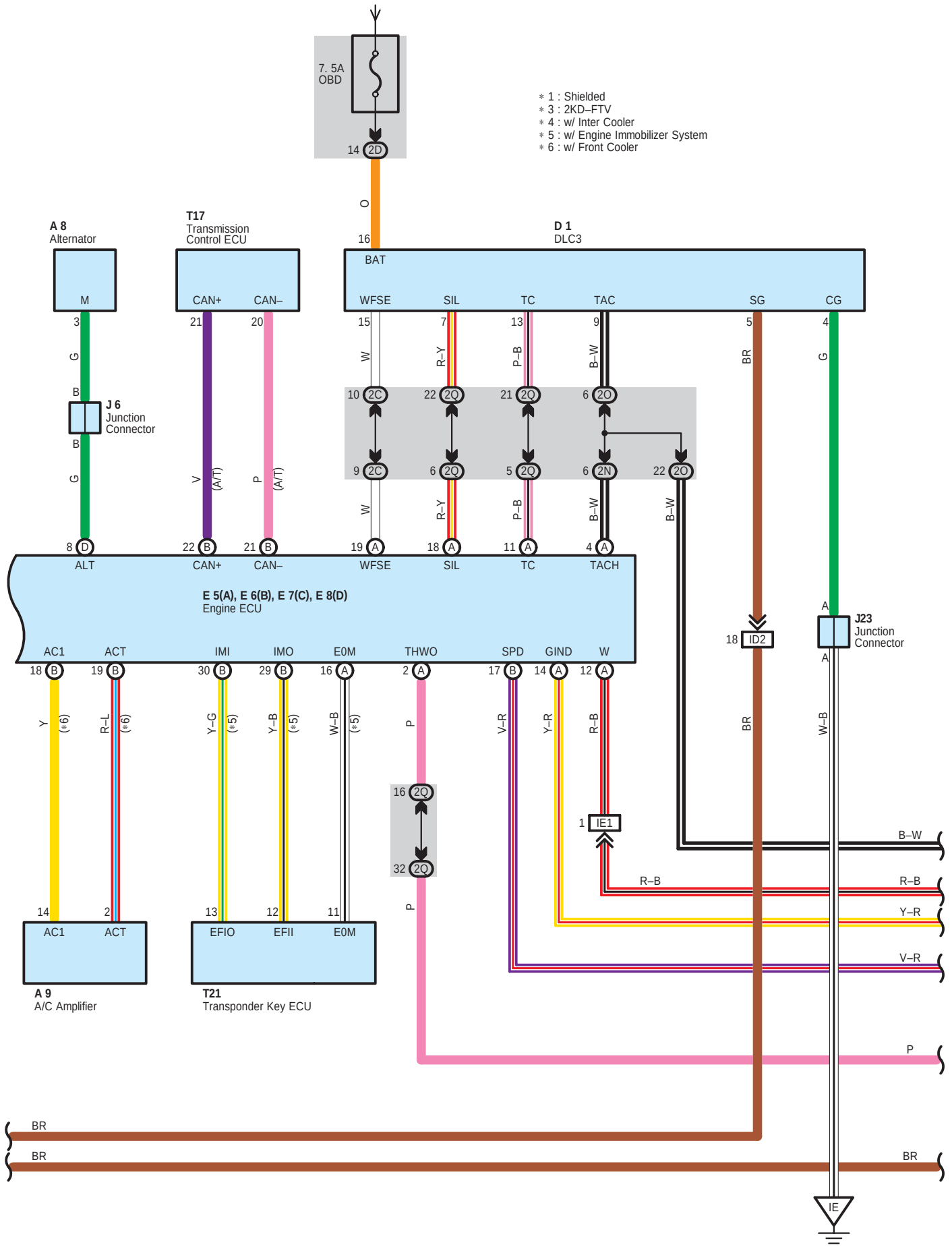




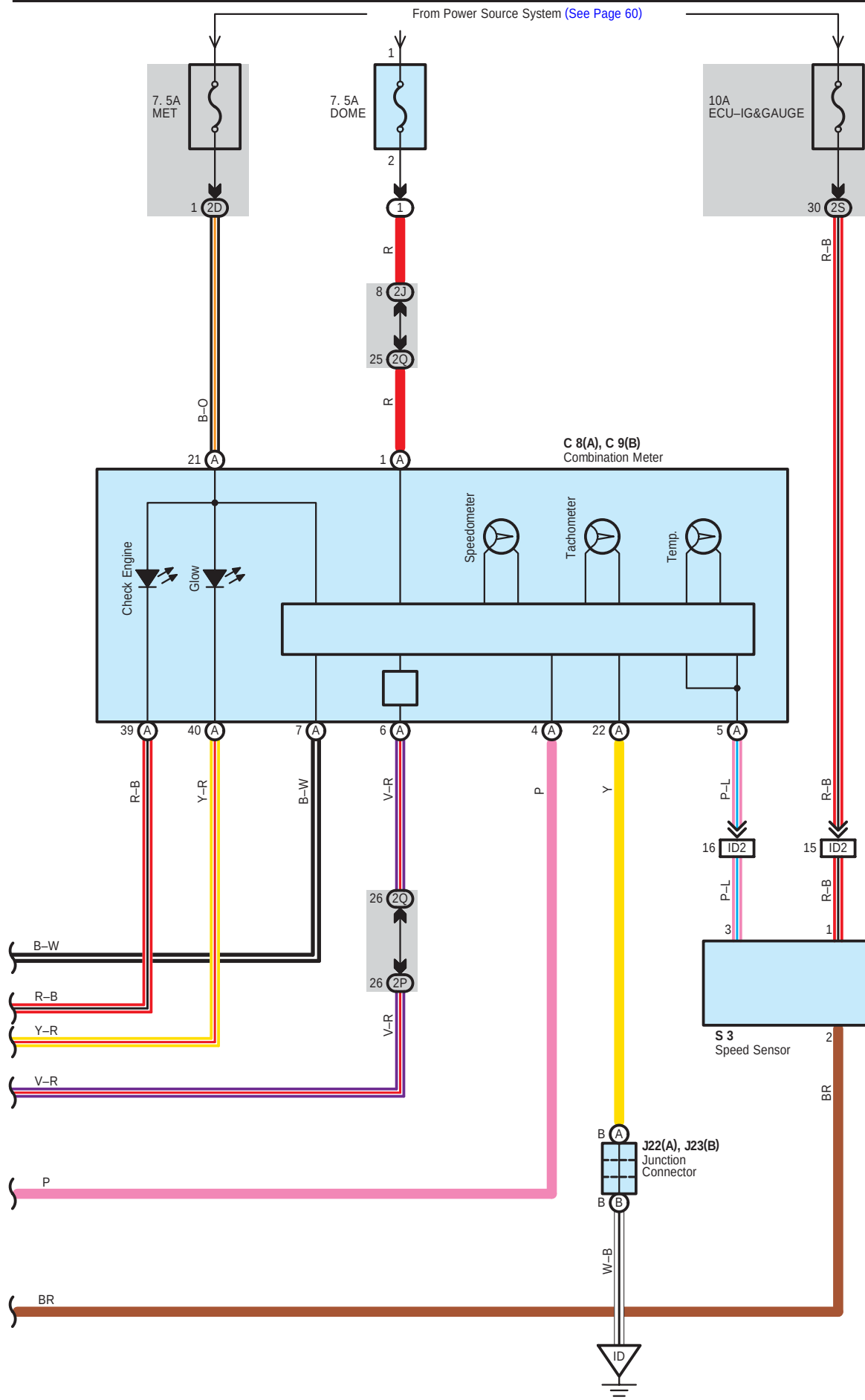


Engine Control





Engine Control



System Outline

This system utilizes an engine ECU and maintains overall control of the engine, transmission and so on. An outline of the engine control is explained here.

1. Input Signals

(1) Water temp. signal circuit

The water temp. sensor detects the engine coolant temp. and has a built-in thermistor with a resistance varies according to the engine coolant temp. Thus the engine coolant temp. is input in the form of a control signal to TERMINAL THW of the engine ECU.

(2) Intake air temp. signal circuit

The intake air temp. sensor is detects the intake air temp. of air cleaner, which is input as a control signal to TERMINAL THA of the engine ECU.

The intake air temp. sensor is detects the intake air temp. of inter cooler, which is input as a control signal to TERMINAL THIA of the engine ECU.

(3) RPM signal circuit

Camshaft position and crankshaft position are detected by the camshaft position sensor and crankshaft position sensor. Camshaft position is input as a control signal to TERMINAL G+ of the engine ECU, and engine RPM is input into TERMINAL NE+.

(4) Throttle signal circuit

The accelerator position sensor detects the accelerator pedal opening angle, which is input as a control signal to TERMINALS VPA and VPA2 of the engine ECU.

(5) Vehicle speed signal circuit

The speed sensor detects the vehicle speed and inputs a control signal to TERMINAL SPD of the engine ECU via the combination meter.

(6) Battery signal circuit

Voltage is constantly applied to TERMINAL BATT of the engine ECU. When the ignition SW is turned to on, voltage for engine ECU operation is applied via the MAIN relay to TERMINAL +B of the engine ECU.

(7) Started signal circuit

To confirm that the engine is cranking, the voltage applied to the starter motor during cranking is detected and is input as a control signal to TERMINAL STA of the engine ECU.

(8) Fuel temp. signal circuit

The fuel temp. sensor is detects the fuel temp., which is input as a control signal to TERMINAL THF of the engine ECU.

(9) Intake air vacuum pressure signal system

Intake air vacuum pressure is detected by the turbo pressure sensor and is input as a control signal to TERMINAL PIM of the engine ECU.

2. Control System

- * EGR control

The EGR control system detects the signals from each sensor, then the current is output to the TERMINAL EGR to control the VRV (EGR).

- * Common rail pressure control

The target rail pressure is calculated according to the engine status (Accelerator opening, engine speed) and environmental change detected by sensors. The fuel amount to be pressure-fed from the supply pump is calculated so as to match the indicated value of rail pressure sensor with the target value and the signal is sent to the intake amount adjusting valve of the supply pump in order to control the rail pressure.

- * Fuel injection timing control

The fuel injection timing is controlled by calculating the basic fuel injection timing based on the engine status (Accelerator opening, engine speed), making corrections according to environmental change detected by sensors, then sending a signal to the solenoid control valve of the injector via the injector driver (EDU).

- * Fuel injection volume control

The fuel injection volume is controlled by calculating the basic fuel injection volume based on the engine status (Accelerator opening, engine speed), making corrections according to environmental change detected by sensors and the inside pressure conditions in the rail, then sending a signal to the solenoid control valve of the injector via the injector driver (EDU).

- * Pilot injection control

The fuel injection volume and timing are controlled by calculating the pilot injection volume/timing based on the engine status (Accelerator opening, engine speed), making corrections according to environmental change detected by sensors, then sending a signal to the solenoid control valve of the injector via the injector driver (EDU).

3. Diagnosis System

With the diagnosis system, when there is a malfunctioning in the engine ECU signal system, the malfunction system is recorded in the memory. The malfunctioning system can be found by reading the display (Code) of the check engine warning light.

4. Fail-Safe System

When a malfunction occurs in any system, if there is a possibility of engine trouble being caused by continued control based on the signals from that system, the fail-safe system either controls the system by using data (Standard values) recorded in the engine ECU memory or else stops the engine.

Service Hints

E5 (A), E6 (B), E7 (C), E8 (D) Engine ECU

- BATT-E1 : Always 9.0–14.0 volts
- +B-E1 : 9.0–14.0 volts (Ignition SW at ON position)
- VC, VCPA, VCP2-E2 : 4.5–5.5 volts (Ignition SW at ON position)
- STA-E1 : 6.0 volts or more (Engine cranking)
- MREL-E1 : 9.0–14.0 volts (Ignition SW at ON position)
0–1.5 volts (More than 10 seconds passed away after the ignition SW was turned to OFF)
- IREL-E1 : 0–1.5 volts (Engine idling)
9.0–14.0 volts (Ignition SW at OFF position)
- GREL-E1 : 9.0–14.0 volts (Engine cranking)
0–1.5 volts (Engine idling (More than 600 seconds passed away after the engine started to run))
- NE+ -NE- : Pulse generation (Engine idling)
- G+ -G- : Pulse generation (Engine idling)
- SPD-E1 : Pulse generation (In driving at about 20km/h)
- VPA-EPA : 0.5–1.1 volts (Ignition SW on and accelerator pedal fully closed)
3.0–4.6 volts (Ignition SW on and accelerator pedal fully opened)
- VPA2-EPA2 : 0.9–2.3 volts (Ignition SW on and accelerator pedal fully closed)
3.4–5.0 volts (Ignition SW on and accelerator pedal fully opened)
- STP-E1 : 7.5–14.0 volts (Ignition SW on and brake pedal depressed)
0–1.5 volts (Ignition SW on and brake pedal released)
- ST1- -E1 : 0–1.5 volts (Ignition SW on and brake pedal depressed)
7.5–14.0 volts (Ignition SW on and brake pedal released)
- PIM-E2 : 1.2–1.6 volts (When a negative pressure of 40 kPa (300 mmHg) is applied)
1.3–1.9 volts (During air release)
3.2–3.8 volts (When a pressure of 69 kPa (0.7 kg/cm³) is applied)
- THW-E2 : 0.2–1.0 volts (During warm-up and coolant temp. 60°C, 140°F–120°C, 248°F)
- THA-E2 : 0.5–3.4 volts (During warm-up and intake temp. 0°C, 32°F–80°C, 176°F)
- THIA-E2 : 0.5–3.4 volts (During warm-up and intake air temp. 0°C, 32°F–80°C, 176°F)
- THF-E2 : 0.5–3.4 volts (During cooling with the ignition SW kept at the ON position)
- VG-E2 : 0.5–3.4 volts (Engine idling)
- PCR1-E2 : 1.3–1.8 volts (Engine idling)
- ALT-E1 : Pulse generation (Engine idling)
- PCV+ -PCV- : Pulse generation (Engine idling)
- #1, #2, #3, #4-E1 : Pulse generation (Engine idling)
- INJF-E1 : Pulse generation (Engine idling)
- EGR-E1 : 9.0–14.0 volts (Ignition SW at ON position)
Pulse generation (EGR at ON position (Staying at 1500r/min after warm-up))
- EGLS-E2 : 0.6–1.4 volts (Ignition SW at ON position)
- AC1-E1 : 0–1.5 volts (A/C SW on (Magnetic clutch ON))
7.5–14.0 volts (A/C SW off)
- ACT-E1 : 3.5–14.0 volts (Ignition SW at ON position)
0–1.5 volts (During air conditioner's operation cut
(For 5 seconds during driving at 30km/h or slower at full throttle))
- W-E1 : 0–3.0 volts (Check engine warning light lights up and ignition SW at ON position)
9.0–14.0 volts (Engine idling and except check engine warning light lights up)
- GIND-E1 : 0–3.0 volts (Glow indicator light lights up and ignition SW at ON position)
9.0–14.0 volts (Engine idling)
- TACH-E1 : Pulse generation (Engine idling)
- THWO-E1 : Pulse generation (Engine idling)
- TC-E1 : 9.0–14.0 volts (Ignition SW at ON position)
0–3.0 volts (The DLC3 is shorted between the TERMINALS TC and CG)
- CAN+ -CAN- : 54.0–69.0 Ω (More than 10 seconds passed away after the ignition SW was turned to OFF)
- VNTO-E1 : Pulse generation (Engine idling)
- VNTI-E1 : Pulse generation (Engine idling)
- NSW-E1 : 0–0.3 volts (Shift position in P or N position)
9–14.0 volts (Other shift position in P or N position)

Engine Control

: Parts Location

Code	See Page	Code	See Page	Code	See Page
A4	36	F9	36	J23	38
A8	36	F11	36	N1	37
A9	38	G1	36	S3	37
A13	38	I1	36	S8	37
C1	36	I2	36	S9	37
C4	36	I3	36	S14	39
C8	38	I4	36	T1	37
C9	38	J1	37	T2	37
D1	38	J4	38	T8	39
E2	36	J5	38	T9	37
E5	38	J6	38	T10	37
E6	38	J7	38	T11	37
E7	38	J8	38	T17	39
E8	38	J9	38	T21	39
F5	36	J10	38	V1	37
F6	36	J11	38	V2	37
F7	36	J12	38	W1	37
F8	36	J22	38		

: Relay Blocks

Code	See Page	Relay Blocks (Relay Block Location)
1	22	Engine Room R/B (Engine Compartment Left)

: Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
1B	23	Engine Room Main Wire and Engine Room J/B (Engine Compartment Left)
1G		
1H	24	
1J		
2C	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2D		
2F	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2J		
2M		
2N	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2O		
2P		
2Q		
2S		

: Connector Joining Wire Harness and Wire Harness

Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
EA1	46	Engine Wire and Engine Room Main Wire (Inside of Engine Room R/B and Engine Room J/B)
EA2		
ID2	48	Engine Wire and Instrument Panel Wire (Behind the Glove Box)
IE1	48	Instrument Panel Wire and Instrument Panel Wire (Instrument Panel Reinforcement LH)
IF2	50	Engine Room Main Wire and Instrument Panel Wire (Behind the Driver Side J/B)
IF3		

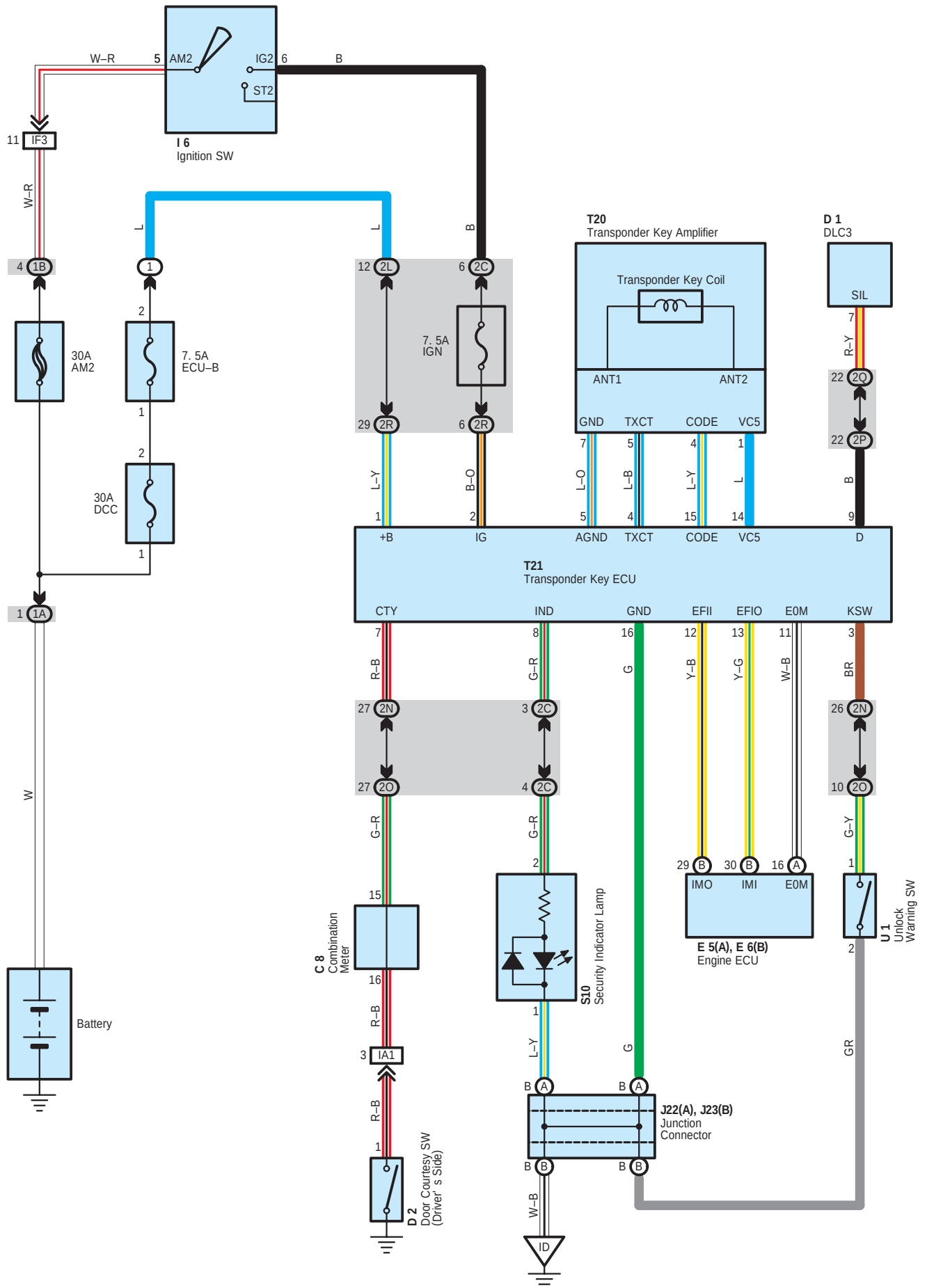
**: Ground Points**

Code	See Page	Ground Points Location
EB	46	Left Side of the Fender Apron
EC	46	Rear Side of the Cylinder Block
ID	48	Instrument Panel Reinforcement Center
IE	48	Instrument Panel Reinforcement RH

**: Splice Points**

Code	See Page	Wire Harness with Splice Points	Code	See Page	Wire Harness with Splice Points
E1	46	Engine Wire	I1	50	Engine Wire
E2					

Engine Immobilizer System



Service Hints

T21 Transponder Key ECU

- 1—Ground : Always approx. 12 volts
- 16—Ground : Always continuity
- 2—Ground : Approx. 12 volts with the ignition SW at ON position

○ : Parts Location

Code	See Page	Code		See Page	Code	See Page
C8	38	E5	A	38	S10	39
D1	38	E6	B	38	T20	39
D2	40 (Double Cab)	I6		38	T21	39
	42 (Extra Cab)	J22	A	38	U1	39
	44 (Single Cab)	J23	B	38		

○ : Relay Blocks

Code	See Page	Relay Blocks (Relay Block Location)
1	22	Engine Room R/B (Engine Compartment Left)

○ : Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
1A	23	Engine Room Main Wire and Engine Room J/B (Engine Compartment Left)
1B		
2C	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2L	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2N	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2O		
2P		
2Q		
2R		

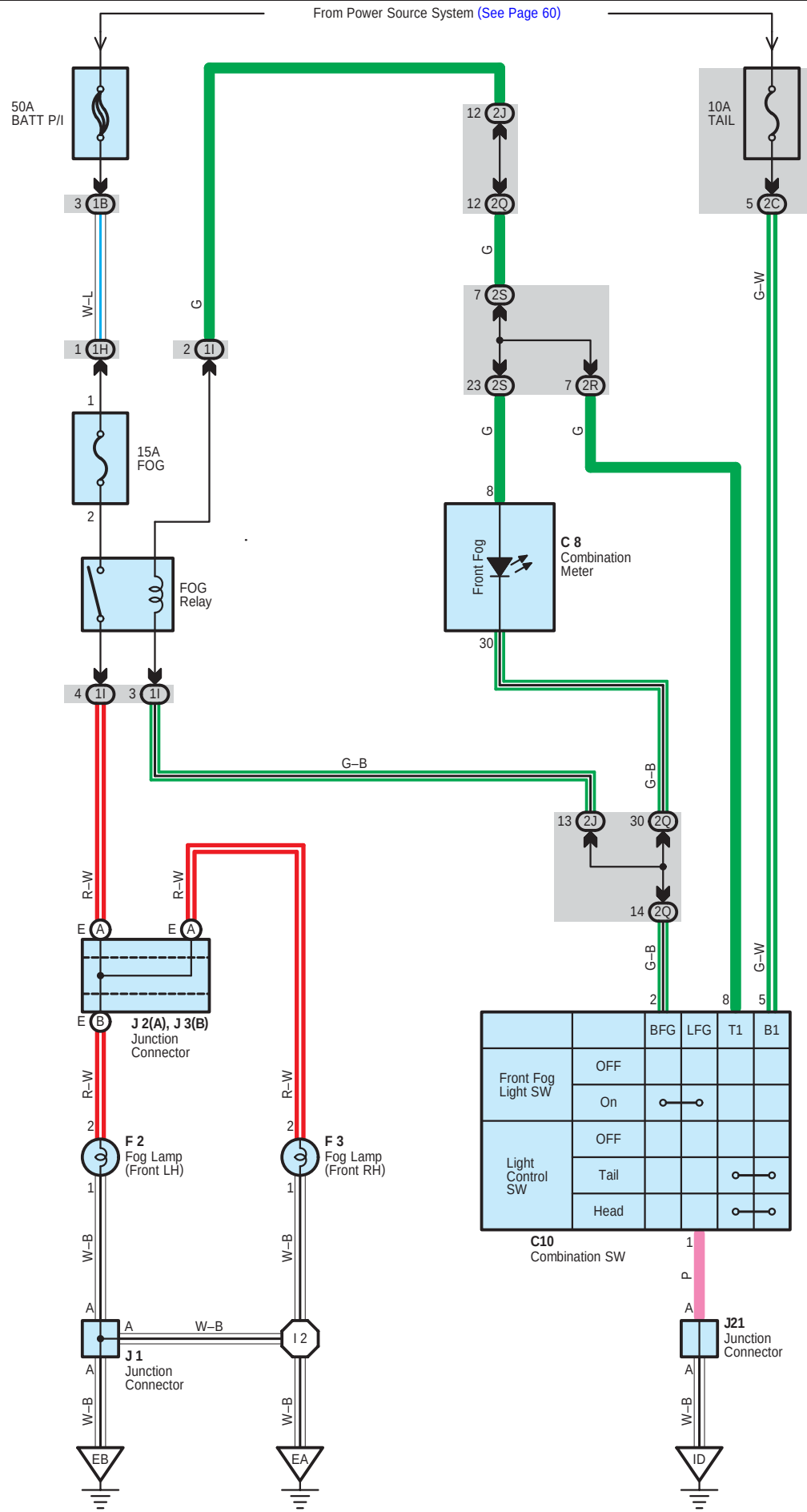
□ : Connector Joining Wire Harness and Wire Harness

Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
IA1	48	Instrument Panel Wire and Floor Wire (Left Kick Panel)
IF3	50	Engine Room Main Wire and Instrument Panel Wire (Behind the Driver Side J/B)

▽ : Ground Points

Code	See Page	Ground Points Location
ID	48	Instrument Panel Reinforcement Center

Front Fog Light



Service Hints

F2, F3 Fog Lamp (Front LH), (Front RH)

2-Ground : Approx. 12 volts with the light control SW at TAIL or HEAD position, dimmer SW at LOW position and front fog light SW at ON position

○ : Parts Location

Code	See Page	Code	See Page	Code	See Page
C8	38	F3	36	J3 B	38
C10	38	J1	37	J21	38
F2	36	J2 A	38		

○ : Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
1B	23	Engine Room Main Wire and Engine Room J/B (Engine Compartment Left)
1H	24	
1I		
2C	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2J	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2Q	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2R		
2S		

▽ : Ground Points

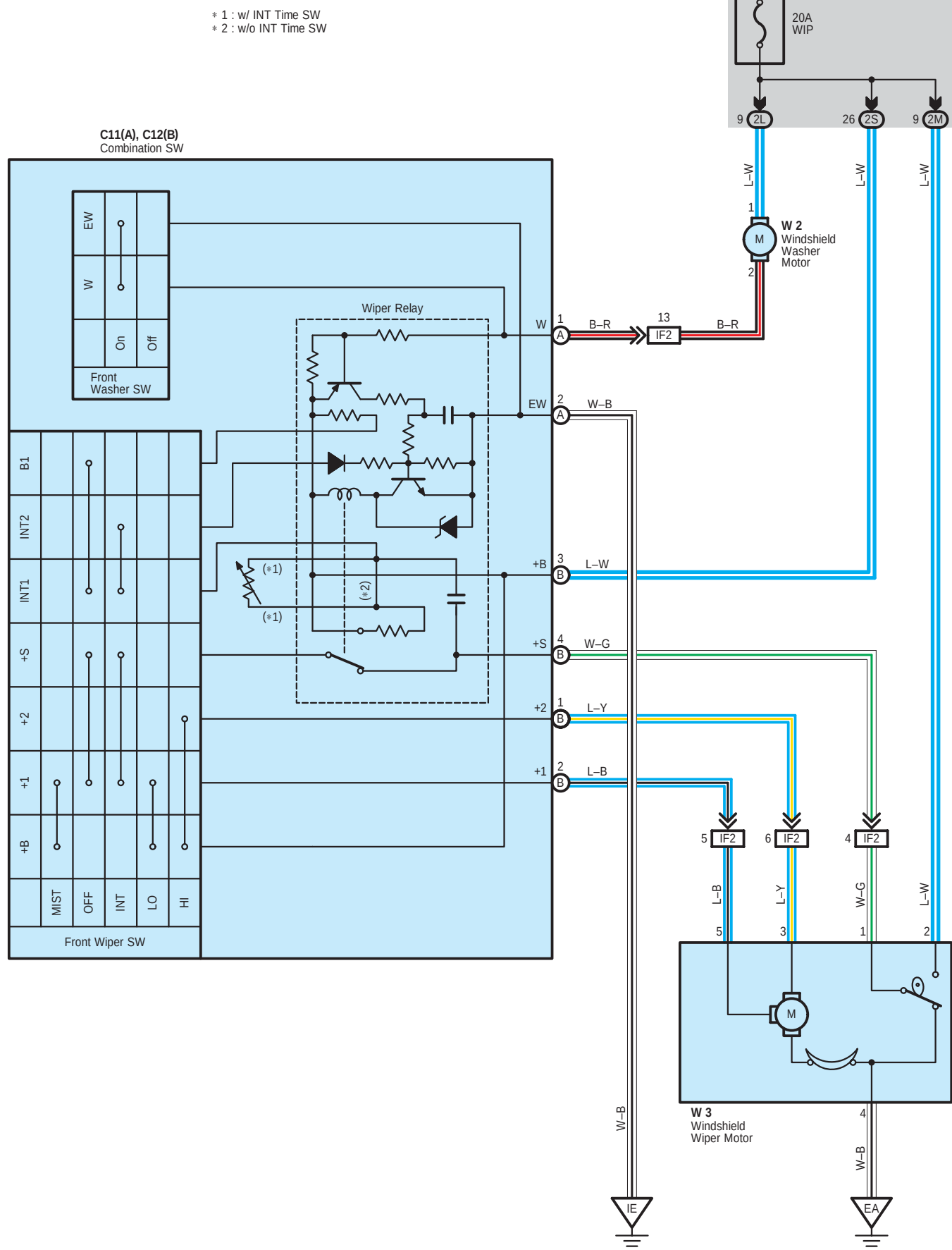
Code	See Page	Ground Points Location
EA	46	Right Side of the Fender Apron
EB	46	Left Side of the Fender Apron
ID	48	Instrument Panel Reinforcement Center

○ : Splice Points

Code	See Page	Wire Harness with Splice Points	Code	See Page	Wire Harness with Splice Points
I2	50	Engine Wire			

Front Wiper and Washer

From Power Source System (See Page 60)



System Outline

With the ignition SW turned on, the current flows to TERMINAL (B) 3 of the front wiper and washer SW, TERMINAL 1 of the windshield washer motor and TERMINAL 2 of the windshield wiper motor through the WIP fuse.

1. Low Speed Position

With the front wiper SW turned to LO position, the current flows from TERMINAL (B) 3 of the front wiper and washer SW to TERMINAL (B) 2 to TERMINAL 5 of the windshield wiper motor to TERMINAL 4 to GROUND and causes the windshield wiper motor to run at low speed.

2. High Speed Position

With the front wiper SW turned to HI position, the current flows from TERMINAL (B) 3 of the front wiper and washer SW to TERMINAL (B) 1 to TERMINAL 3 of the windshield wiper motor to TERMINAL 4 to GROUND and causes the windshield wiper motor to run at high speed.

3. INT Position

With the front wiper SW turned to INT position, the wiper relay operates and current flows from TERMINAL (B) 3 of the front wiper and washer SW to TERMINAL (A) 2 to GROUND. This activates the intermittent circuit and the current flows from TERMINAL (B) 3 of the front wiper and washer SW to TERMINAL (B) 2 to TERMINAL 5 of the windshield wiper motor to TERMINAL 4 to GROUND and the wiper operates.

The intermittent operation is controlled by the charge/discharge function of the condenser installed in the relay, and the intermittent time is controlled by a time control SW to change the charging time of the condenser (w/ INT time SW).

Intermittent operation is controlled by a condenser charge and discharge function in the relay (w/o INT time SW).

4. Mist Position

With the front wiper SW turned to MIST position, the current flows from TERMINAL (B) 3 of the front wiper and washer SW to TERMINAL (B) 2 to TERMINAL 5 of the windshield wiper motor to TERMINAL 4 to GROUND and causes the windshield wiper motor to run at low speed.

5. Washer Interlocking Operation

With the washer SW pulled to ON position, the current flows from the WIP fuse to TERMINAL 1 of the windshield washer motor to TERMINAL 2 to TERMINAL (A) 1 of the front wiper and washer SW to TERMINAL (A) 2 to GROUND and causes the windshield washer motor to run and the window washer to spray. Simultaneously, current flows from the WIP fuse to TERMINAL (B) 3 of the front wiper and washer SW to TERMINAL (B) 2 to TERMINAL 5 of the windshield wiper motor to TERMINAL 4 to GROUND, causing the wiper to function.

Service Hints

C11 (A), C12 (B) Combination SW

(A) 2–Ground : Always continuity

(B) 3–Ground : Approx. 12 volts with the ignition SW at ON position

(B) 2–Ground : Approx. 12 volts with the ignition SW on and the front wiper SW at LO or MIST position
Approx. 12 volts intermittently with the ignition SW on and the front wiper SW at INT position

(B) 4–Ground : Approx. 12 volts with the ignition SW on and unless the windshield wiper motor at STOP position

(B) 1–Ground : Approx. 12 volts with the ignition SW on and the front wiper SW at HI position

W3 Windshield Wiper Motor

2–1 : Closed unless the windshield wiper motor at STOP position

○ : Parts Location

Code	See Page	Code	See Page	Code	See Page
C11	A	38	W2	37	
C12	B	38	W3	37	

□ : Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
2L	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2M		
2S	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)

□ : Connector Joining Wire Harness and Wire Harness

Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
IF2	50	Engine Room Main Wire and Instrument Panel Wire (Behind the Driver Side J/B)

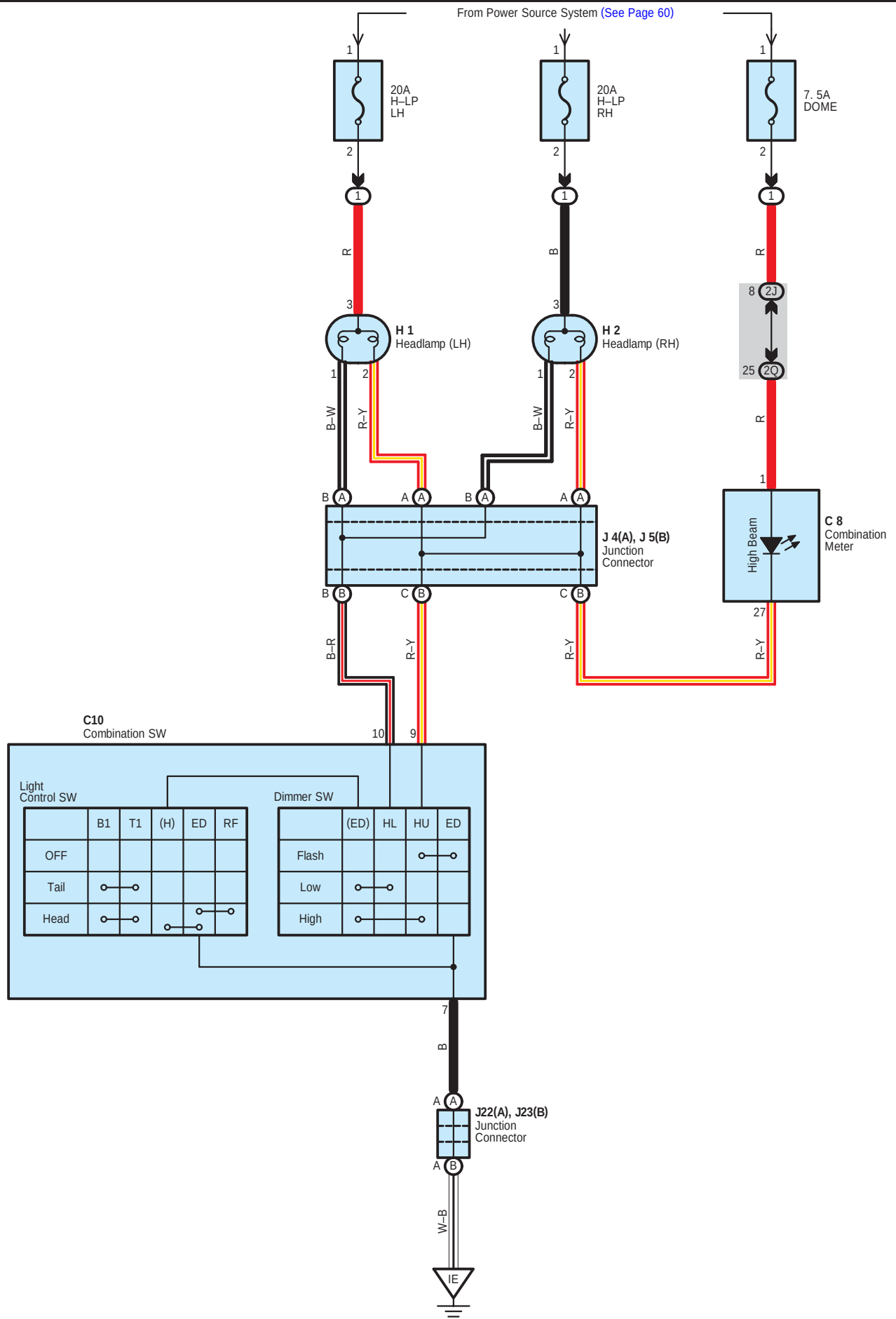
Front Wiper and Washer



: Ground Points

Code	See Page	Ground Points Location
EA	46	Right Side of the Fender Apron
IE	48	Instrument Panel Reinforcement RH

Headlight



Service Hints

C10 Combination SW

- 10-7 : Continuity with the dimmer SW at LOW position and the light control SW at HEAD position
 9-7 : Continuity with the dimmer SW at FLASH position or the dimmer SW at HIGH position and the light control SW at HEAD position

○ : Parts Location

Code	See Page	Code	See Page	Code	See Page
C8	38	H2	36	J22 A	38
C10	38	J4 A	38	J23 B	38
H1	36	J5 B	38		

○ : Relay Blocks

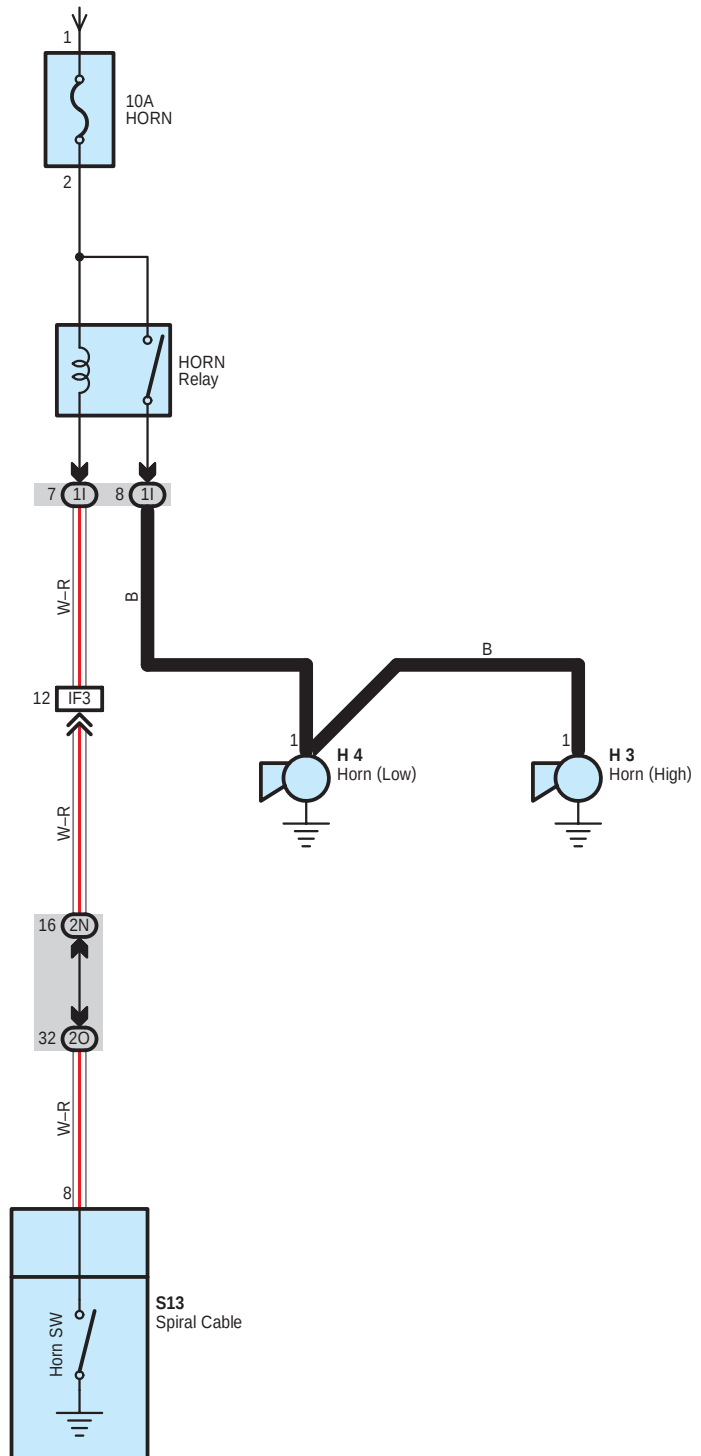
Code	See Page	Relay Blocks (Relay Block Location)
1	22	Engine Room R/B (Engine Compartment Left)

○ : Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
2J	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2Q	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)

▽ : Ground Points

Code	See Page	Ground Points Location
IE	48	Instrument Panel Reinforcement RH



Service Hints

H3, H4 Horn (High), (Low)

1-Ground : Approx. 12 volts with the horn SW on

: Parts Location

Code	See Page	Code	See Page	Code	See Page
H3	36	H4	36	S13	39

: Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
1I	24	Engine Room Main Wire and Engine Room J/B (Engine Compartment Left)
2N	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2O		

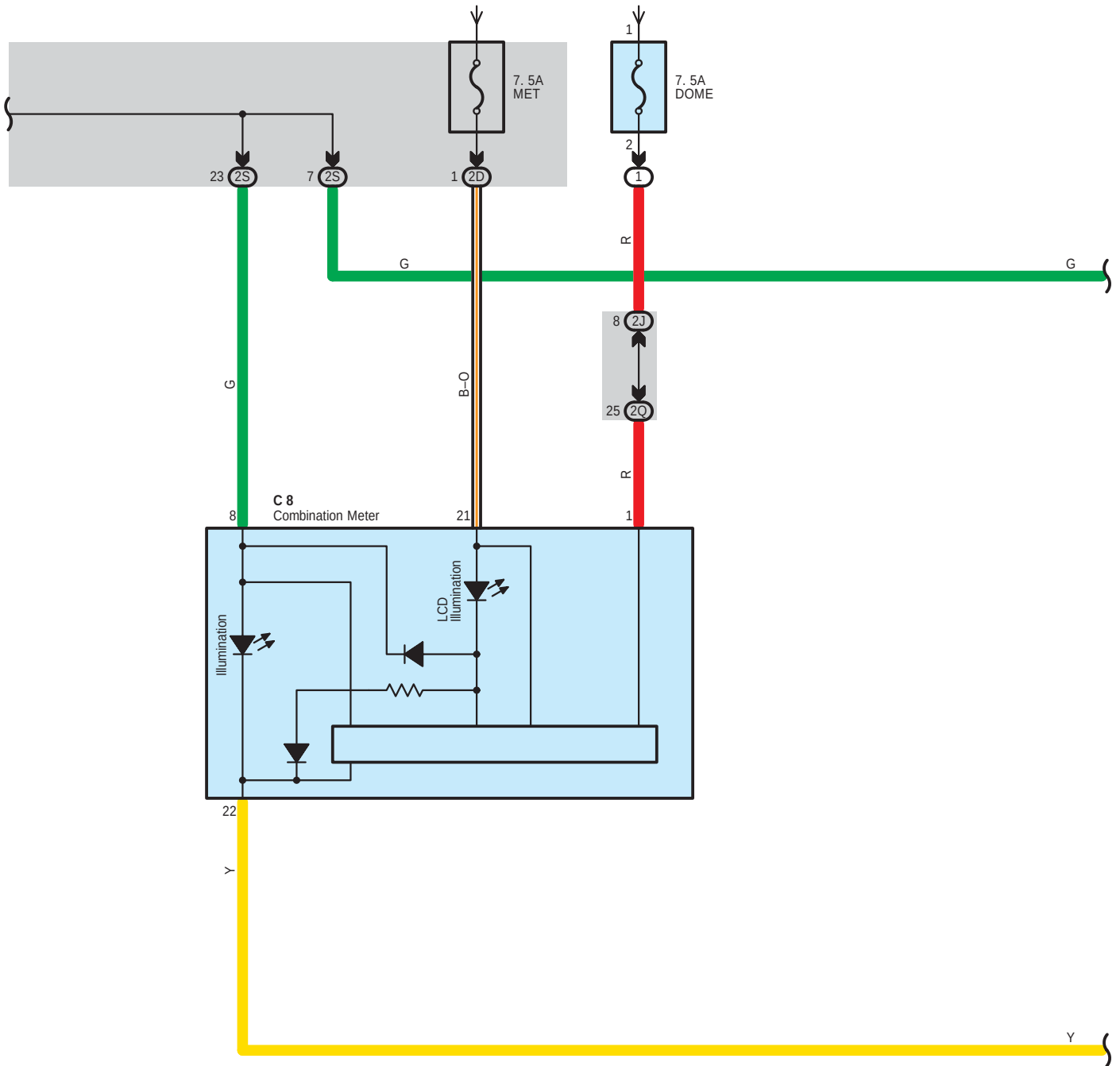
: Connector Joining Wire Harness and Wire Harness

Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
IF3	50	Engine Room Main Wire and Instrument Panel Wire (Behind the Driver Side J/B)

From Power Source System (See Page 60)



From Power Source System (See Page 60)





Service Hints

C10 Combination SW

5-8 : Continuity with the light control SW at HEAD or TAIL position

○ : Parts Location

Code	See Page	Code	See Page	Code	See Page
A12	38	J15	38	R1	39
A16	38	J21	38	R3	39
B6	38	J22	A 38	R4	41 (Double Cab)
C2	36	J23	B 38		43 (Extra Cab)
C3	36	L1	40 (Double Cab)		45 (Single Cab)
C6	38		42 (Extra Cab)	R5	41 (Double Cab)
C8	38		44 (Single Cab)		43 (Extra Cab)
C10	38	L2	40 (Double Cab)		45 (Single Cab)
H5	38		42 (Extra Cab)		
J1	37		44 (Single Cab)		

○ : Relay Blocks

Code	See Page	Relay Blocks (Relay Block Location)
1	22	Engine Room R/B (Engine Compartment Left)

○ : Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
2C	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2D		
2J	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2K	28	Floor Wire and Driver Side J/B (Instrument Panel Brace RH)
2L	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2M		
2P	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2Q		
2R		
2S		

□ : Connector Joining Wire Harness and Wire Harness

Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
BA1	52 (Double Cab)	Frame Wire and Floor Wire (Under the Front Seat LH)
	54 (Extra Cab)	
	56 (Single Cab)	

▽ : Ground Points

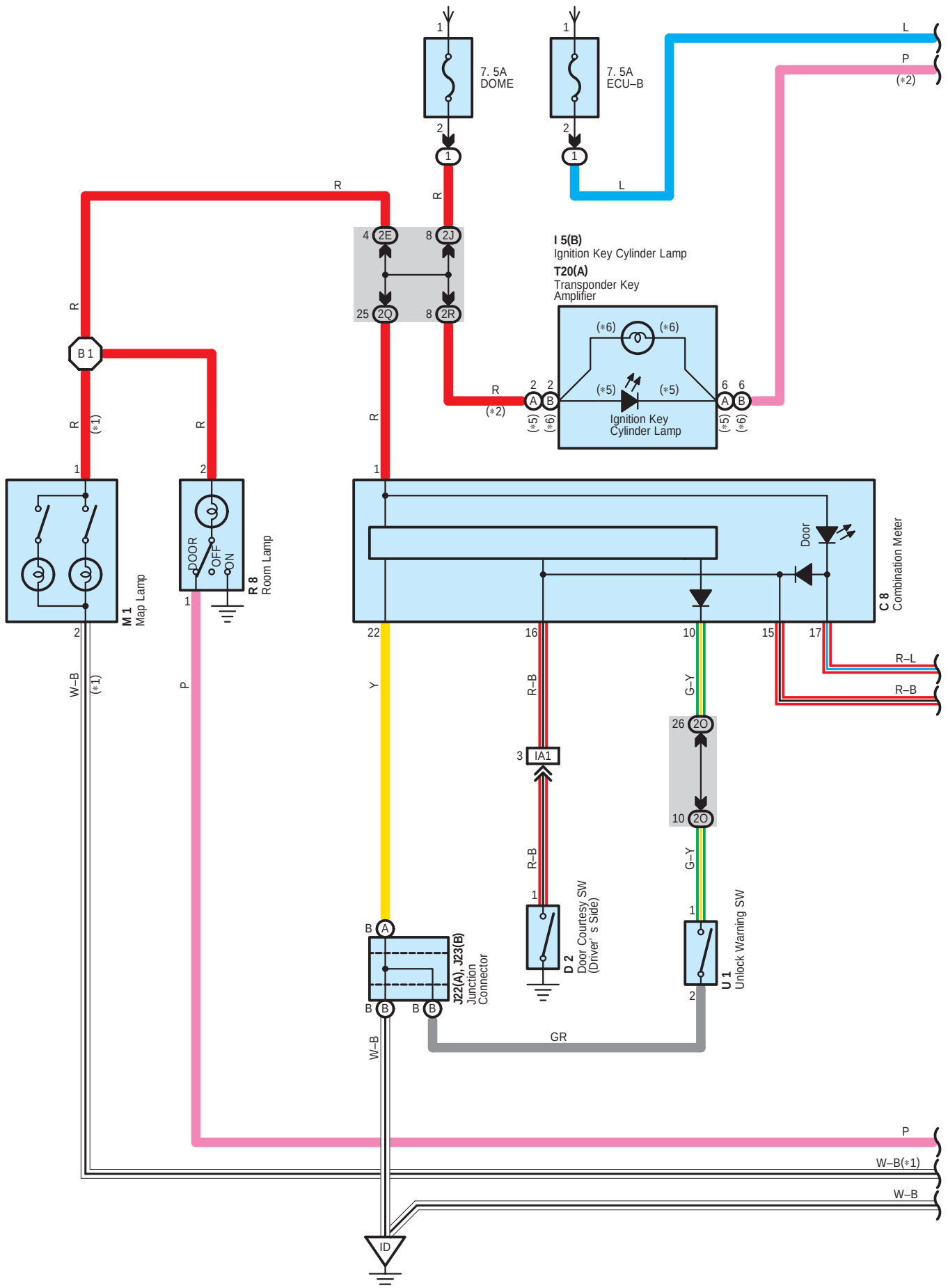
Code	See Page	Ground Points Location
EA	46	Right Side of the Fender Apron
EB	46	Left Side of the Fender Apron
ID	48	Instrument Panel Reinforcement Center
IE	48	Instrument Panel Reinforcement RH
IG	48	Cowl Side Panel LH

○ : Splice Points

Code	See Page	Wire Harness with Splice Points	Code	See Page	Wire Harness with Splice Points
I2	50	Engine Room Main Wire	B3	54 (Extra Cab)	Frame Wire
B3	52 (Double Cab)	Frame Wire		56 (Single Cab)	

Interior Light

From Power Source System (See Page 60)



Interior Light

Service Hints

D2, D3, D4, D5 Door Courtesy SW (Driver's Side), (Front Passenger's Side), (Rear LH), (Rear RH)
 1–Ground : Continuity with each of the door open

○ : Parts Location

Code	See Page	Code	See Page	Code	See Page
C8	38	D4	40 (Double Cab)	M1	42 (Extra Cab)
D2	40 (Double Cab)	D5	40 (Double Cab)	R8	41 (Double Cab)
	42 (Extra Cab)	I5 B	38		43 (Extra Cab)
	44 (Single Cab)	J22 A	38		45 (Single Cab)
D3	40 (Double Cab)	J23 B	38	T20 A	39
	42 (Extra Cab)	J29	38	U1	39
	44 (Single Cab)	M1	40 (Double Cab)		

○ : Relay Blocks

Code	See Page	Relay Blocks (Relay Block Location)
1	22	Engine Room R/B (Engine Compartment Left)

○ : Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
2D	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2E		
2J	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2L		
2N	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2O		
2Q		
2R		

□ : Connector Joining Wire Harness and Wire Harness

Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
IA1	48	Instrument Panel Wire and Floor Wire (Left Kick Panel)

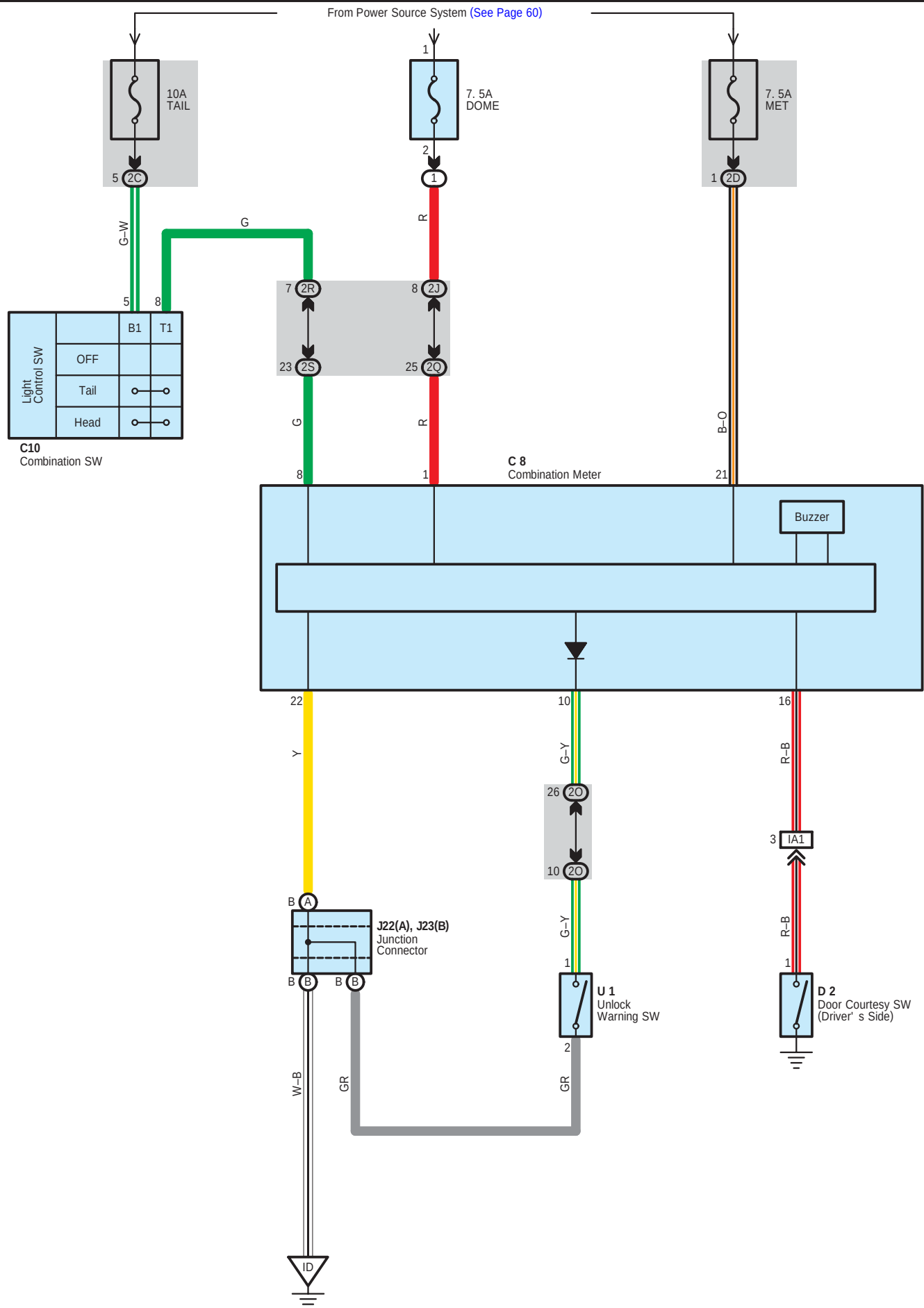
▽ : Ground Points

Code	See Page	Ground Points Location
ID	48	Instrument Panel Reinforcement Center
IE	48	Instrument Panel Reinforcement LH

○ : Splice Points

Code	See Page	Wire Harness with Splice Points	Code	See Page	Wire Harness with Splice Points
B1	52 (Double Cab)	Roof Wire	B1	54 (Extra Cab)	Roof Wire

Key Reminder and Light Reminder



System Outline

The current is applied at all times to TERMINAL 1 of the combination meter through the DOME fuse.

When the ignition SW is turned to ON position, the current flows to TERMINAL 21 of the combination meter through the MET fuse. When the light control SW is turned to TAIL or HEAD position, current is applied to TERMINAL 8 of the combination meter through the TAIL fuse.

1. Key Reminder System

When the driver door is opened with the ignition SW off and ignition key remaining in the key cylinder (Unlock warning SW on), a signal is input from the unlock warning SW to TERMINAL 10 of the combination meter, and from the door courtesy SW (Driver's side) to TERMINAL 16 of the combination meter. As a result, the buzzer in the combination meter goes on and warns the driver that the key is remaining in the key cylinder.

2. Light Reminder System

When the light control SW is in TAIL or HEAD position, the ignition SW turned to OFF from ON position, ignition key is not in the key cylinder and the driver's door opened (Door courtesy SW on), the current flows to TERMINAL 21 of the combination meter stops. As a result, the combination meter is activated and current flows from TERMINAL 1 of the combination meter, the buzzer in the combination meter goes on to remind the light is lighting up.

Service Hints

C8 Combination Meter

21-Ground : Approx. 12 volts with the ignition SW at ON position

8-Ground : Approx. 12 volts with the light control SW at TAIL or HEAD position

16-Ground : Continuity with the driver's door open

22-Ground : Always continuity

D2 Door Courtesy SW (Driver's Side)

1-Ground : Closed with the driver's door open

U1 Unlock Warning SW

1-2 : Closed with the ignition key in cylinder

○ : Parts Location

Code	See Page	Code	See Page	Code	See Page
C8	38	D2	42 (Extra Cab)	J23 B	38
C10	38		44 (Single Cab)	U1	39
D2	40 (Double Cab)	J22 A	38		

○ : Relay Blocks

Code	See Page	Relay Blocks (Relay Block Location)
1	22	Engine Room R/B (Engine Compartment Left)

○ : Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
2C	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2D		
2J	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2O	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2Q		
2R		
2S		

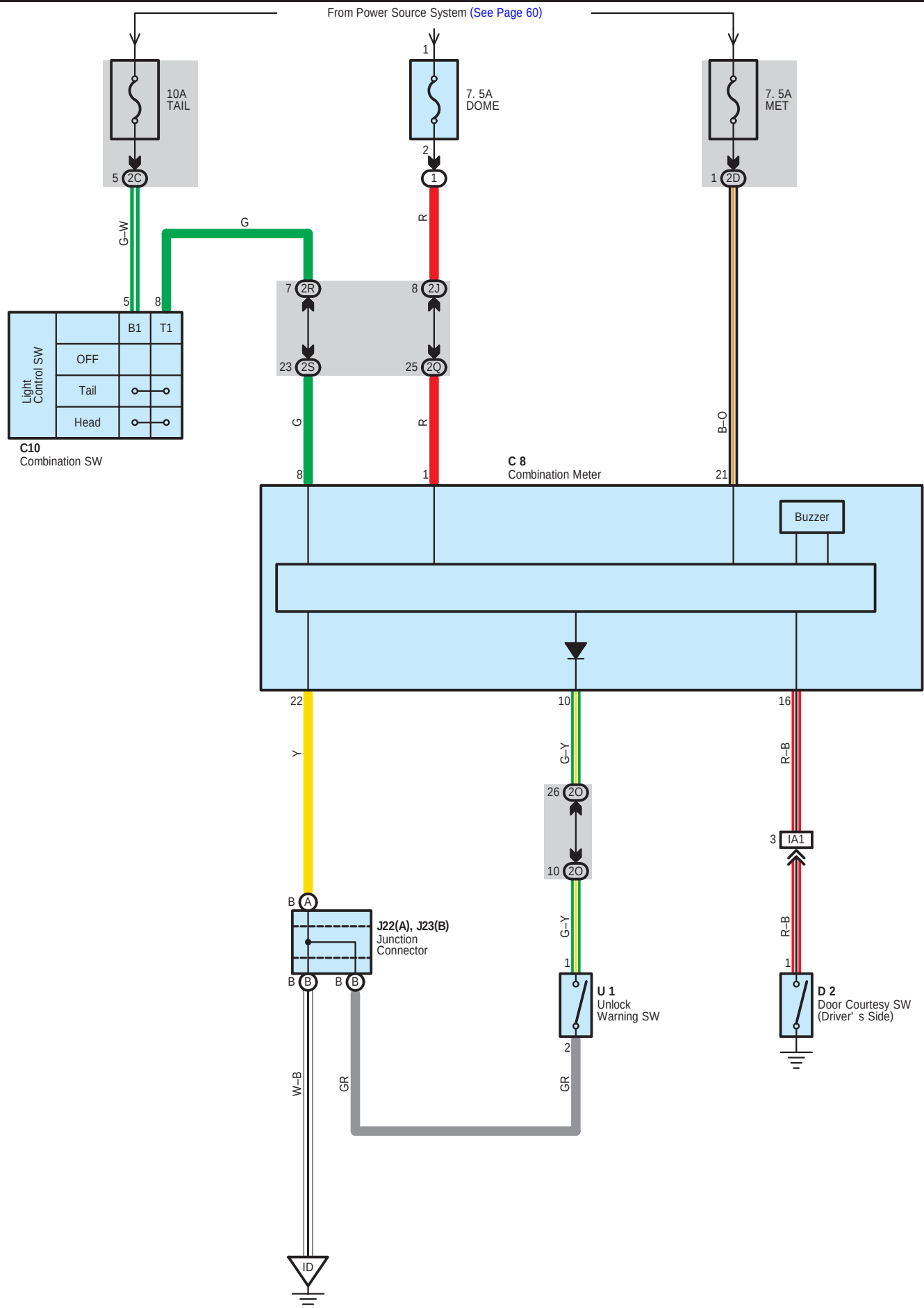
□ : Connector Joining Wire Harness and Wire Harness

Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
IA1	48	Instrument Panel Wire and Floor Wire (Left Kick Panel)

▽ : Ground Points

Code	See Page	Ground Points Location
ID	48	Instrument Panel Reinforcement Center

Key Reminder and Light Reminder



System Outline

The current is applied at all times to TERMINAL 1 of the combination meter through the DOME fuse.
When the ignition SW is turned to ON position, the current flows to TERMINAL 21 of the combination meter through the MET fuse. When the light control SW is turned to TAIL or HEAD position, current is applied to TERMINAL 8 of the combination meter through the TAIL fuse.

1. Key Reminder System

When the driver door is opened with the ignition SW off and ignition key remaining in the key cylinder (Unlock warning SW on), a signal is input from the unlock warning SW to TERMINAL 10 of the combination meter, and from the door courtesy SW (Driver's side) to TERMINAL 16 of the combination meter. As a result, the buzzer in the combination meter goes on and warns the driver that the key is remaining in the key cylinder.

2. Light Reminder System

When the light control SW is in TAIL or HEAD position, the ignition SW turned to OFF from ON position, ignition key is not in the key cylinder and the driver's door opened (Door courtesy SW on), the current flows to TERMINAL 21 of the combination meter stops. As a result, the combination meter is activated and current flows from TERMINAL 1 of the combination meter, the buzzer in the combination meter goes on to remind the light is lighting up.

Service Hints

C8 Combination Meter

- 21-Ground : Approx. 12 volts with the ignition SW at ON position
- 8-Ground : Approx. 12 volts with the light control SW at TAIL or HEAD position
- 16-Ground : Continuity with the driver's door open
- 22-Ground : Always continuity

D2 Door Courtesy SW (Driver's Side)

- 1-Ground : Closed with the driver's door open

U1 Unlock Warning SW

- 1-2 : Closed with the ignition key in cylinder

○ : Parts Location

Code	See Page	Code	See Page	Code	See Page
C8	38	D2	42 (Extra Cab)	J23 B	38
C10	38		44 (Single Cab)	U1	39
D2	40 (Double Cab)	J22 A	38		

○ : Relay Blocks

Code	See Page	Relay Blocks (Relay Block Location)
1	22	Engine Room R/B (Engine Compartment Left)

○ : Junction Block and Wire Harness Connector

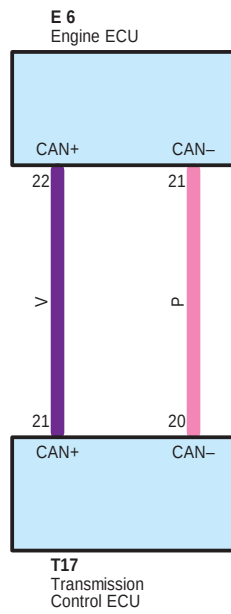
Code	See Page	Junction Block and Wire Harness (Connector Location)
2C	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2D		
2J	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2O	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2Q		
2R		
2S		

□ : Connector Joining Wire Harness and Wire Harness

Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
IA1	48	Instrument Panel Wire and Floor Wire (Left Kick Panel)

▽ : Ground Points

Code	See Page	Ground Points Location
ID	48	Instrument Panel Reinforcement Center



System Outline

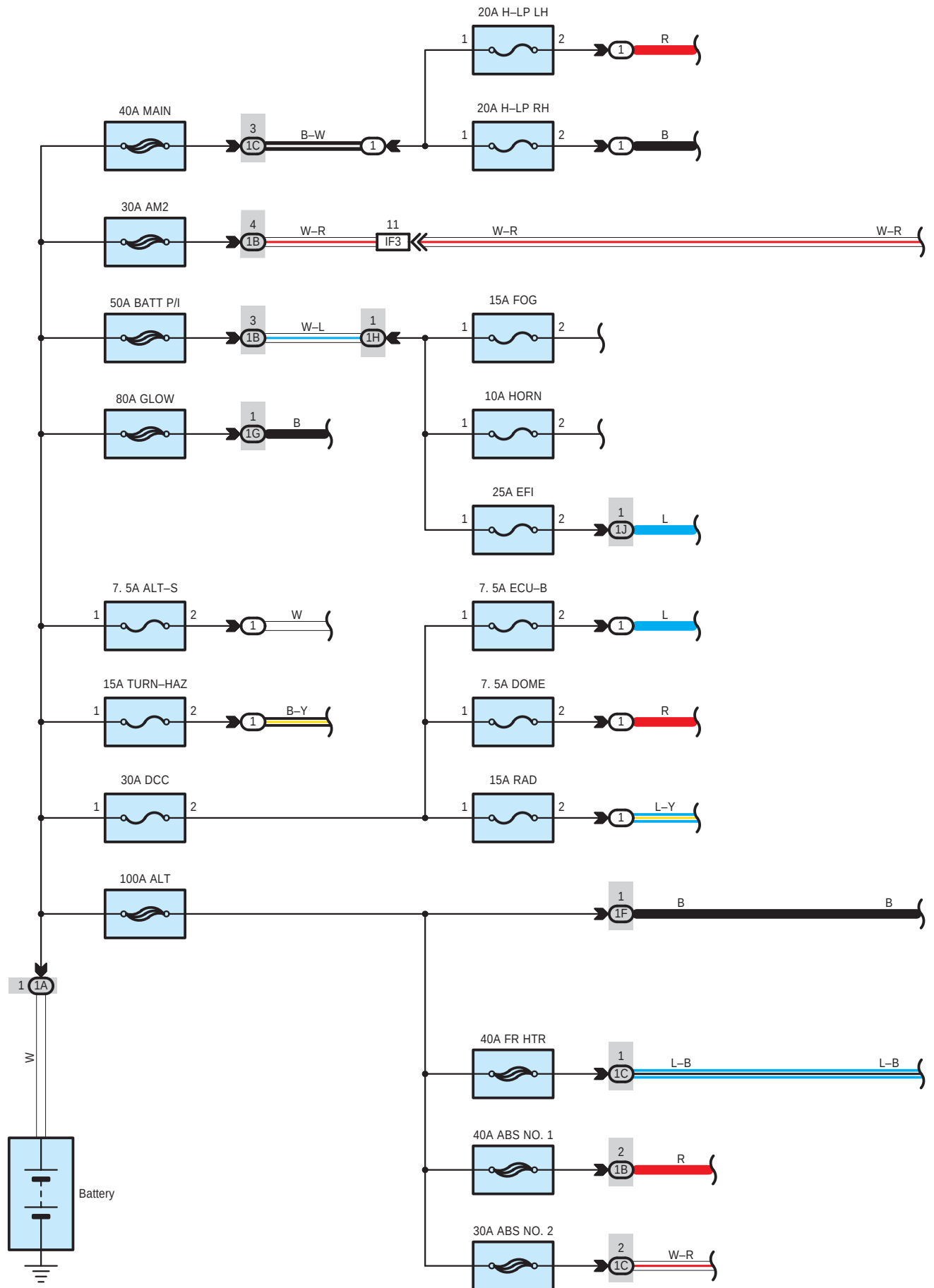
Multiplex communication system (CAN) uses a serial communication protocol and communicates with a differential voltage. In this network system, TERMINALS CAN+ and CAN- are used for communication between the ECUs and sensors, and excellent data communication speed and communication error detecting facility are provided. This system is working for the following systems:

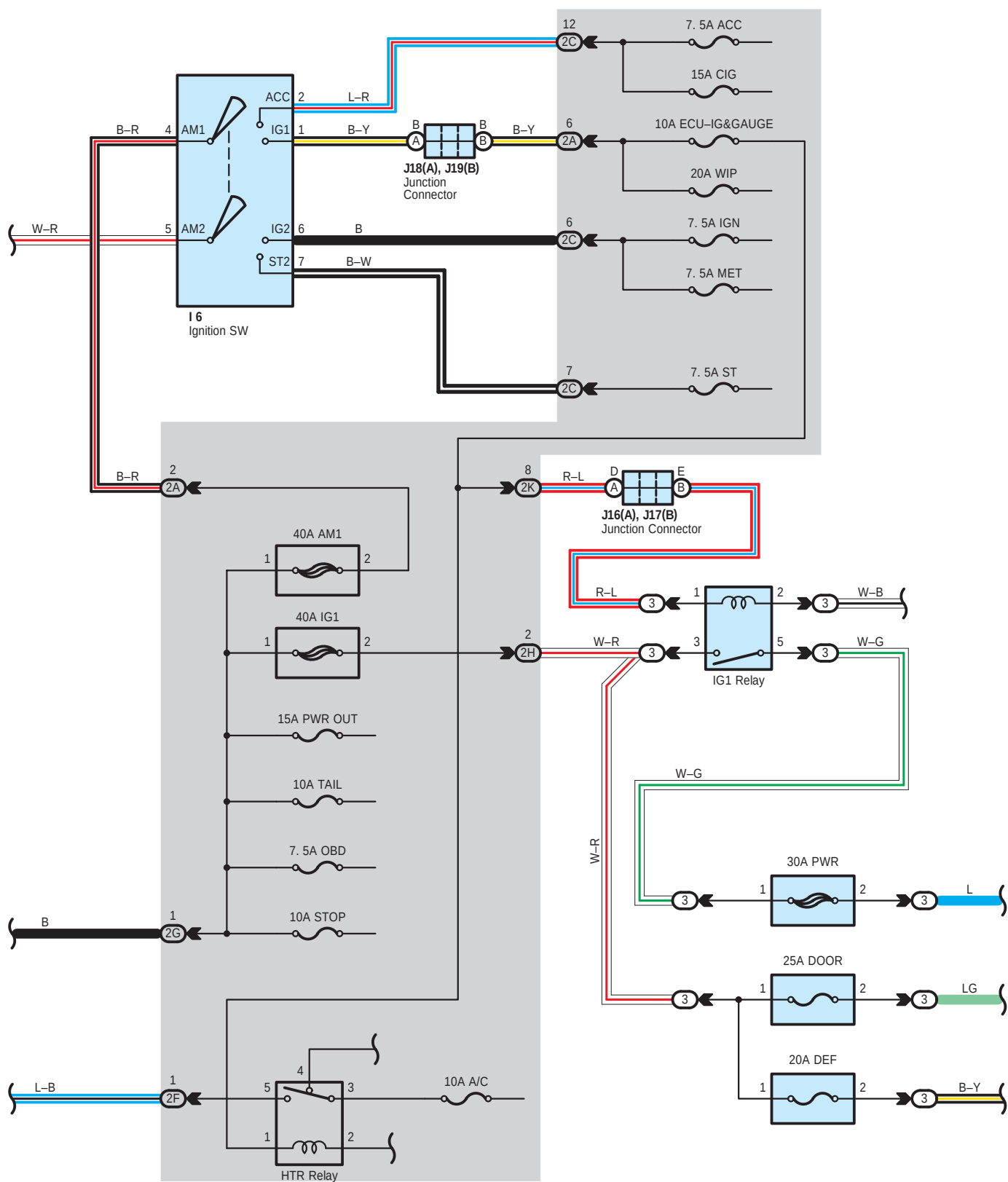
- * Engine Control
- * ECT

○ : Parts Location

Code	See Page	Code	See Page	Code	See Page
E6	38	T17	39		

Power Source





Power Source

Service Hints

I6 Ignition SW

- 4-2 : Closed with the ignition SW at ACC or ON position
- 4-1 : Closed with the ignition SW at ON or ST position
- 5-6 : Closed with the ignition SW at ON or ST position
- 5-7 : Closed with the ignition SW at ST position

: Parts Location

Code	See Page	Code	See Page	Code	See Page
I6	38	J17	B	38	
J16	A	38		J19	B
		J18	A	38	

: Relay Blocks

Code	See Page	Relay Blocks (Relay Block Location)
1	22	Engine Room R/B (Engine Compartment Left)
3	35	R/B No.3 (Cowl Side Panel LH)

: Junction Block and Wire Harness Connector

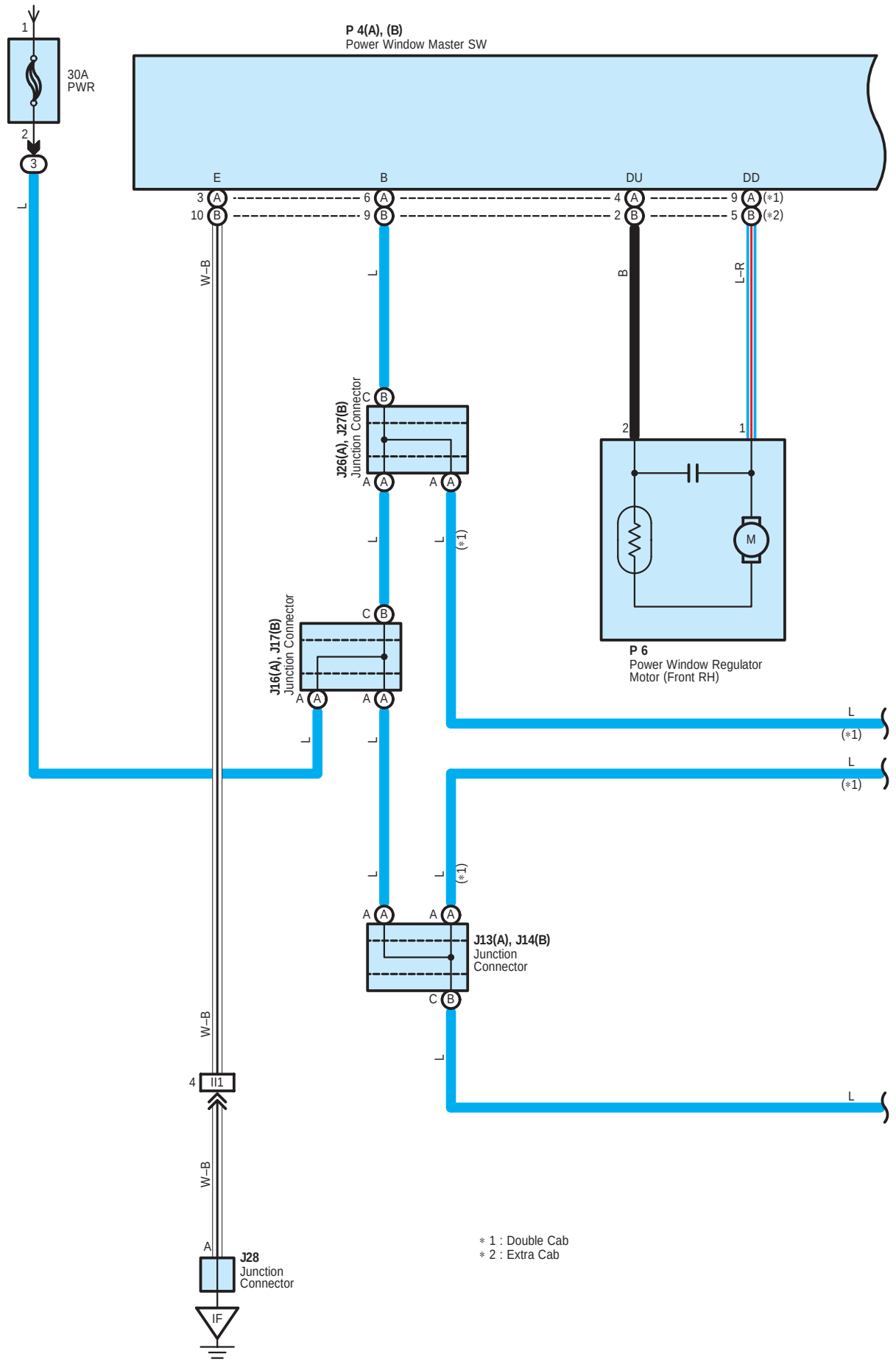
Code	See Page	Junction Block and Wire Harness (Connector Location)
1A	23	Engine Room Main Wire and Engine Room J/B (Engine Compartment Left)
1B		
1C		
1F		
1G		
1H	24	
1J		
2A	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2C		
2F	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2G		
2H	28	Floor Wire and Driver Side J/B (Instrument Panel Brace RH)
2K		

: Connector Joining Wire Harness and Wire Harness

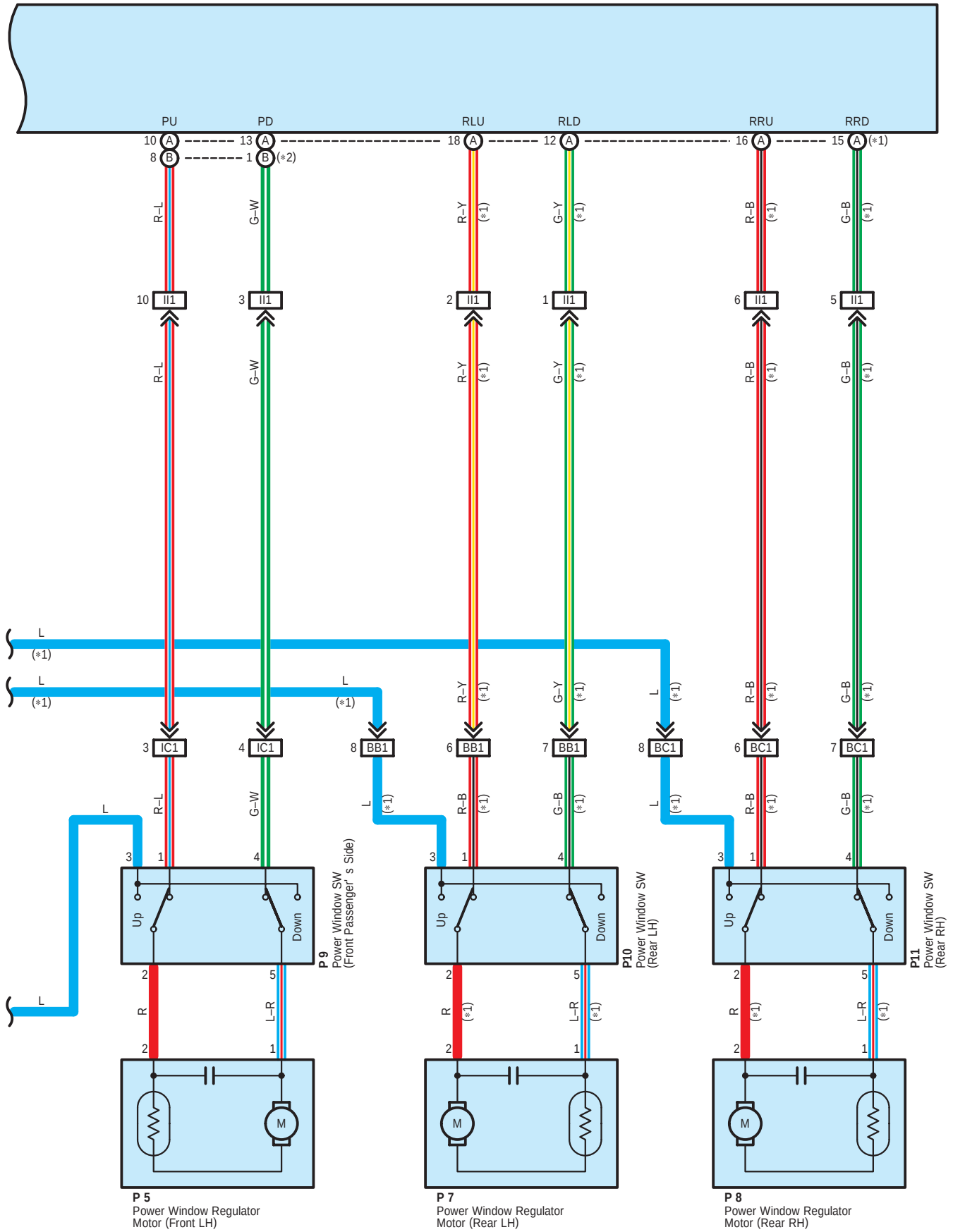
Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
IF3	50	Engine Room Main Wire and Instrument Panel Wire (Behind the Driver Side J/B)

Power Window

From Power Source System (See Page 60)



P 4(A), (B)
Power Window Master SW



Power Window

System Outline

1. Manual Down or Up Operation

When the power window master SW is pushed one step, the motor rotates to open the window.

When the power window master SW is pulled up one step, the motor rotates in the opposite direction, to close the window.

All the other windows can be opened/closed as well, by the operation of the power window master SW or respective power window SW.

When the window lock SW is pushed to the lock side, the ground circuit to the passenger's window becomes open. As a result, even if Open/Close operation of the passenger's window is attempted, the current from TERMINAL E of the power window master SW is not grounded and the motor does not rotate, so the passenger's window can not be operated and window lock occurs.

2. Auto Down Operation (Driver's Window)

When the power window master SW is pushed two steps, the motor rotates to open the window automatically.

3. Stopping of Auto Down Operation (Driver's Window)

Auto operation can be stopped in mid-course with switching power window master SW one step upward during the down operation.

Service Hints

P4 (A), (B) Power Window Master SW

B-Ground : Approx. 12 volts with the ignition SW at ON position

E-Ground : Always continuity

DU-Ground : Approx. 12 volts with the ignition SW on and the master SW (Driver's window) at UP or AUTO UP position

DD-Ground : Approx. 12 volts with the ignition SW on and the master SW (Driver's window) at DOWN or AUTO DOWN position

○ : Parts Location

Code	See Page	Code	See Page	Code	See Page
J13	A 38	P4	B 42 (Extra Cab)	P8	40 (Double Cab)
J14	B 38	P5	40 (Double Cab)	P9	40 (Double Cab)
J16	A 38		42 (Extra Cab)		42 (Extra Cab)
J17	B 38		44 (Single Cab)		44 (Single Cab)
J26	A 38		40 (Double Cab)	P10	40 (Double Cab)
J27	B 38	P6	42 (Extra Cab)	P11	40 (Double Cab)
J28	38		44 (Single Cab)		
P4	A 40 (Double Cab)	P7	40 (Double Cab)		

○ : Relay Blocks

Code	See Page	Relay Blocks (Relay Block Location)
3	35	R/B No.3 (Cowl Side Panel LH)

□ : Connector Joining Wire Harness and Wire Harness

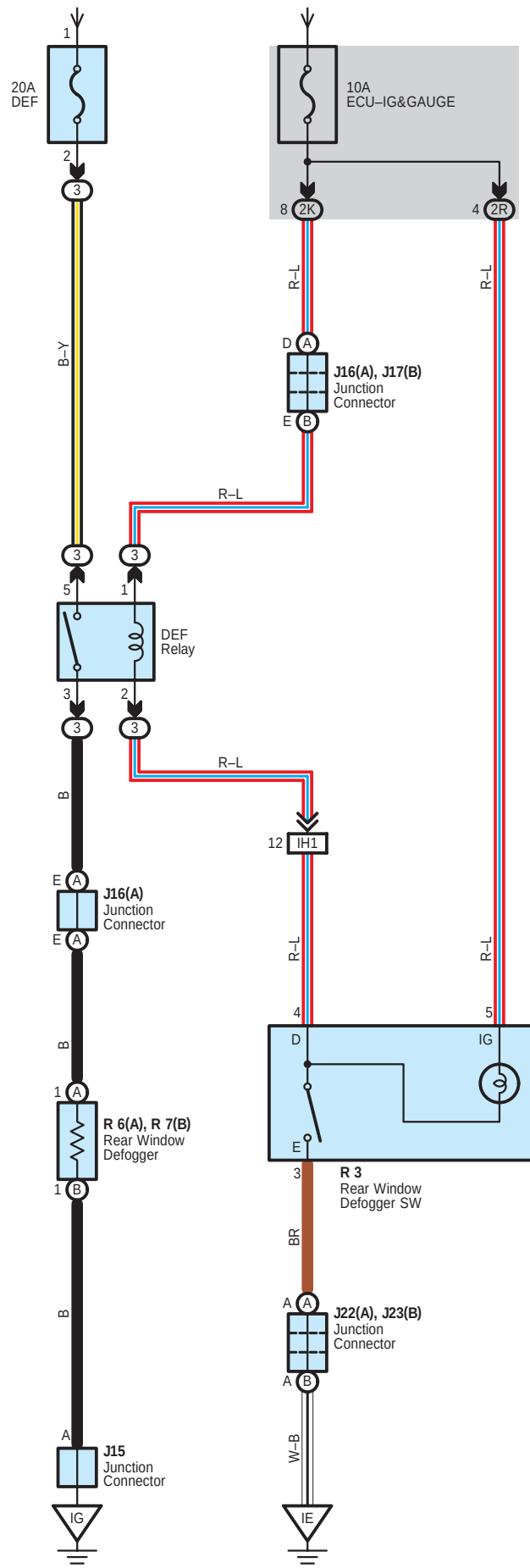
Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
IC1	48	Front Door LH Wire and Floor Wire (Left Kick Panel)
II1	50	Front Door RH Wire and Floor Wire (Right Kick Panel)
BB1	52 (Double Cab)	Rear Door No.1 Wire LH and Floor Wire (Center Pillar LH)
BC1	52 (Double Cab)	Rear Door No.1 Wire RH and Floor Wire (Center Pillar RH)

▽ : Ground Points

Code	See Page	Ground Points Location
IF	48	Cowl Side Panel RH

Rear Window Defogger

From Power Source System (See Page 60)



Service Hints

DEF Relay

5-3 : Closed with the ignition SW at ON position and the rear window defogger SW at ON position

: Parts Location

Code	See Page	Code	See Page	Code	See Page
J15	38	J23	B	38	41 (Double Cab)
J16	A	38	R3	39	43 (Extra Cab)
J17	B	38	R6	A	41 (Double Cab)
J22	A	38			43 (Extra Cab)

: Relay Blocks

Code	See Page	Relay Blocks (Relay Block Location)
3	35	R/B No.3 (Cowl Side Panel LH)

: Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
2K	28	Floor Wire and Driver Side J/B (Instrument Panel Brace RH)
2R	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)

: Connector Joining Wire Harness and Wire Harness

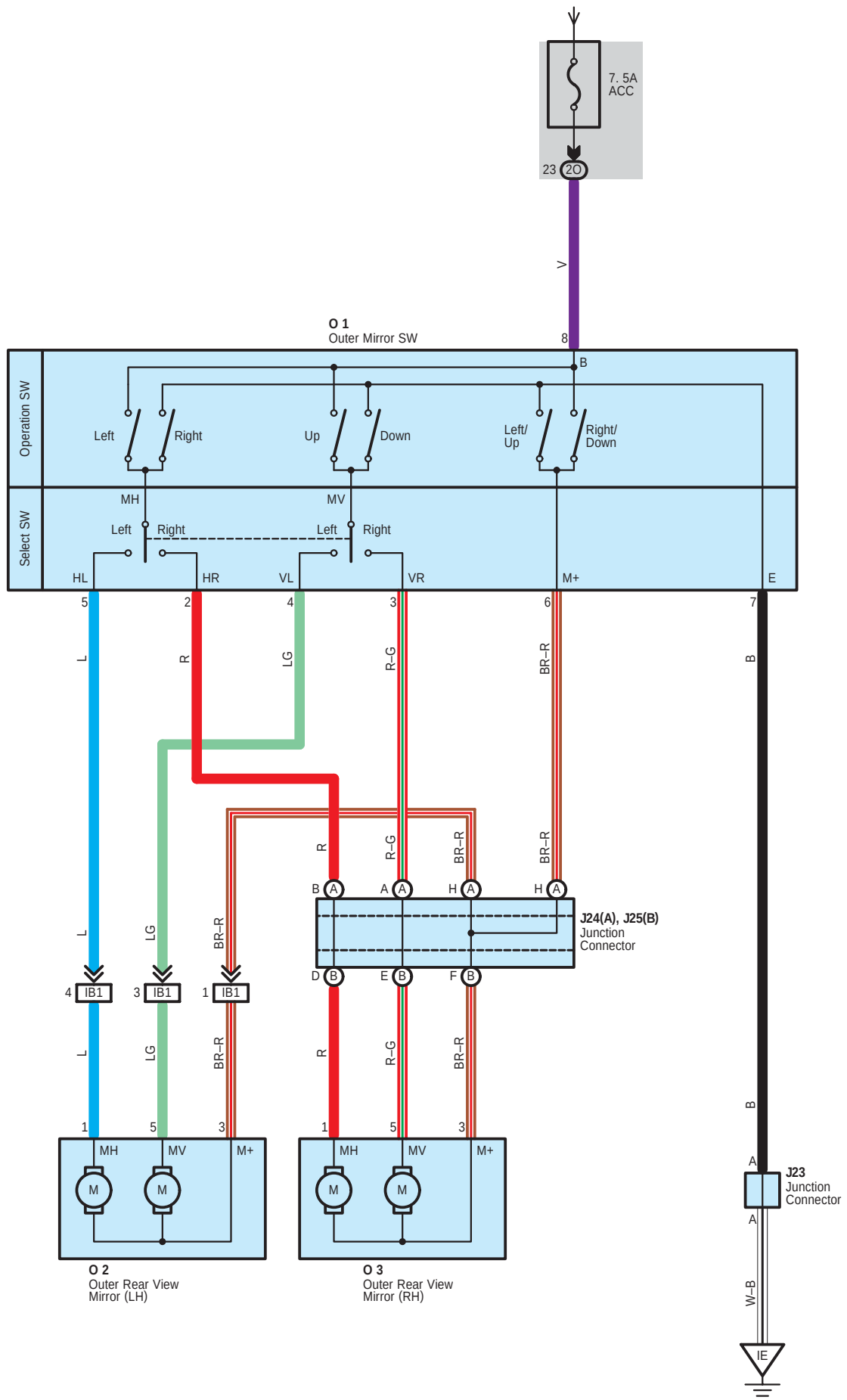
Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
IH1	50	Instrument Panel Wire and Floor Wire (Right Kick Panel)

: Ground Points

Code	See Page	Ground Points Location
IE	48	Instrument Panel Reinforcement RH
IG	48	Cowl Side Panel LH

Remote Control Mirror

From Power Source System (See Page 60)



Service Hints

O1 Outer Mirror SW

8–Ground : Approx. 12 volts with the ignition SW at ACC or ON position

7–Ground : Always continuity

6–7 : Continuity with the operation SW at LEFT or UP position

8–6 : Continuity with the operation SW at RIGHT or DOWN position

: Parts Location

Code	See Page	Code	See Page	Code	See Page
J23	38	O2	40 (Double Cab)	O3	42 (Extra Cab)
J24 A	38		42 (Extra Cab)		44 (Single Cab)
J25 B	38		44 (Single Cab)		
O1	39	O3	40 (Double Cab)		

: Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
2O	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)

: Connector Joining Wire Harness and Wire Harness

Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
IB1	48	Front Door LH Wire and Instrument Panel Wire (Left Kick Panel)

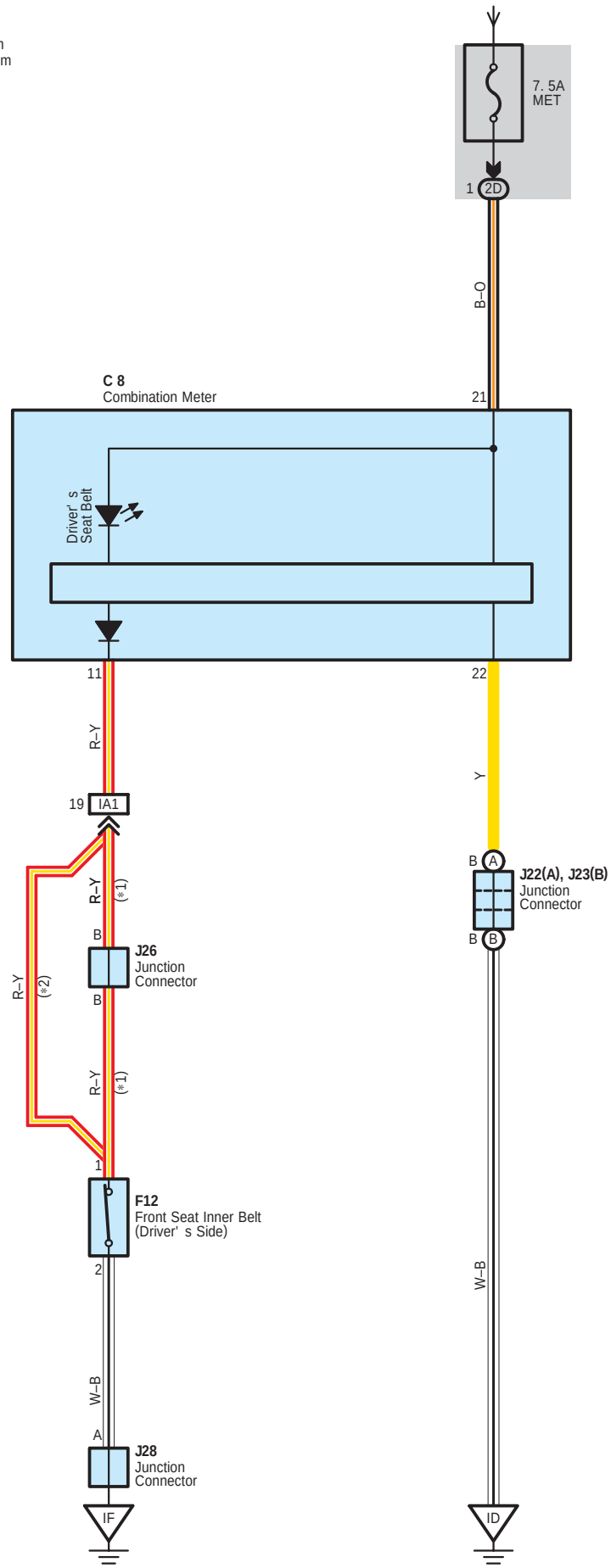
: Ground Points

Code	See Page	Ground Points Location
IE	48	Instrument Panel Reinforcement RH

Seat Belt Warning

From Power Source System (See Page 60)

- * 1 : w/ Door Lock Control System
- * 2 : w/o Door Lock Control System



System Outline

When the ignition SW is turned to ON position, the current from the MET fuse flows to TERMINAL 21 of the combination meter.

Seat Belt Warning System

When the ignition SW turned on, a signal is input to the combination meter. To determine whether the driver has fastened the seat belt, a signal is input from the front seat inner belt (Driver's side) to TERMINAL 11 of the combination meter. When the seat belt is not fastened, the driver's seat belt warning light in the combination meter blinks.

Service Hints

F12 Front Seat Inner Belt (Driver's Side)

1-2 : Open with the driver's seat belt in use

C8 Combination Meter

21-Ground : Approx. 12 volts with the ignition SW at ON position

22-Ground : Always continuity

○ : Parts Location

Code	See Page	Code	See Page	Code	See Page
C8	38	F12	44 (Single Cab)	J26	38
F12	40 (Double Cab)	J22	A	J28	38
	42 (Extra Cab)	J23	B		

□ : Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
2D	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)

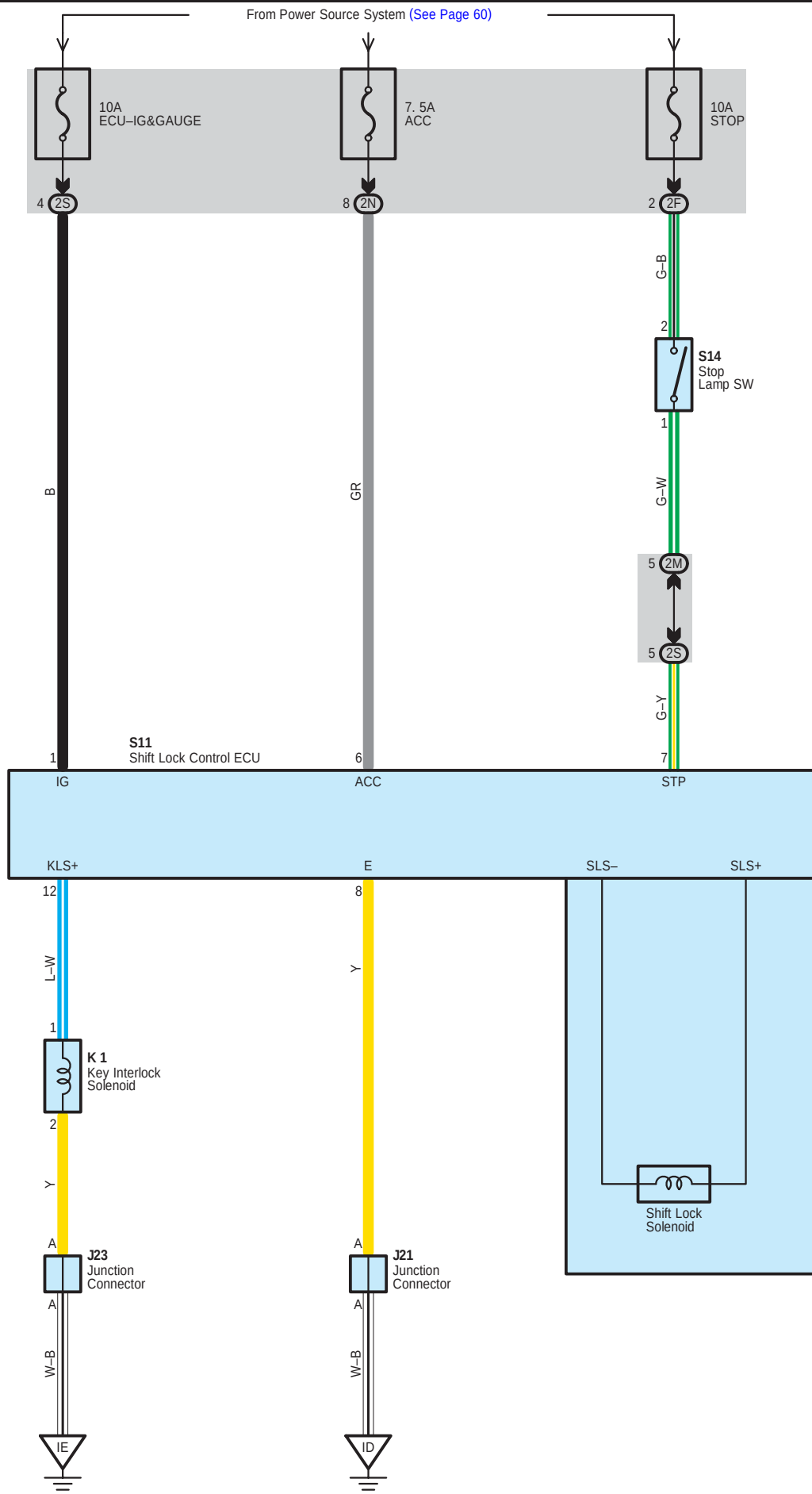
□ : Connector Joining Wire Harness and Wire Harness

Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
IA1	48	Instrument Panel Wire and Floor Wire (Left Kick Panel)

▽ : Ground Points

Code	See Page	Ground Points Location
ID	48	Instrument Panel Reinforcement Center
IF	48	Cowl Side Panel RH

Shift Lock



System Outline

When the ignition SW is turned to ACC position the current from the ACC fuse flows to TERMINAL 6 of the shift lock control ECU. When the ignition SW is turned to ON position, the current from the ECU-IG&GAUGE fuse flows to TERMINAL 1 of the shift lock control ECU.

1. Shift Lock Mechanism

If the brake pedal is depressed with the ignition SW set at ON (The stop lamp SW is on), the shift lock control ECU is activated, allowing the driver to change the shift lever to a position other than the P position.

2. Key Interlock Mechanism

With the ignition SW at ON or ACC position, when the shift lever is put in P position, the current flowing from TERMINAL 12 of the shift lock control ECU to key interlock solenoid is cut off. This causes the key interlock solenoid to turn off (Lock lever disengages from LOCK position) and the ignition key can be turned from ACC to LOCK position.

Service Hints

S11 Shift Lock Control ECU

- 6-Ground : Approx. 12 volts with the ignition SW at ACC or ON position
- 1-Ground : Approx. 12 volts with the ignition SW at ON position
- 7-Ground : Approx. 12 volts with the brake pedal depressed
- 8-Ground : Always continuity

S14 Stop Lamp SW

- 2-1 : Closed with the brake pedal depressed

○ : Parts Location

Code	See Page	Code	See Page	Code	See Page
J21	38	K1	39	S14	39
J23	38	S11	39		

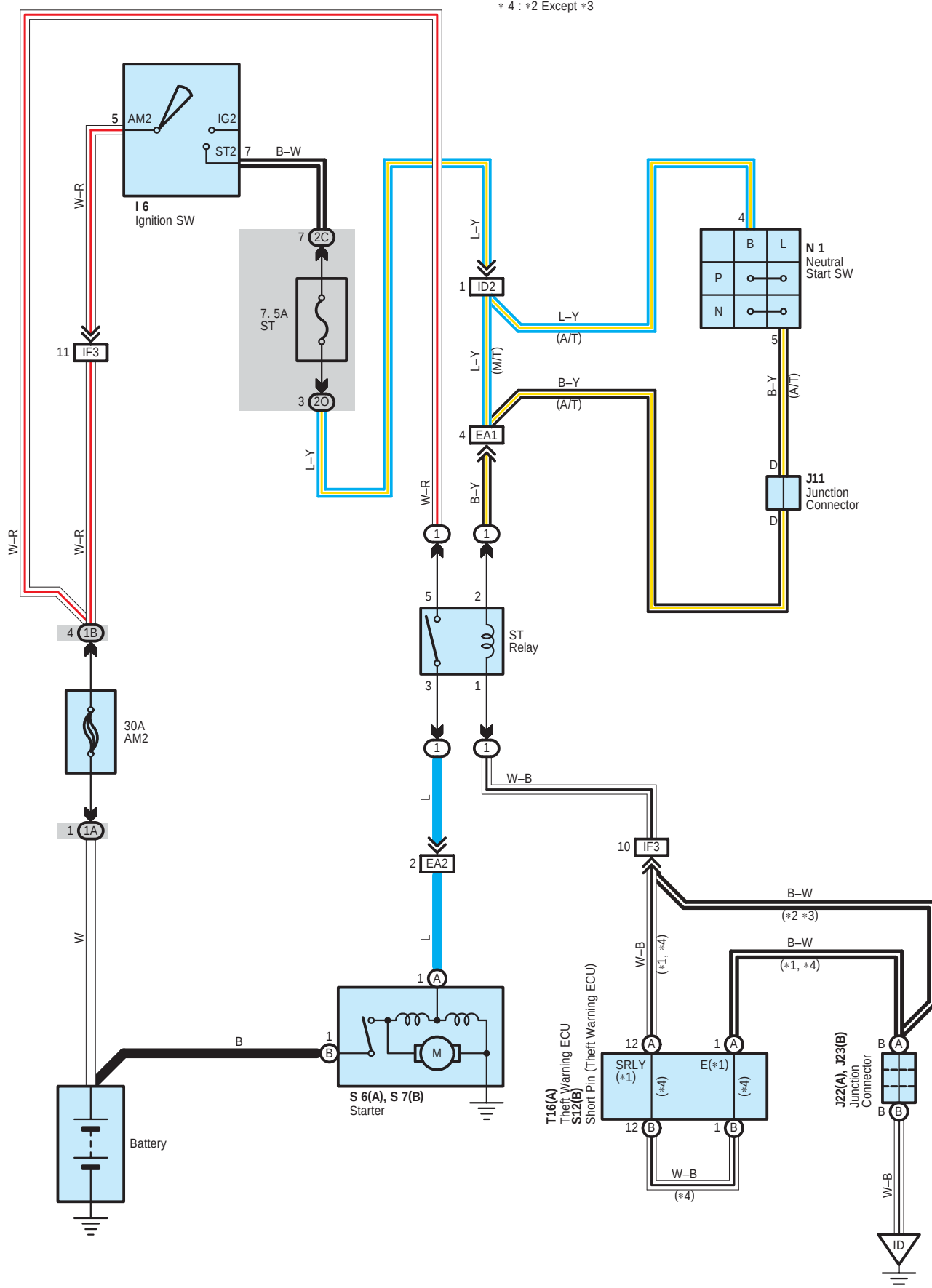
○ : Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
2F	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2M		
2N	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2S		

▽ : Ground Points

Code	See Page	Ground Points Location
ID	48	Instrument Panel Reinforcement Center
IE	48	Instrument Panel Reinforcement RH

- * 1 : w/ Theft Deterrent System
- * 2 : w/o Theft Deterrent System
- * 3 : M/T Single Cab, Double Cab 2KD-FTV w/o Inter Cooler
- * 4 : *2 Except *3



Service Hints

S6 (A), S7 (B) Starter

Point closed with the neutral start SW at P or N position and the ignition SW at ST position (A/T)

Point closed with the ignition SW at ST position (M/T)

I6 Ignition SW

5-7 : Closed with the ignition SW at ST position

N1 Neutral Start SW

4-5 : Closed with the neutral start SW at P or N position

ST Relay

5-3 : Closed with the neutral start SW at P or N position and the ignition SW at ST position (A/T)

5-3 : Closed with the ignition SW at ST position (M/T)

○ : Parts Location

Code		See Page		Code		See Page		Code		See Page	
I6		38		J23	B	38		S7	B	37	
J11		38		N1		37		S12	B	39	
J22	A	38		S6	A	37		T16	A	39	

○ : Relay Blocks

Code	See Page	Relay Blocks (Relay Block Location)
1	22	Engine Room R/B (Engine Compartment Left)

○ : Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
1A	23	Engine Room Main Wire and Engine Room J/B (Engine Compartment Left)
1B		
2C	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2O	29	

□ : Connector Joining Wire Harness and Wire Harness

Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
EA1	46	Engine Wire and Engine Room Main Wire (Inside of Engine Room R/B and Engine Room J/B)
EA2		
ID2	48	Engine Wire and Instrument Panel Wire (Behind the Glove Box)
IF3	50	Engine Room Main Wire and Instrument Panel Wire (Behind the Driver Side J/B)

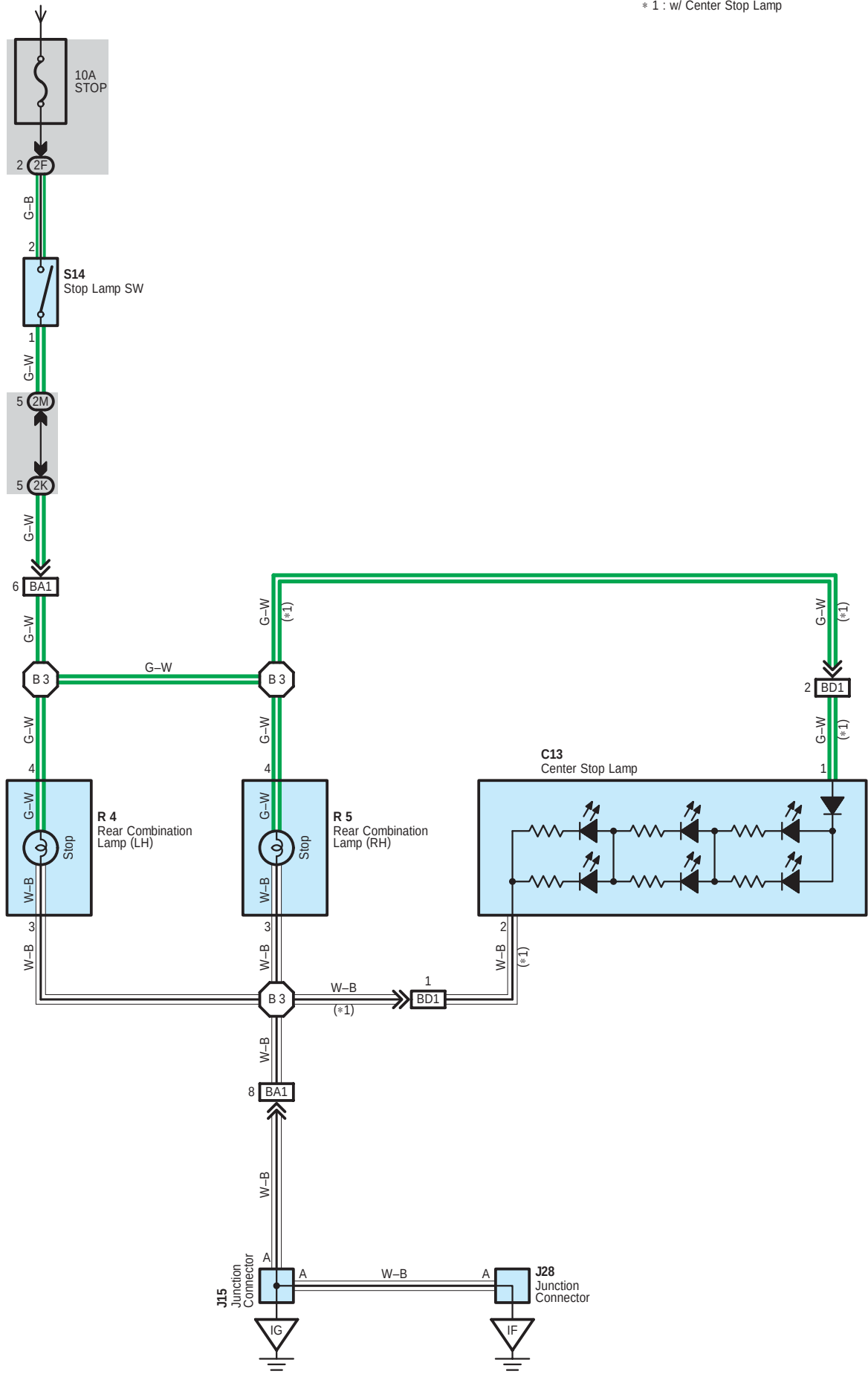
▽ : Ground Points

Code	See Page	Ground Points Location
ID	48	Instrument Panel Reinforcement Center

Stop Light

From Power Source System (See Page 60)

* 1 : w/ Center Stop Lamp



Service Hints

S14 Stop Lamp SW

2-1 : Closed with the brake pedal depressed

○ : Parts Location

Code	See Page	Code	See Page	Code	See Page
C13	40 (Double Cab)	J28	38	R5	41 (Double Cab)
	42 (Extra Cab)	R4	41 (Double Cab)		43 (Extra Cab)
	44 (Single Cab)		43 (Extra Cab)		45 (Single Cab)
J15	38		45 (Single Cab)	S14	39

○ : Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
2F	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2K	28	Floor Wire and Driver Side J/B (Instrument Panel Brace RH)
2M	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)

□ : Connector Joining Wire Harness and Wire Harness

Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
BA1	52 (Double Cab)	Frame Wire and Floor Wire (Under the Front Seat LH)
	54 (Extra Cab)	
	56 (Single Cab)	
BD1	52 (Double Cab)	Frame No.2 Wire and Frame Wire (Under the Back Panel LH)
	54 (Extra Cab)	
	56 (Single Cab)	

▽ : Ground Points

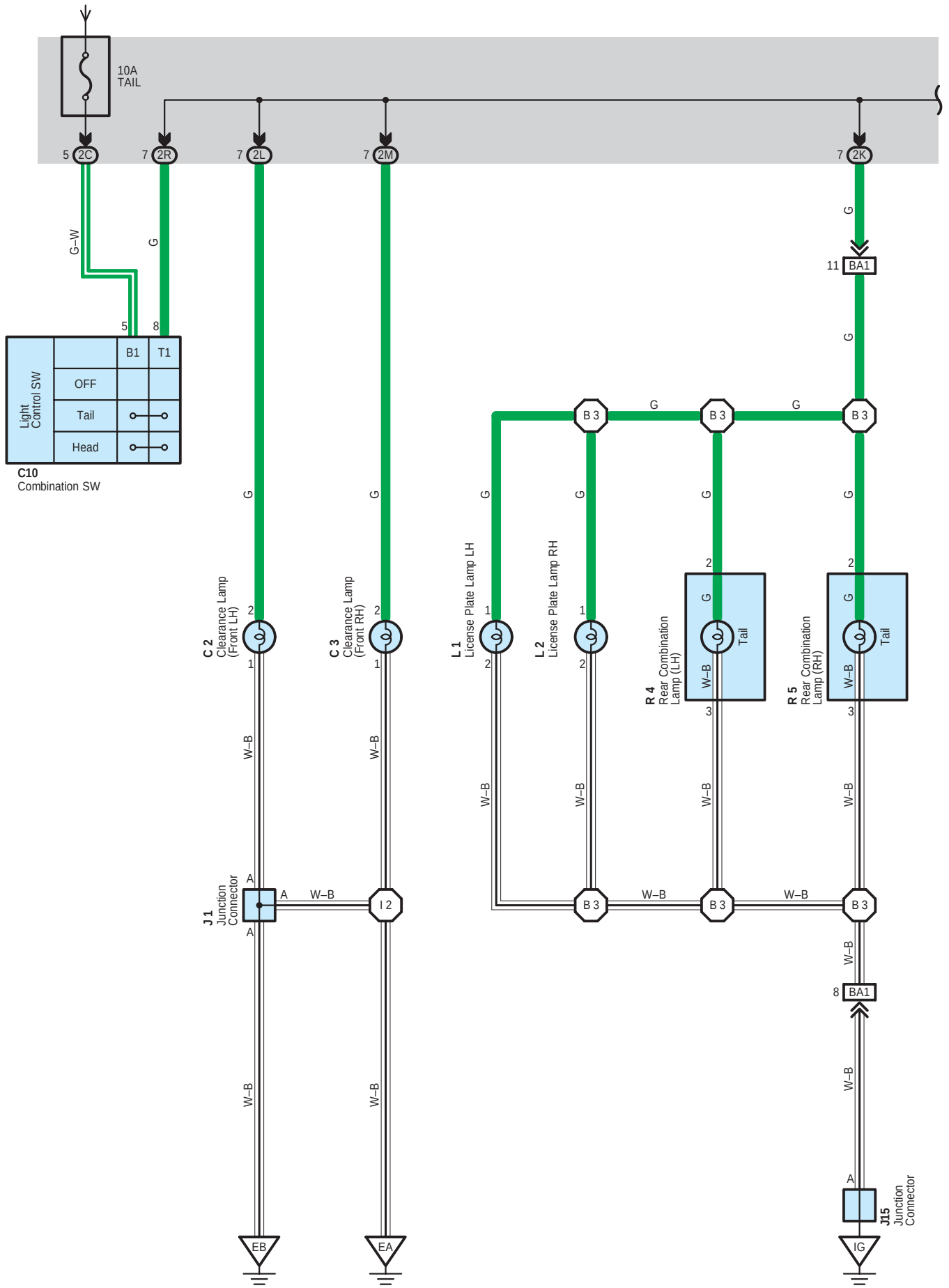
Code	See Page	Ground Points Location
IF	48	Cowl Side Panel RH
IG	48	Cowl Side Panel LH

○ : Splice Points

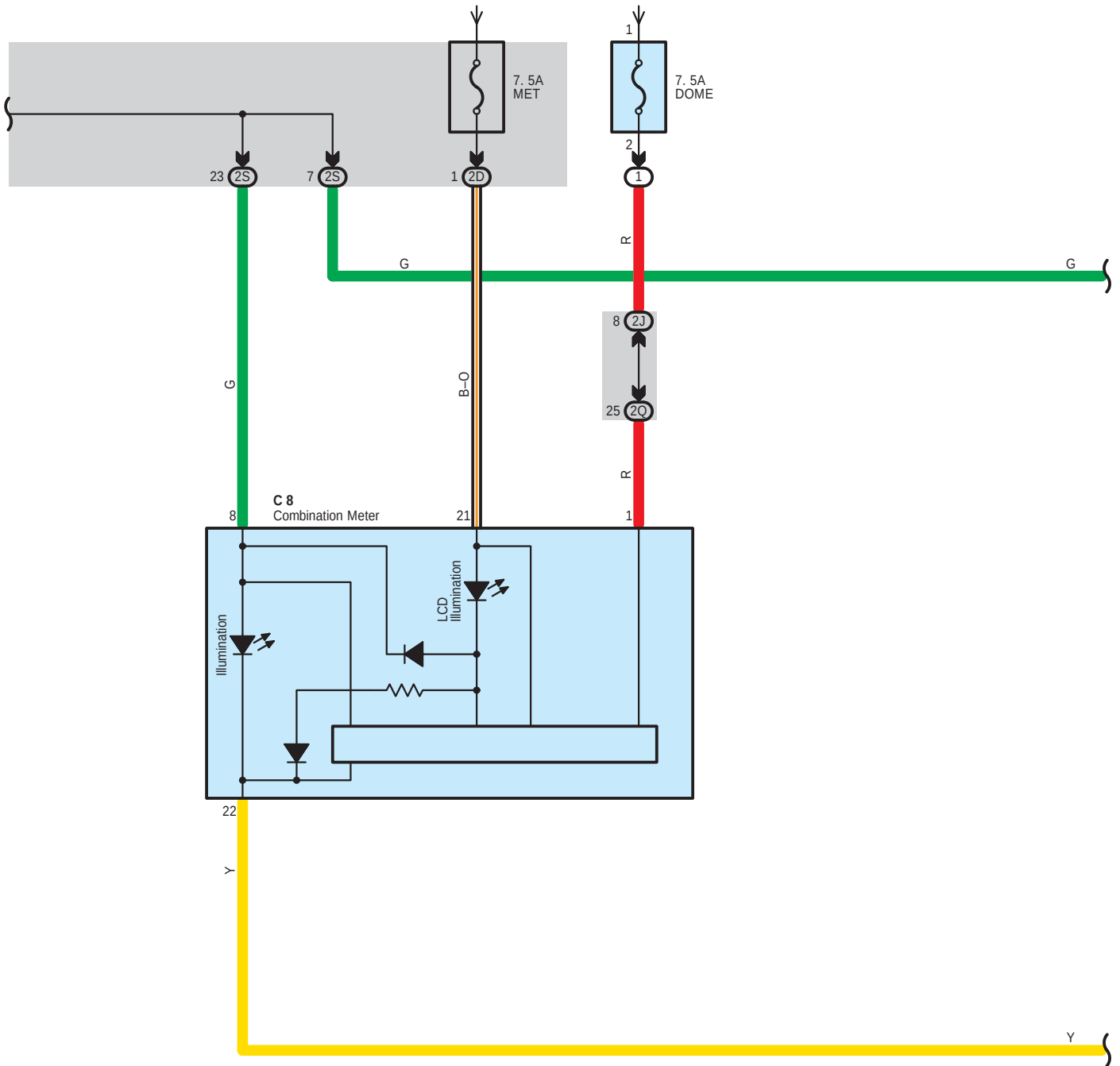
Code	See Page	Wire Harness with Splice Points	Code	See Page	Wire Harness with Splice Points
B3	52 (Double Cab)	Frame Wire	B3	56 (Single Cab)	Frame Wire
	54 (Extra Cab)				

Taillight and Illumination

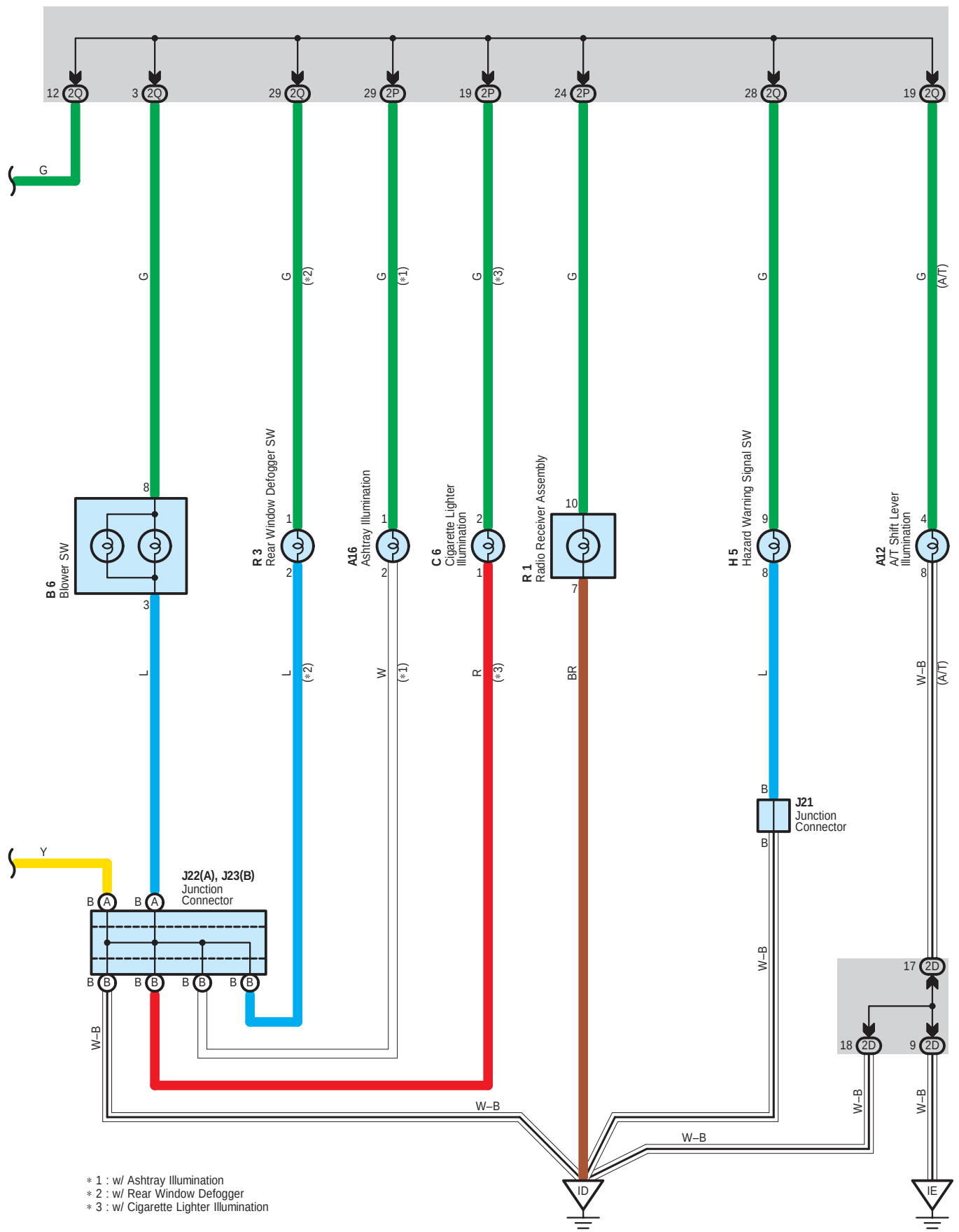
From Power Source System (See Page 60)



From Power Source System (See Page 60)



Taillight and Illumination



Service Hints

C10 Combination SW

5-8 : Continuity with the light control SW at HEAD or TAIL position

○ : Parts Location

Code	See Page	Code	See Page	Code	See Page
A12	38	J15	38	R1	39
A16	38	J21	38	R3	39
B6	38	J22	A 38	R4	41 (Double Cab)
C2	36	J23	B 38		43 (Extra Cab)
C3	36	L1	40 (Double Cab)		45 (Single Cab)
C6	38		42 (Extra Cab)	R5	41 (Double Cab)
C8	38		44 (Single Cab)		43 (Extra Cab)
C10	38	L2	40 (Double Cab)		45 (Single Cab)
H5	38		42 (Extra Cab)		
J1	37		44 (Single Cab)		

○ : Relay Blocks

Code	See Page	Relay Blocks (Relay Block Location)
1	22	Engine Room R/B (Engine Compartment Left)

○ : Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
2C	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2D		
2J	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2K	28	Floor Wire and Driver Side J/B (Instrument Panel Brace RH)
2L	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2M		
2P	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2Q		
2R		
2S		

□ : Connector Joining Wire Harness and Wire Harness

Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
BA1	52 (Double Cab)	Frame Wire and Floor Wire (Under the Front Seat LH)
	54 (Extra Cab)	
	56 (Single Cab)	

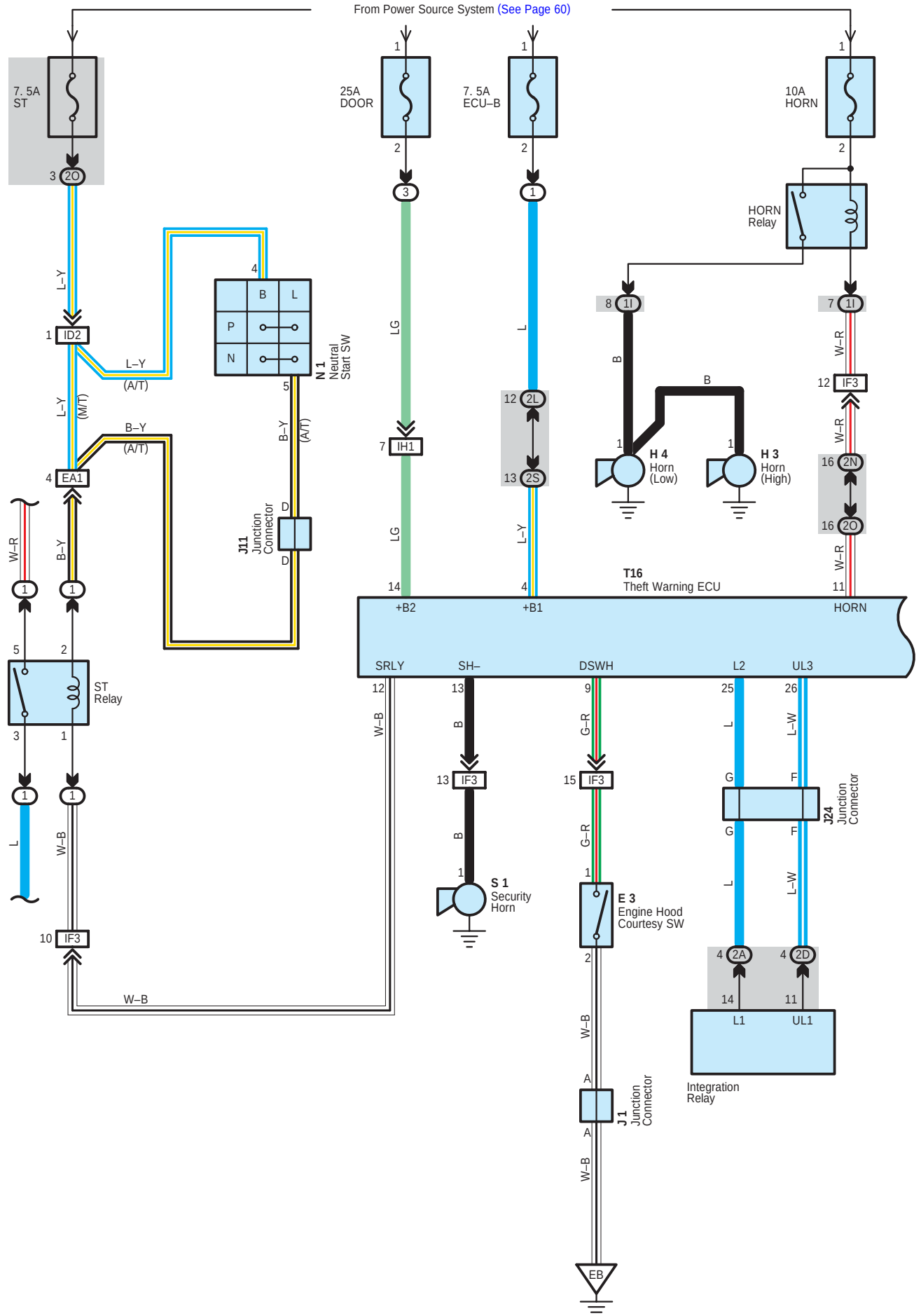
▽ : Ground Points

Code	See Page	Ground Points Location
EA	46	Right Side of the Fender Apron
EB	46	Left Side of the Fender Apron
ID	48	Instrument Panel Reinforcement Center
IE	48	Instrument Panel Reinforcement RH
IG	48	Cowl Side Panel LH

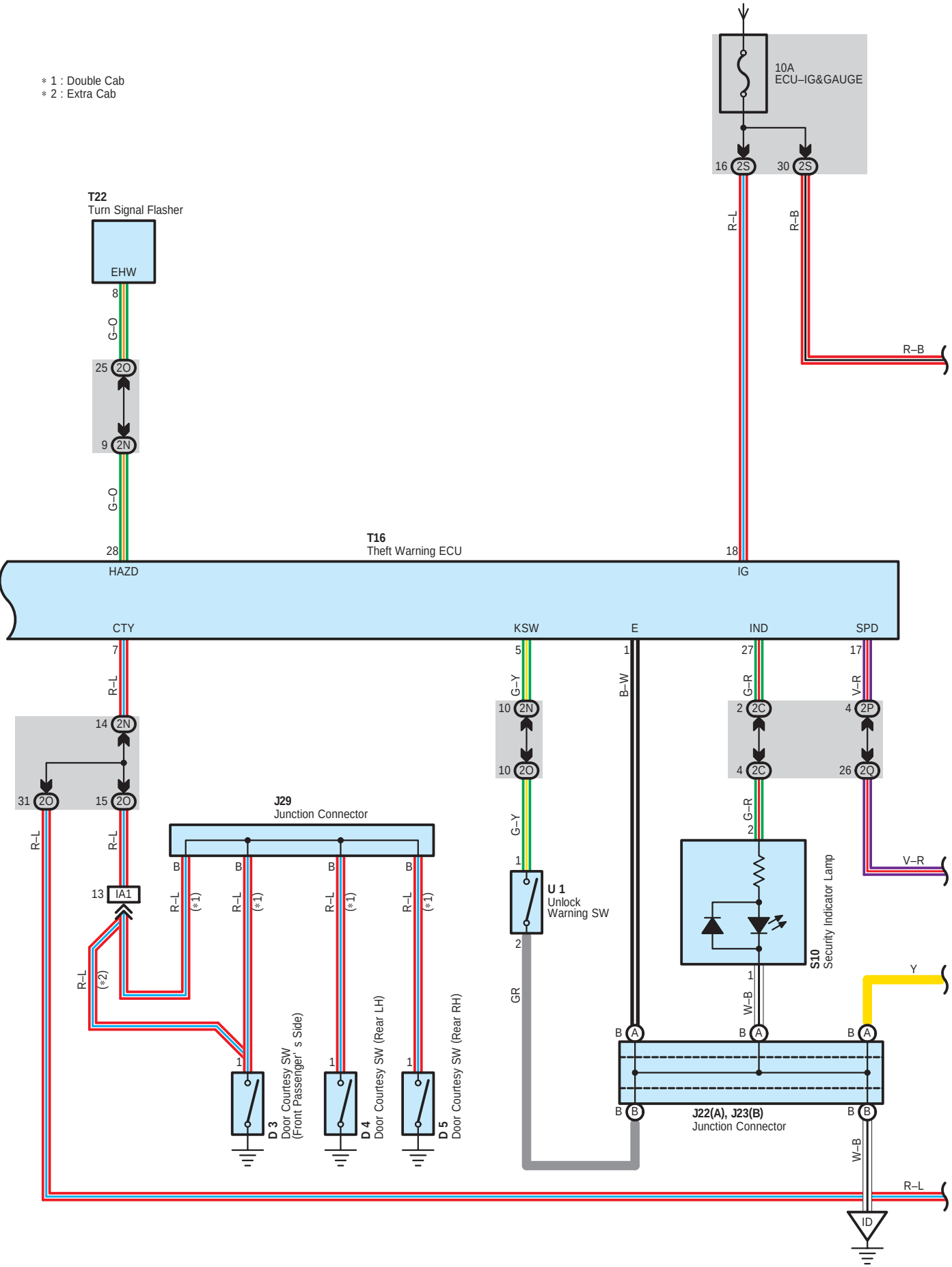
○ : Splice Points

Code	See Page	Wire Harness with Splice Points	Code	See Page	Wire Harness with Splice Points
I2	50	Engine Room Main Wire	B3	54 (Extra Cab)	Frame Wire
B3	52 (Double Cab)	Frame Wire		56 (Single Cab)	

Theft Deterrent

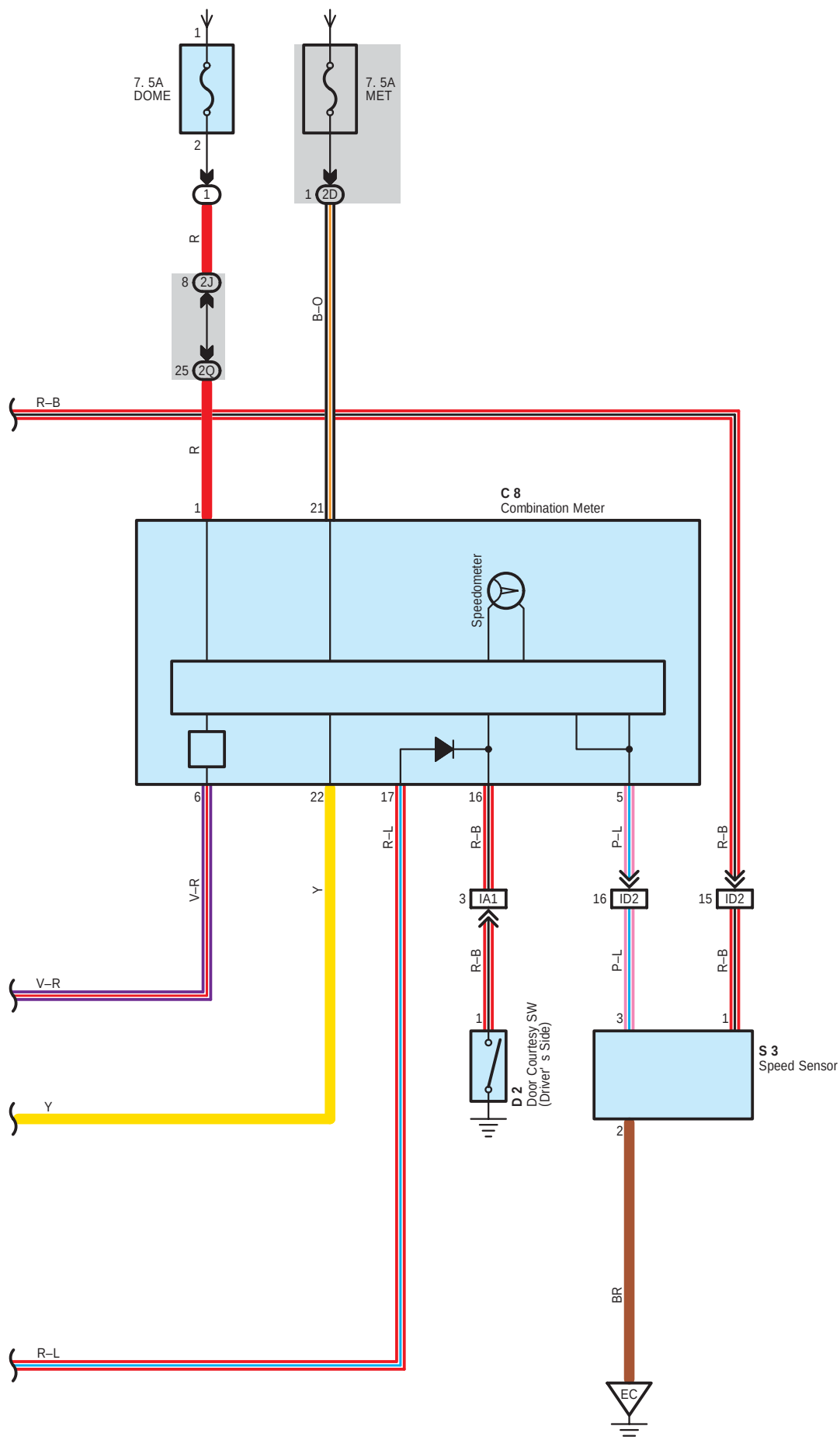


- * 1 : Double Cab
- * 2 : Extra Cab



Theft Deterrent

From Power Source System (See Page 60)



Service Hints

D2, D3, D4, D5 Door Courtesy SW (Driver's Side), (Front Passenger's Side), (Rear LH), (Rear RH)

1–Ground : Continuity with each of the door open

E3 Engine Hood Courtesy SW

1–2 : Open with the engine hood open

T16 Theft Warning ECU

9–Ground : Continuity with the engine hood close

4, 14–Ground : Always approx. 12 volts

1–Ground : Always continuity

18–Ground : Approx. 12 volts with the ignition SW at ON position

: Parts Location

Code	See Page	Code	See Page	Code	See Page
C8	38	E3	36	J29	38
D2	40 (Double Cab)	H3	36	N1	37
	42 (Extra Cab)	H4	36	S1	37
	44 (Single Cab)	J1	37	S3	37
D3	40 (Double Cab)	J11	38	S10	39
	42 (Extra Cab)	J22	A 38	T16	39
	44 (Single Cab)	J23	B 38	T22	39
D4	40 (Double Cab)	J24	A 38	U1	39
D5	40 (Double Cab)	J25	B 38		

: Relay Blocks

Code	See Page	Relay Blocks (Relay Block Location)
1	22	Engine Room R/B (Engine Compartment Left)
3	35	R/B No.3 (Cowl Side Panel LH)

: Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
1I	24	Engine Room Main Wire and Engine Room J/B (Engine Compartment Left)
2A	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2C		
2D		
2J	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2L		
2N	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2O		
2P		
2Q		
2S		

: Connector Joining Wire Harness and Wire Harness

Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
EA1	46	Engine Wire and Engine Room Main Wire (Inside of Engine Room R/B and Engine Room J/B)
IA1	48	Instrument Panel Wire and Floor Wire (Left Kick Panel)
ID2	48	Engine Wire and Instrument Panel Wire (Behind the Glove Box)
IF3	50	Engine Room Main Wire and Instrument Panel Wire (Behind the Driver Side J/B)
IH1	50	Instrument Panel Wire and Floor Wire (Right Kick Panel)

Theft Deterrent

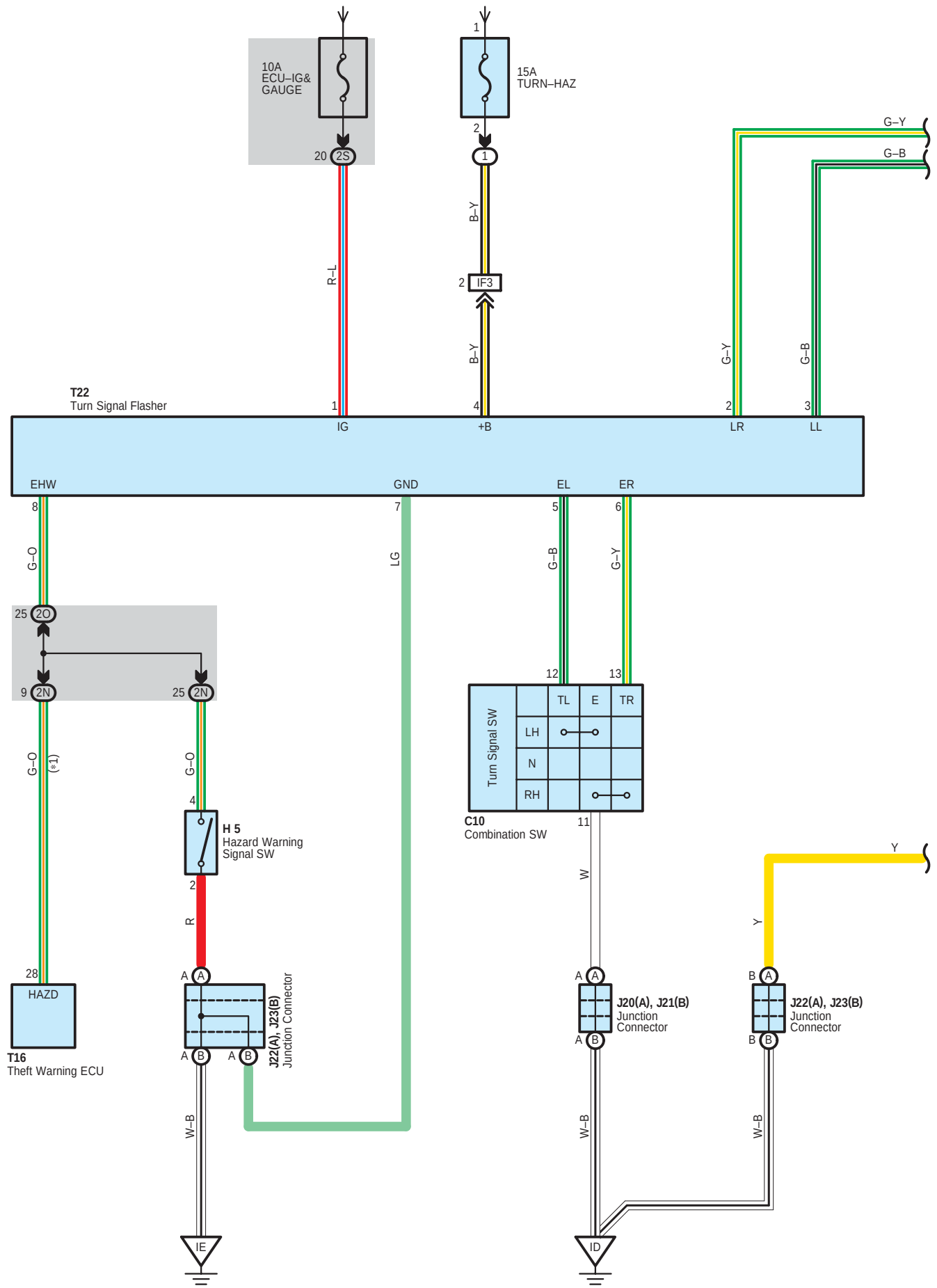


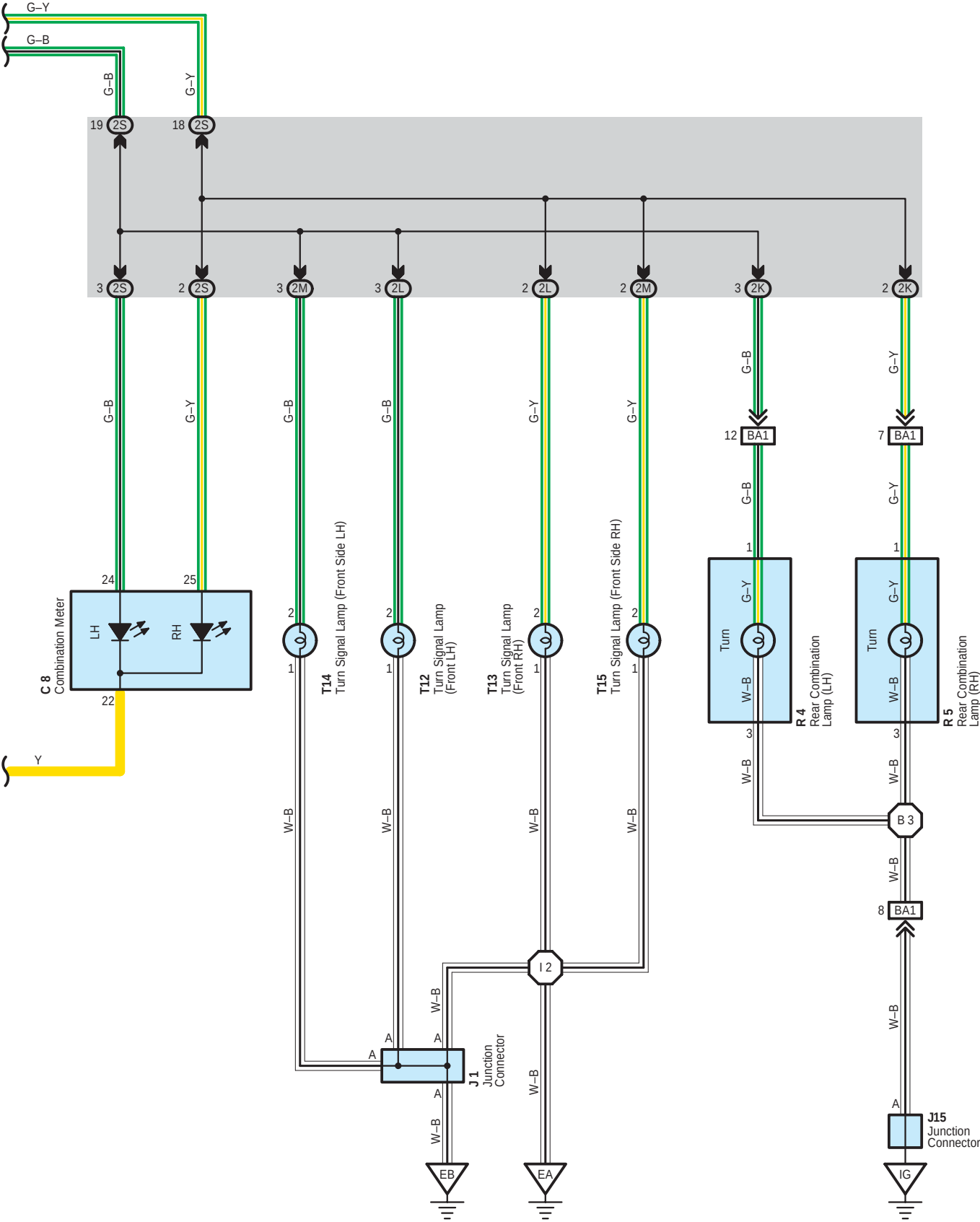
: Ground Points

Code	See Page	Ground Points Location
EB	46	Left Side of the Fender Apron
EC	46	Rear Side of the Cylinder Block
ID	48	Instrument Panel Reinforcement Center

Turn Signal and Hazard Warning Light

From Power Source System (See Page 60)





Turn Signal and Hazard Warning Light

Service Hints

T22 Turn Signal Flasher

- 1-Ground : Approx. 12 volts with the ignition SW at ON position
- 4-Ground : Always approx. 12 volts
- 2, 3-Ground : Changes from approx. 12 to 0 volts with the ignition SW at ON position and the turn signal SW at LEFT or RIGHT position or the hazard warning signal SW at ON position
- 5-Ground : Continuity with the turn signal SW at LEFT position
- 6-Ground : Continuity with the turn signal SW at RIGHT position
- 8-Ground : Continuity with the hazard warning signal SW at ON position
- 7-Ground : Always continuity

○ : Parts Location

Code		See Page		Code		See Page		Code		See Page	
C8		38		J22	A	38		R5	45 (Single Cab)		
C10		38		J23	B	38		T12	37		
H5		38		R4		41 (Double Cab)		T13	37		
J1		37				43 (Extra Cab)		T14	37		
J15		38				45 (Single Cab)		T15	37		
J20	A	38		R5		41 (Double Cab)		T16	39		
J21	B	38				43 (Extra Cab)		T22	39		

○ : Relay Blocks

Code	See Page	Relay Blocks (Relay Block Location)
1	22	Engine Room R/B (Engine Compartment Left)

○ : Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
2K	28	Floor Wire and Driver Side J/B (Instrument Panel Brace RH)
2L	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2M		
2N	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2O		
2S		

□ : Connector Joining Wire Harness and Wire Harness

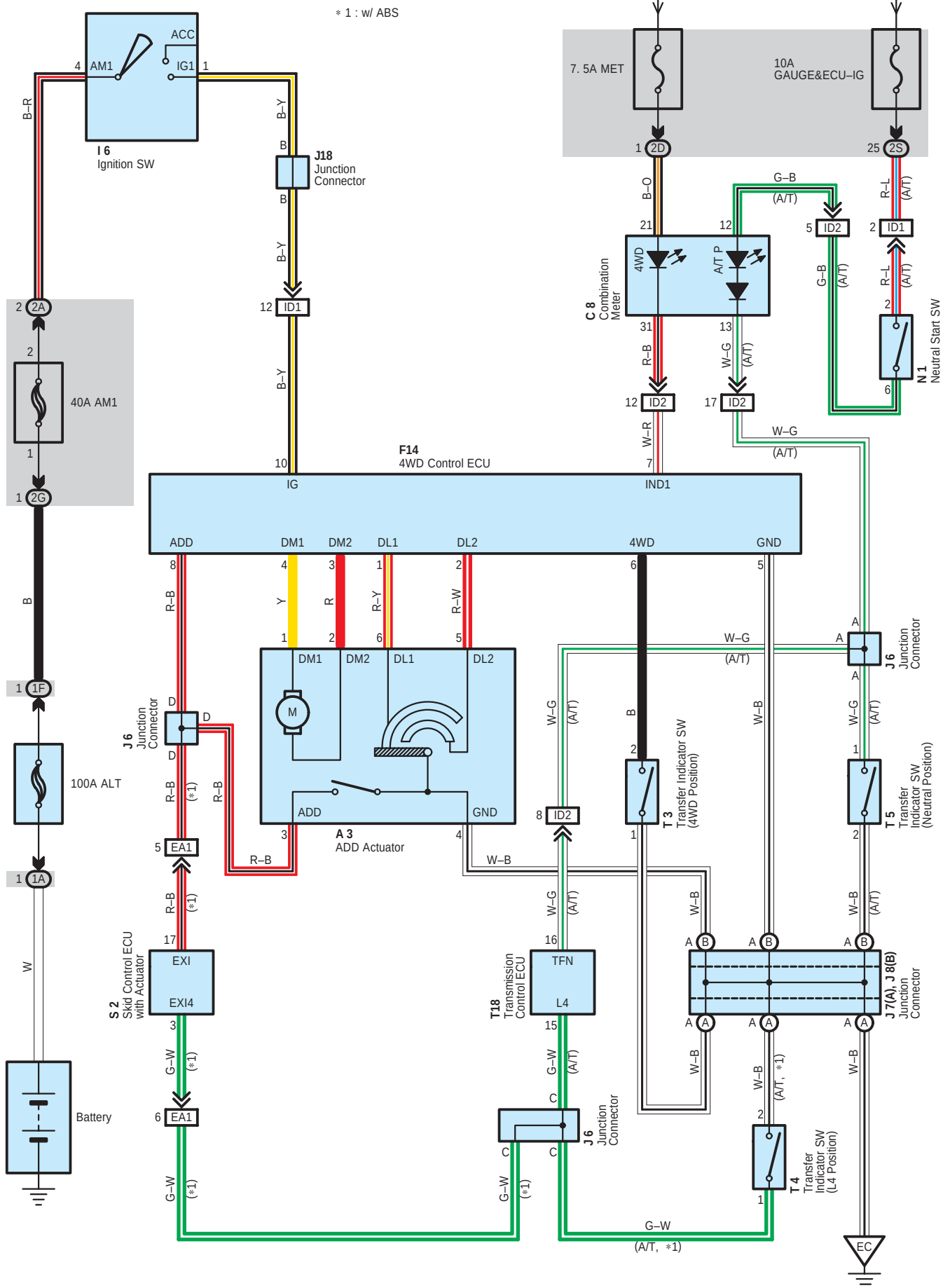
Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
IF3	50	Engine Room Main Wire and Instrument Panel Wire (Behind the Driver Side J/B)
BA1	52 (Double Cab)	Frame Wire and Floor Wire (Under the Front Seat LH)
	54 (Extra Cab)	
	56 (Single Cab)	

▽ : Ground Points

Code	See Page	Ground Points Location
EA	46	Right Side of the Fender Apron
EB	46	Left Side of the Fender Apron
ID	48	Instrument Panel Reinforcement Center
IE	48	Instrument Panel Reinforcement RH
IG	48	Cowl Side Panel LH

○ : Splice Points

Code	See Page	Wire Harness with Splice Points	Code	See Page	Wire Harness with Splice Points
I2	50	Engine Wire	B3	54 (Extra Cab)	Frame Wire
B3	52 (Double Cab)	Frame Wire		56 (Single Cab)	



System Outline

Driving mode is switched to H2 or H4 or L4 when the driver operates the transfer shift lever. Transfer indicator SW (4WD position) is turned on when the transfer shift lever is shifted in 4WD (H4 or L4).

1. 2-4 Select System

(1) Shifting from H4 to H2

When the transfer shift lever is changed from H4 to H2 position, transfer indicator SW (4WD position) detects OFF signal to activate the ADD motor, which results in changing the transfer to 2WD (H2 position).

At this time, the transfer indicator SW (4WD position) are off.

(2) Shifting from H2 to H4

When the transfer shift lever is changed from H2 to H4 position, transfer indicator SW (4WD position) detects ON signal to activate the ADD motor, which results in changing the transfer to 4WD (H4 position) to light up 4WD indicator.

(3) Shifting from H4 to L4

When the transfer shift lever is changed from H4 to L4 position, the shift fork shaft moves to change the transfer to 4WD (L4 position), which results in turning ON transfer indicator SW (L4 position). ADD motor does not operate since it is only the lever operation.

(4) Shifting from L4 to H4

When the transfer shift lever is changed from L4 to H4 position, the shift fork shaft moves. When transfer indicator SW (L4 position) is turned OFF, the transfer changes to 4WD (H4 position). ADD motor does not operate since it is only the lever operation.

2. Other Control

When abnormality is found in the 4WD system, the driver is warned by 4WD indicator light's flashing.

Service Hints

F14 4WD Control ECU

10-Ground : Approx. 12 volts with the ignition SW at ON position

5-Ground : Always continuity

T3 Transfer Indicator SW (4WD Position)

2-1 : Closed with the transfer shift lever at H4 position

T4 Transfer Indicator SW (L4 Position)

2-1 : Closed with the transfer shift lever at L4 position

○ : Parts Location

Code	See Page	Code	See Page	Code	See Page
A3	36	J7	A	T3	37
C8	38	J8	B	T4	37
F14	39	J18		T5	37
I6	38	N1		T18	39
J6	38	S2			

○ : Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
1A	23	Engine Room Main Wire and Engine Room J/B (Engine Compartment Left)
1F		
2A	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2D		
2G	28	Engine Room Main Wire and Driver Side J/B (Instrument Panel Brace RH)
2S	29	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)

□ : Connector Joining Wire Harness and Wire Harness

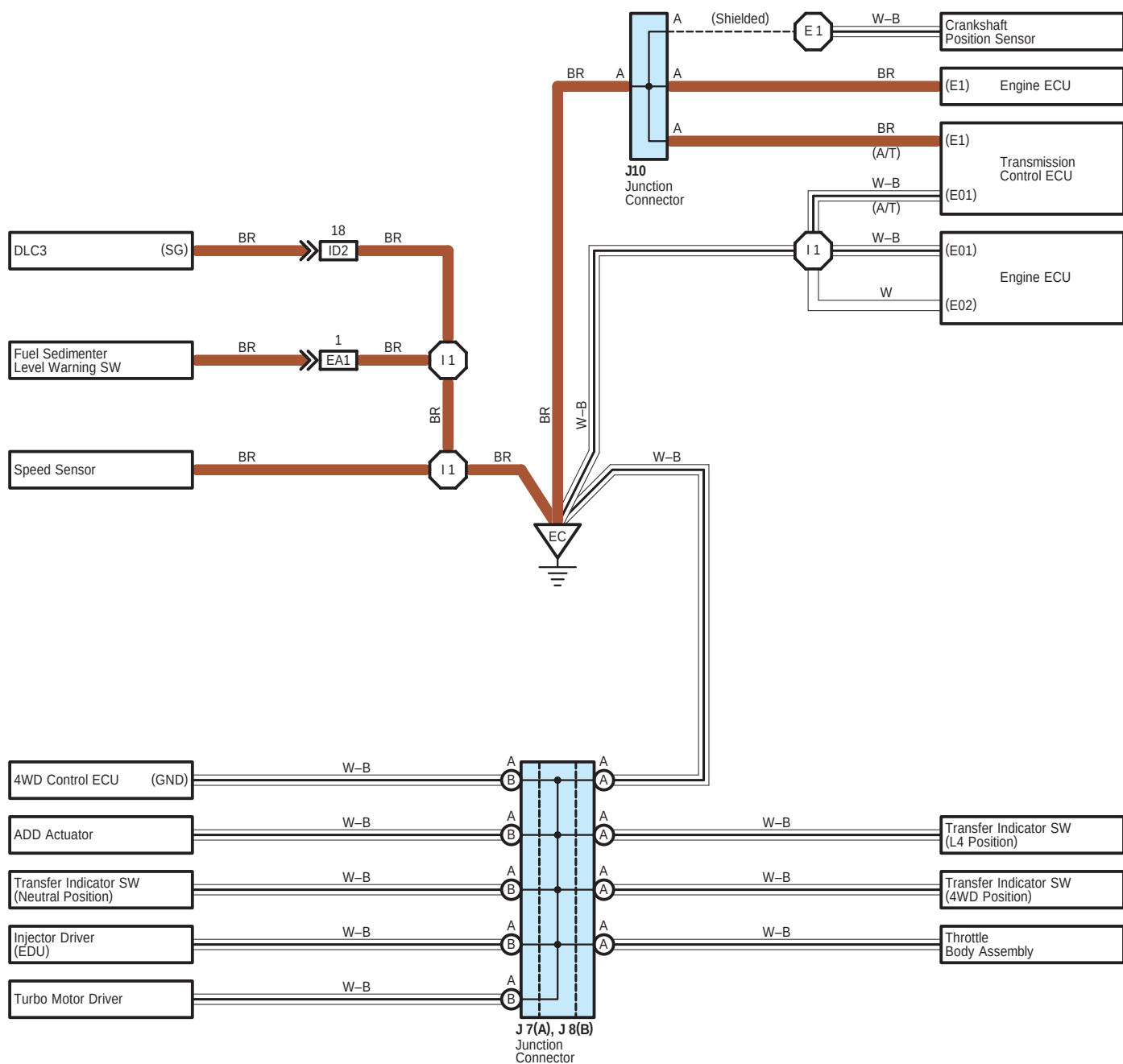
Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
EA1	46	Engine Wire and Engine Room Main Wire (Inside of Engine Room R/B and Engine Room J/B)
ID1	48	Engine Wire and Instrument Panel Wire (Behind the Glove Box)
ID2		

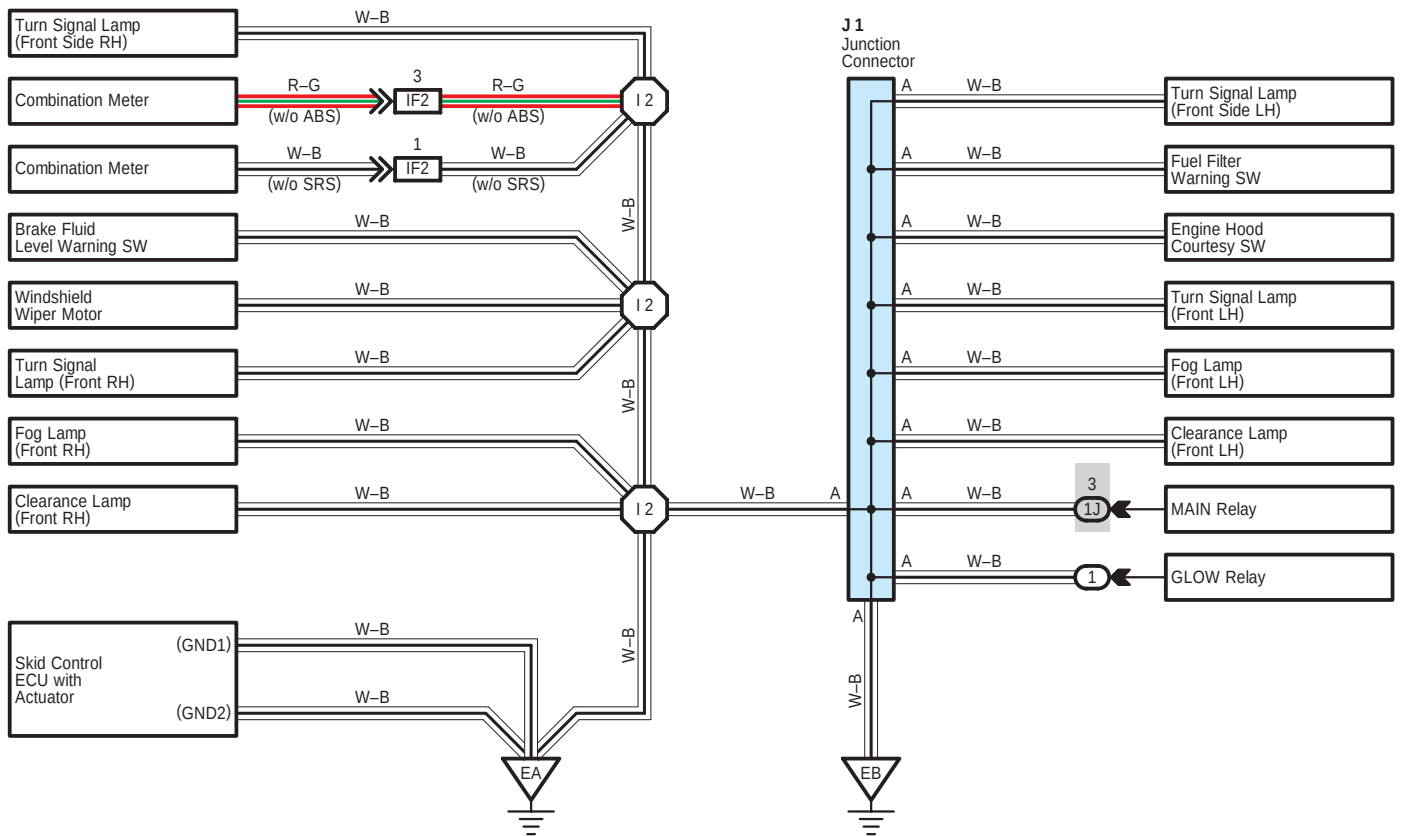


: Ground Points

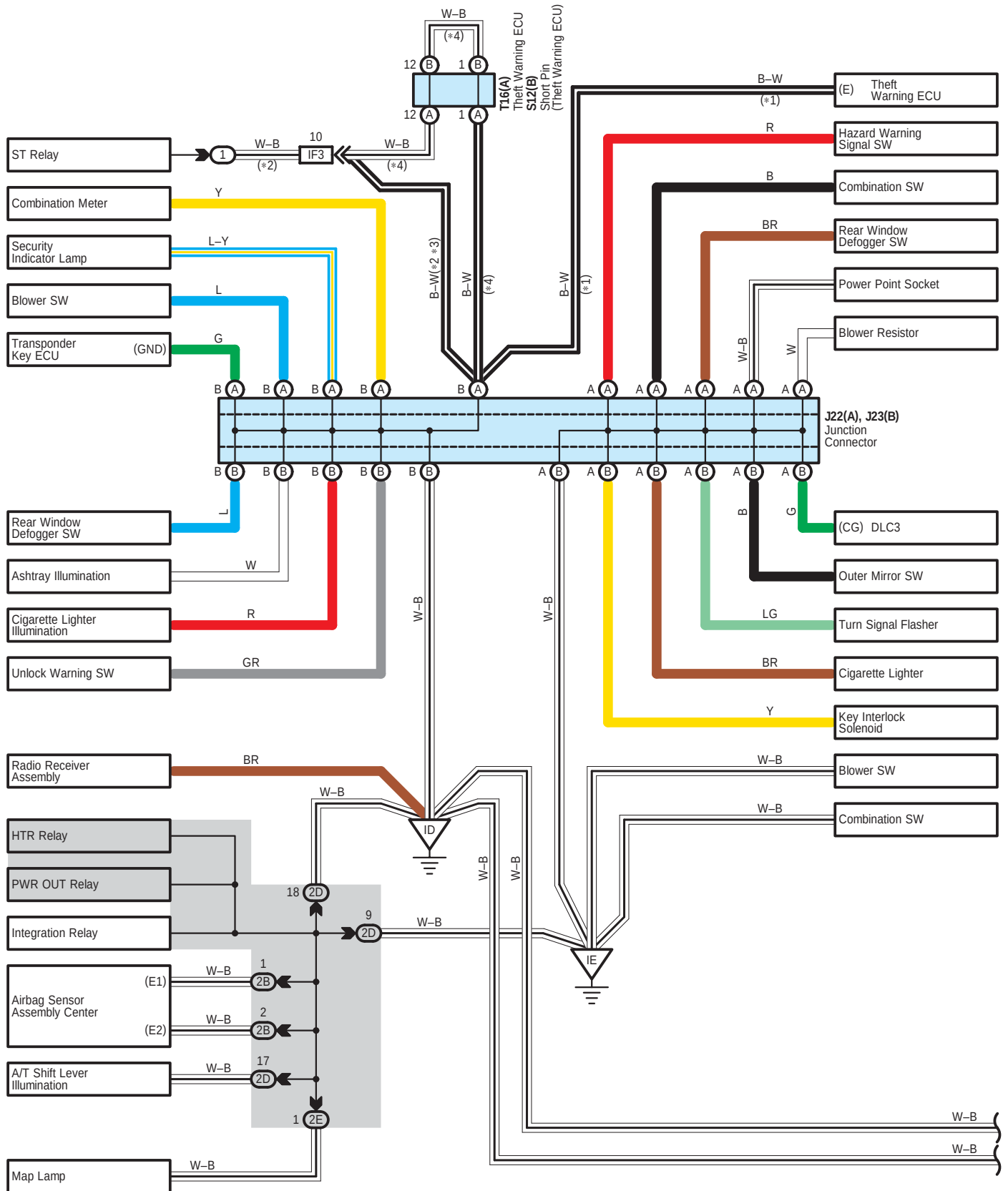
Code	See Page	Ground Points Location
EC	46	Rear Side of the Cylinder Block

I GROUND POINT

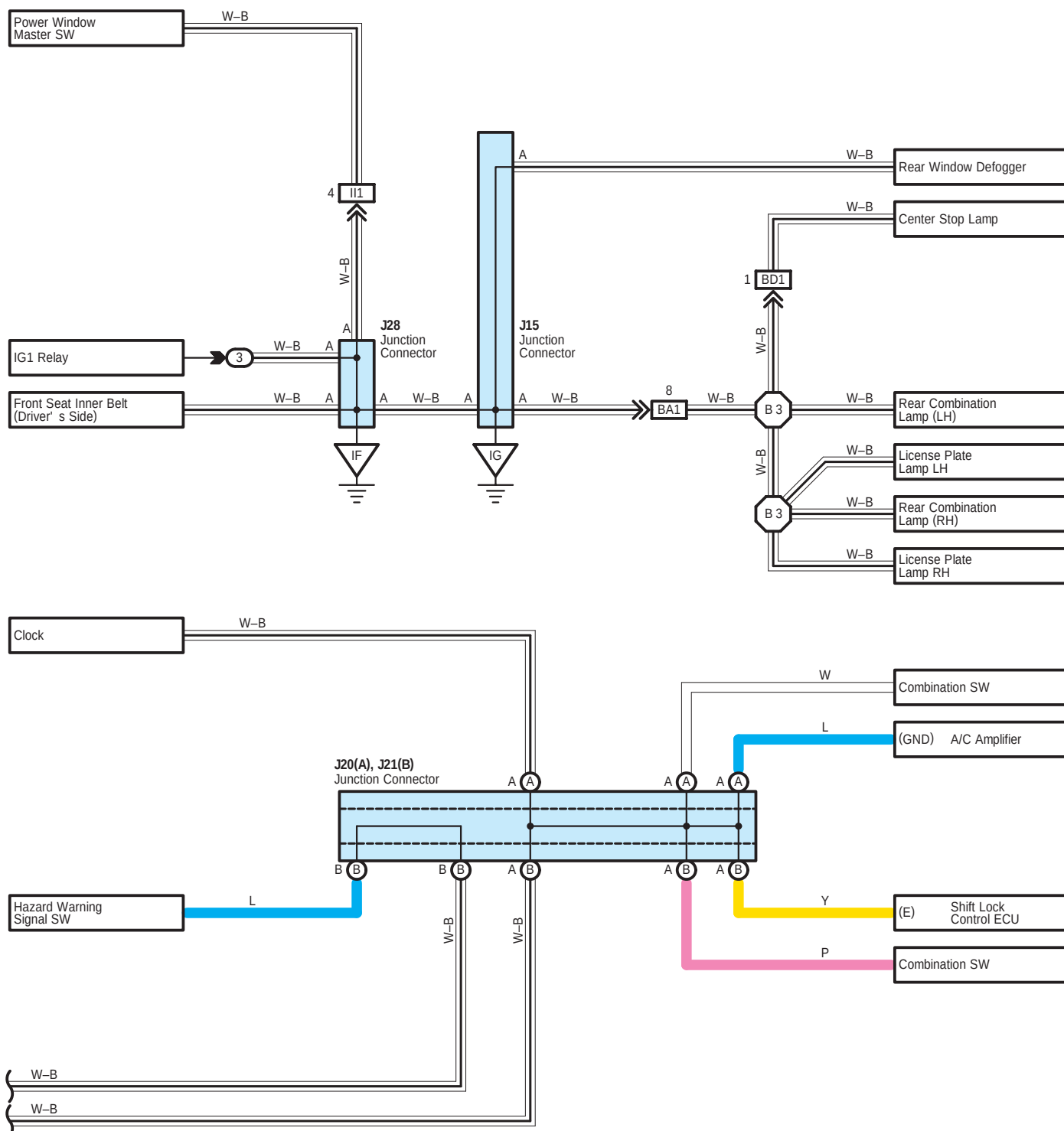




I GROUND POINT



- * 1 : w/ Theft Deterrent System
- * 2 : w/o Theft Deterrent System
- * 3 : M/T Single Cab, Double Cab 2KD-FTV w/o Inter Cooler
- * 4 : * 2 Except *3



I GROUND POINT

: Parts Location

Code	See Page	Code	See Page	Code	See Page
J1	37	J15	38	J23	B 38
J7	A 38	J20	A 38	J28	38
J8	B 38	J21	B 38	S12	B 39
J10	38	J22	A 38	T16	A 39

: Relay Blocks

Code	See Page	Relay Blocks (Relay Block Location)
1	22	Engine Room R/B (Engine Compartment Left)
3	35	R/B No.3 (Cowl Side Panel LH)

: Junction Block and Wire Harness Connector

Code	See Page	Junction Block and Wire Harness (Connector Location)
1J	24	Engine Room Main Wire and Engine Room J/B (Engine Compartment Left)
2B	28	Instrument Panel Wire and Driver Side J/B (Instrument Panel Brace RH)
2D		
2E		

: Connector Joining Wire Harness and Wire Harness

Code	See Page	Joining Wire Harness and Wire Harness (Connector Location)
EA1	46	Engine Wire and Engine Room Main Wire (Inside of Engine Room R/B and Engine Room J/B)
ID2	48	Engine Wire and Instrument Panel Wire (Behind the Glove Box)
IF2	50	Engine Room Main Wire and Instrument Panel Wire (Behind the Driver Side J/B)
IF3		
II1	50	Front Door RH Wire and Floor Wire (Right Kick Panel)
BA1	52 (Double Cab)	Frame Wire and Floor Wire (Under the Front Seat LH)
	54 (Extra Cab)	
	56 (Single Cab)	
BD1	52 (Double Cab)	Frame No.2 Wire and Frame Wire (Under the Back Panel LH)
	54 (Extra Cab)	
	56 (Single Cab)	

: Ground Points

Code	See Page	Ground Points Location
EA	46	Right Side of the Fender Apron
EB	46	Left Side of the Fender Apron
EC	46	Rear Side of the Cylinder Block
ID	48	Instrument Panel Reinforcement Center
IE	48	Instrument Panel Reinforcement RH
IF	48	Cowl Side Panel RH
IG	48	Cowl Side Panel LH

: Splice Points

Code	See Page	Wire Harness with Splice Points	Code	See Page	Wire Harness with Splice Points
E1	46	Engine Wire	B3	52 (Double Cab)	Frame Wire
I1	50			54 (Extra Cab)	
I2				56 (Single Cab)	